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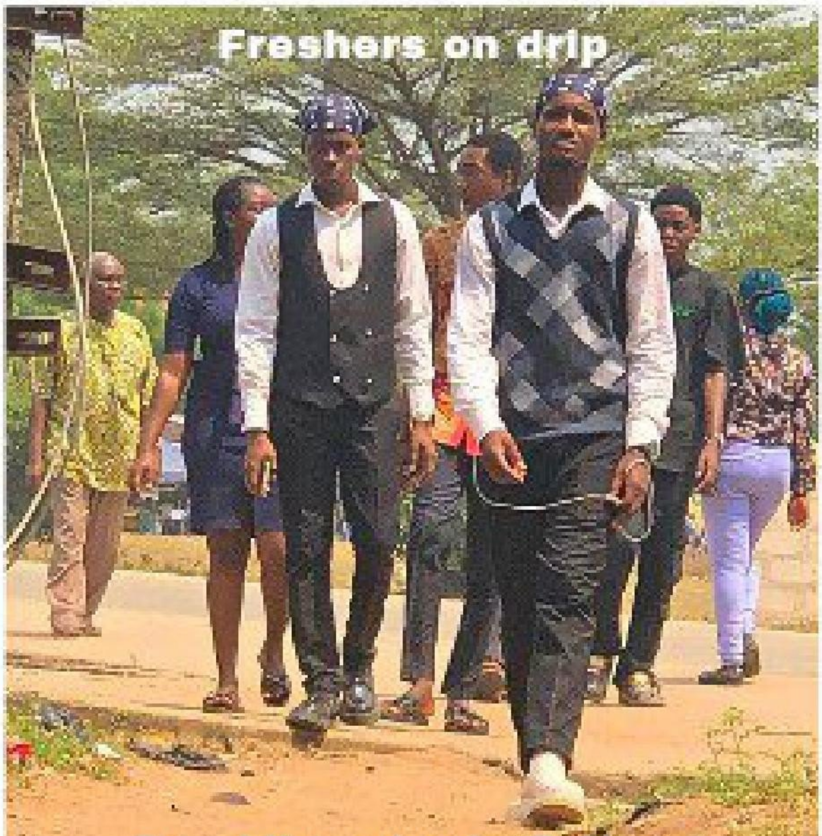

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JAY'S CRIBB

STA 211: TABLE OF CONTENT/COURSE OUTLINE

STA 211: INTRODUCTION

THE MEANING AND NATURE OF STATISTICS

Statistics is defined as the study or mathematical body that pertains to the collection, observation or organization, presentation, analysis, interpretation or explanation of data. In the narrow sense, the term statistics is used to **denote the data themselves or number derived from the data.**

POPULATION AND SAMPLE

In collecting data concerning the characteristics of a group of individuals or objects, weight or height in the different variations, we observed in two ways; population and sample.

i. **Population:** is the set of all possible observations of a defined type (e.g Population Census) and describing the characteristics based on the population value is known as <u>*Parameter*</u> e.g μ, σ, σ^2.
ii. **Sample:** is only a part or subset of a population (e.g Sample Survey) and describing the characteristics based on the sample value is known as <u>*Statistic*</u> e.g $\bar{x}, S, S^2$.

METHOD OF DATA COLLECTION

What is Data? Data is simplified as the set off information based on the background characteristics of the observations. Characteristics of a variable can be defined as the properties of the observation that is being measured or observed. The characteristics of observations in the data collected can be classified into two;

i. **Qualitative form:** the properties of the observation are being observed e.g Christianity, Islamic, rich, poor etc.
ii. **Quantitative form:** the properties of the observation are being measured (numbered) e.g Weight, Height, Age of person etc.

LEVEL OF MEASUREMENTS

There are four main type of level of measurement used in statistics;

1) Nominal Scale (Words, 0/1- Yes/No, True/False, region- western, eastern, etc)
2) Ordinal Scale (1st, 2nd, 3rd, etc)
3) Interval Scale
4) Ratio Scale

SOURCE OF DATA

The data collected could be classified in two categories; Primary source and Secondary source.

1. **Primary source:** the source may include observations, mail questionnaire and personal interview. Primary data are original data that have been collected specifically for the purpose in the mind of the researcher,
2. **Secondary source:** refers to data from documents taken as second hands or pre-existing information from figures originally collected by someone else. The data gotten from the secondary source is called Secondary data. The secondary data can be obtained from published Statistical data, records of activities of government.

BASIC TERMINOLOGY

I. **Survey:** this refers to the collection of data when no special control is exercised over any of the factors influencing the variable of interest.
II. **Experiment:** this refers to the collection of data when control is exercised over any of the factors influencing the variable of interest.
III. **Variable:** is when the characteristics change from one observation to another.
IV. This involves the direct measurement of an ongoing activity.

PRESENTATION OF DATA

In this, we would like to consider the various form of presenting the collected data. There are 3 main forms in which data can be presented – tabular form, graphical form and form of equations.

TABULAR FORM

The type of table to be constructed will depend on the nature of the data and most tables have common features; titles, rows and columns, indications of unit of measurement and probably the source of data.

FREQUENCY DISTRIBUTION OF BOTH GROUP AND UNGROUP DATA

Frequency Distribution is the arrangement of numerical data to show the number of times each event occurred in a data set. Frequency distribution is derived from raw data and array data.

e.g

Raw Data	Array Data
20 44 91 92 58	20 20 23 24 24
58 48 24 48 20	29 30 44 44 48
29 44 24 23 30	48 58 58 91 92

When summarizing large data, it is often useful to distribute the data into classes or categories and to determine the number of individuals belonging to each class (e.g Age class) is called Class Frequency. Frequency distribution is a tabular arrangement of data by classes together with the corresponding class frequencies is known as Frequency table.

Example; given the distribution is a tabular arrangement of age of mothers

Table: Distribution of Age of mothers

Age	Class frequency (f)
20	2
23	1
24	2
29	1
30	1
44	2
48	2
58	2
91	1
92	1
Total	15 *observations*

Ungroup Data

GROUP DATA

The data is organized and summarized from the frequency distribution that falls within the same interval, for example;

Table: Age of patients examined at FMC Owerri

Age group	f
5 – 9	3
10 – 14	33
15 – 19	21
20 – 24	28
25 – 29	15
Total	97

To construction of a frequency distribution table one has to put into consideration the appropriate terms like;

a. **Class Limit:** this is the value of the upper limit and lower limit of a particular class. For example, we have 60 – 69,60 is the lower class limit and 69 is the upper class limit.
b. **Class Interval:** this is the range divided the number of classes.

$$class\ interval = \frac{Range}{k}$$

Where $k = 1 + 3.22\log n =$ the number of classes

c. **Class Mark:** It is known as the midpoint of a class limit. It can be calculated as;

$$C_k = \frac{UCl + LCl}{2}$$

Where UCl is the Upper Class Limit and LCl is the Lower Class Limit

d. **Class Boundary:** It represent the decimal point of the modal class limit and it is obtained by dividing the size of the gap between the consecutive classes by two and subtracting the quotient from the lower class limit of each class and adding the quotient of the upper class limit of the corresponding class. For example, in Table, if given

$$20 - 24;\ (20 - 0.5) - (24 + 0.5)$$
$$= 19.5 - 24.5 \text{ is the class boundary}$$

e. **Class Size:** $C = [(UCl - LCl) + 1]$ or $(UCb - LCb)$
f. **The Cumulative Frequency** of a class is obtained by adding its frequency to the sum of the frequencies of the class before it.
Example; in the table, given the first class $= 5$
The second class $= 5 + 9 = 14$
The third class $= 5 + 9 + 13 = 14 + 13 = 27$
etc.

g. **The Relative Frequency** of a class is the frequency of that class divided by the total frequency.

$$R_f = \frac{f}{N} = \frac{class\ frequency}{total\ frequency}$$

h. **The Percentage Relative Frequency** is the relative frequency of a particular class multiplied by 100. $\%R_f = 100 \times R_f$

EXAMPLE:

In a survey of monthly rents payable in a certain town, the following sample data (in Naira) were collected for 48 residential flats in the town

200 250 335 230 270 190 270 490
255 240 445 290 256 245 310 480
340 580 280 235 274 220 435 525
330 260 310 172 405 295 385 535
160 250 190 210 390 230 280 460
315 370 345 355 232 170 270 362

SOLUTION:
If we agree to use a class interval of ₦50,

$$Range = 580 - 160 = 420$$

$class\ interval = \frac{Range}{k}$, then making k the subject of the formula;

$$k = \frac{Range}{class\ interval} = \frac{420}{50} = 8.4 \sim 9\ classes$$

Table: Frequency distribution table for group data

Class limit (Cl)	f	Class boundary (Cb)	Cum. Freq. (C_f)	Relative frequency (R_f)	Percentage Relative frequency ($\%R_f$)
150 – 199	5	149.5 – 199.5	5	0.1042	10.42
200 – 249	9	199.5 – 249.5	14	0.1875	18.75
250 – 299	13	249.5 – 299.5	27	0.2708	27.08
300 – 349	7	299.5 – 349.5	34	0.1458	14.58
350 – 399	5	349.5 – 399.5	39	0.1042	10.42
400 – 449	3	399.5 – 449.5	42	0.0625	6.25
450 – 499	3	349.5 – 499.5	45	0.0625	6.25
500 – 549	2	499.5 – 549.5	47	0.0417	4.17
550 – 599	1	549.5 – 599.5	48	0.0208	2.08
Total	48			1.0000	100.00

GRAPHICAL FORM

Data can be represented graphically using different types of graphs or pictorial representations. Among the most common used of graphs are; Histograms, Bar charts, Pie charts, Frequency polygons, Cumulative frequency polygon and Scatter diagram.

Histogram: This is a graph of a frequency distribution which is obtained by plotting frequency against class interval. If the group widths are equal, frequency is plotted along the vertical axis and the variables on horizontal axis. Thus a Histogram consists of set of rectangles.

Example; given the distribution of examination score

Table: Frequency distribution of examination score

Scores	f	Cb
10 – 19	3	9.5 – 19.5
20 – 29	7	19.5 – 29.5
30 – 39	10	29.5 – 39.5
40 – 49	15	39.5 – 49.5
50 – 59	12	49.5 – 59.5
60 – 69	9	59.5 – 69.5
70 – 79	6	69.5 – 79.5
80 – 89	4	79.5 – 89.5

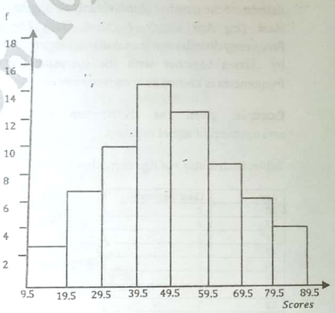


Figure: Histogram for examination score

Bar charts: In the construction of Bar charts, bars are drawn proportional to the magnitude of the values under consideration. The bars are normally separated by equal intervals and their heights are drawn proportional to the size of the corresponding observation. There are various types of bar charts in use; Simple bar charts, Component bar charts, Compound bar charts.

Illustrative Examples

(i) Simple Bar Chart:

Table: School Enrolment

School	School Enrolment		Total
Year	Male	Female	
1991	1500	1000	2500
1992	1800	1300	3100
1993	1900	1500	3400
1994	1700	1100	2800
1995	2000	1200	3200

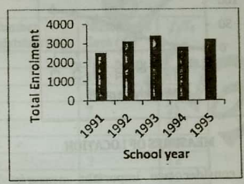


Figure: Simple bar Chart

(ii) Compound (Composite or Multiple) Bar Chart

In the construction of composite bar charts the sub-total are used instead of the totals. The sub-total bars for each year or time have common boundaries but are separated from the bars of the other years.

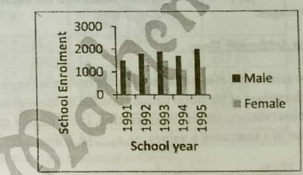


Figure: Composite Bar chart

(iii) Component Bar Chart:

In the case each bar is sub-divided into parts or components in proportion to the size or magnitude of the items of interest. For example, Table represents the tax collated (in million naira).

Example; Table represents the tax collected in million ₦

Table: Revenue Collection

Fiscal	Revenue (₦ million)			
Year	PAYE	RT	DT	TOTAL
1981	6.5	4.8	3.2	14.5
1982	8.4	5.6	3.8	17.8
`1983	7.8	5.2	3.4	16.4
1984	8.2	5.8	3.9	17.9

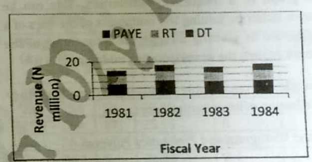


Figure: Component Bar Chart

Pie Chart: To construct pie chart a circle of convenient radius is sub-divided into sectors in proportion to the relative contributions of the items under consideration.

Example; Table: production (in coded units)

Items	Quantity	Degrees
Gas	17.0	85^0
Kerosene	12.0	60^0
Gasoline	15.1	76^0
Coal	18.2	91^0
Tin	9.7	48^0
Total	72.0	360^0

To calculate the angles of various sector, therefore;

$$Gas = \frac{17}{72} \times 360^0 = 85^0$$

$$Kerosene = \frac{12}{72} \times 360^0 = 60^0$$

In general; $item = \frac{quantity}{Total\ quantity} \times 360^0$

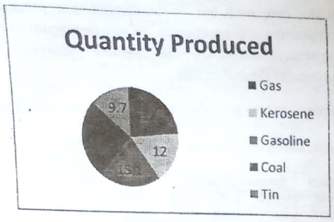
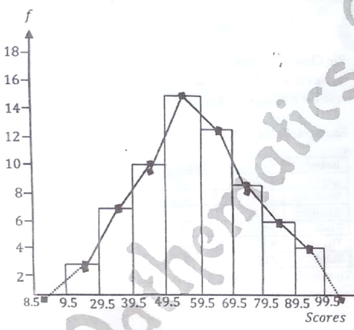


Figure: Pie Chart

Frequency polygon: this is another way of presenting the graph of a frequency distribution. This can be drawn either by joining the midpoints of the bars of a Histogram or plotting directly the frequencies of the classes against the corresponding midpoints.

Example; from Table showing the frequency distribution of examination score.

Figure: Histogram and Frequency Polygon

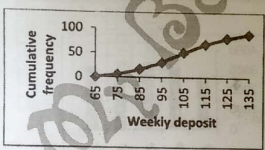


Cumulative Frequency Polygon (Ogive): the cumulative frequency graph of a distribution is called an Ogive. The most common form is constructed by plotting the cumulative frequency of each class against the corresponding upper class boundary.

Example; distribution of weekly deposit (in thousand ₦)

Table: Cumulative frequency

Weekly deposit in ₦'000	f	C_f
65 and under 75	6	6
75 and under 85	10	16
85 and under 95	14	30
95 and under 105	20	50
105 and under 115	18	68
115 and under 125	12	80
125 and under 135	8	88
Total	88	



MEASURES OF LOCATION

We shall consider two important measures; the measure of central tendency and the measure of partition.

MEASURE OF CENTRAL TENDENCY

This indicates the location of a frequency distribution in terms of a value that signifies the central location of a frequency distribution. Among the measures of central position we shall consider in this section are; the Arithmetic Mean, the mode, the median, the Geometric mean, the Harmonic mean and the Moving Averages.

The Arithmetic Mean

The arithmetic mean of a set of observations is the sum of the values of the observations divided by the number of observations under consideration. Let $\bar{X}$ denote the mean of a sample of n values $x_1, x_2, \ldots, x_n$, then

$\bar{X} = \frac{\sum_{i=1}^{n} x_i}{n}$; $for\ i = 1, 2, \ldots, n$

Where $\sum$ denotes summation

Example; the grade point average of 8 students selected from the graduating class are as follows;

2.38, 2.64, 3.45, 3.10, 1.86, 2.52, 2.34, 3.24

$\bar{X} = \frac{1}{8}(2.38 + 2.64 + 3.45 + 3.10 + 1.86 + 2.52 + 2.34 + 3.24)$

$= \frac{1}{8}(21.53) = 2.69$

In practice however, let $\bar{X}$ denotes the mean of a sample size of n observation $x_1, x_2, \ldots, x_n$ with corresponding frequencies $f_1, f_2, \ldots, f_n$, then;

$\bar{X} = \frac{\sum_{i=1}^{n} x_i}{\sum_{i=1}^{n} f_i}$; $for\ i = 1, 2, \ldots, n$

Where $\sum_{i=1}^{n} f_i$ is the total frequency.

Example; Table Distribution of mistakes by a typist

Number mistakes (X_i)	Number of days (f_i)	f_iX_i
0	14	0
1	10	10
2	8	16
3	7	21
4	5	20
5	4	20
6	2	12
Total	50	99

$\bar{X} = \frac{\sum_{i=1}^{n} x_i}{\sum_{i=1}^{n} f_i} = \frac{99}{50} = 1.98$ is the mean number of mistakes

For grouped data

First we obtain the class midpoints. These form the x_i values; and then the computation follows as in the above example;

Example; Table Mean for grouped data

Weight (Kg)	Number of bags (f_i)	Class midpoint (X_i)	f_iX_i
25.5 – 25.6	5	25.55	127.75
25.7 – 25.8	10	25.75	257.50
25.9 – 26.0	17	25.95	441.15
26.1 – 26.2	13	26.15	339.95
26.3 – 26.4	6	26.35	158.10
26.5 – 26.6	4	26.55	106.20
Total	55		1,430.65

For the class midpoint $= \frac{UCI+LCI}{2}$

For the first class $= \frac{25.5+25.6}{2} = 25.55$

Mean, $\bar{X} = \frac{\sum_{i=1}^{n} f_iX_i}{\sum_{i=1}^{n} f_i} = \frac{1,430.65}{55} = 26.012$ (approximately)

Mean by method of change of origin

To avoid handling large values in the process of computing the mean, it is of great advantages, in speed and accuracy, to employ the method of change of origin.

The method of change of origin gives the mean as;

$\bar{X} = A + \frac{\sum_{i=1}^{n} f_id_i}{\sum_{i=1}^{n} f_i}$; $for\ i = 1, 2, \ldots, n$

Where A is a fixed value or known as the assumed mean

$d_i = x_i - A$; the deviation of the fixed value of each value of x_i

Example; Table: Mean by method of change of origin.
Let $A = 25.95kg$

Weight (kg)	f_i	Class midpoints (x_i)	$d_i = x_i - A$	f_id_i
25.5 – 25.6	5	25.55	-0.4	-2
25.7 – 25.8	10	25.75	-0.2	-2
25.9 – 26.0	17	25.95	0	0
26.1 – 26.2	13	26.15	0.2	2.6
26.3 – 26.4	6	26.35	0.4	2.4
26.5 – 26.6	4	26.55	0.6	2.4
Total	55			3.4

With $A = 25.95$, then $d_i = x_i - 25.95$

$\bar{X} = 25.95 + \frac{3.4}{55} = 26.012$ (approximately)

Another computation can be further simplified by taking into account the fact that the data being grouped in classes of equal class interval and so we can define $\bar{X}$ as;

$\bar{X} = A + \frac{\sum_{i=1}^{n} f_iU_i}{\sum_{i=1}^{n} f_i} \times C$; $for\ i = 1, 2, \ldots, n$

C is the class size of the interval

Thus $U_i = \frac{1}{c}(x_i - A)$

Median

This is the observation which occupies the central position in an ordered set of data. It is the 50-percentile or otherwise called the second quartile. If there is an ordered set of data (X_i) with an odd number of n

observation, then $M_e = X_{\left(\frac{n+1}{2}\right)}$ and if the number of observation is even, then

$$M_e = {}^1\!/_2\left(X_{\left(\frac{n}{2}\right)} + X_{\left(\frac{n}{2}\right)+1}\right)$$

Example; the point scored by seven (7) ladies in a beauty contest are;

3, 9, 2, 7, 5, 2, 6 – arranged in ordered set will be; 2, 2, 3, 5, 67, 9

$$M_e = X_{\left(\frac{n+1}{2}\right)} = X_{\left(\frac{7+1}{2}\right)} = X_4 = 5.$$

Example; if there had been 8 ladies and the ordered set was;

2, 2, 3, 5, 6, 7, 8, 9 then,

$$M_e = {}^1\!/_2\left(X_{\left(\frac{n}{2}\right)} + X_{\left(\frac{n}{2}\right)+1}\right) = {}^1\!/_2\left(X_{\left(\frac{8}{2}\right)} + X_{\left(\frac{8}{2}\right)+1}\right)$$

$$= {}^1\!/_2\,(X_4 + X_5) = {}^1\!/_2\,(5 + 6) = 5.5$$

In practice where we have large number of observations and it would be a waste valuable time trying the above method. To locate the central position when the date is organized in a frequency table, we can make use of the cumulative frequency.

Example; Table: Distribution of the numbers of cars by a sample of 90 families.

Number of Cars (X_i)	Number of Families (f_i)	C_f
0	38	38
1	30	68
2	12	80
3	5	85
4	3	88
5	2	90

To determine the median number of cars, then using earlier idea, where 90 observations is even:

Therefore, $M_e = {}^1\!/_2\left(X_{\left(\frac{n}{2}\right)} + X_{\left(\frac{n}{2}\right)+1}\right)$ for $n = 90$

$$= {}^1\!/_2\,(X_{45} + X_{46})$$

From the cumulative frequency, the median values X_{45} and X_{46} falls under $C_f = 68$, so therefore the median is 1 *car*.

FOR GROUPED DATA

The median of a grouped data can be calculated using the formula; $M_e = l_m + \frac{\left(\frac{N}{2}\right) - C_{f_{m-1}}}{f_m} \times C$

Where l_m is the lower class boundary of the median class,
N is the total frequency
$C_{f_{m-1}}$ is the cumulative of the pre-median class
f_m is the frequency of the median class.
C is the class size.
Example; Table: frequency distribution of driving test passes

Age	Number of drivers (f)	Cb	C_f
11 – 15	7	10.5 – 15.5	7
16 – 20	15	15.5 – 20.5	22
21 – 25	32	20.5 – 25.5	54
26 – 30	44	25.5 – 30.5	98
31 – 35	30	30.5 – 35.5	128
36 – 40	23	35.5 – 40.5	151
41 – 45	14	40.5 – 45.5	165
46 – 50	9	45.5 – 50.5	174
51 – 55	2	50.5 – 55.5	176
Total	176		

$C = UCb - LCb = 15.5 - 10.5 = 5.0$

The median in the class lies between X_{88} and X_{89}. So therefore, the median class = (26 – 30)from the C_f = 98; $Cb = (25.5 - 30.5)$; *thus* $l_m = 25.5$; $N = 176$; $C_{f_{m-1}} = 54$; $f_m = 44$.

$$M_e = l_m + \frac{\left(\frac{N}{2}\right) - C_{f_{m-1}}}{f_m} \times C = 25.5 + \frac{\left(\frac{176}{2}\right) - 54}{44} \times 5.0 = 29.364$$

Mode

This the observation with the highest frequency; that is, the most common among a set if data. For grouped data, the mode can be estimated graphically by using Histogram and calculating using a formula.

Example; Given a test score of 12 students in biology lab in ordered set

1, 2, 2, 4, 5, 5, 5, 5, 6, 6, 6, 8

$mode = 5$

MODE BY FORMULA METHOD

The mode (M_o) of a grouped data can be computed by using the formula;

$$M_o = l_m + \left(\frac{D_1}{D_1 + D_2}\right) \times C$$

Where l_m is the lower class boundary of the modal class

D_1 is the frequency of the modal class minus frequency of the pre-modal class, i.e.

$$D_1 = f_m - f_{m-1}$$

D_2 is the frequency of the modal class minus frequency of the post modal class, i.e.

$$D_2 = f_m - f_{m+1}$$

The Geometric mean

Example:

Find the geometric mean of the numbers 3, 5, 6, 6, 7, 10 and 12.

Solution:

Geometric mean $(\sum X)^{\frac{1}{N}}$

$$G = \sqrt[7]{(3)(5)(6)(6)(7)(10)(12)}$$

$$G = \sqrt[7]{453{,}600}$$

$$G = 6.43$$

The Harmonic mean

Example:

Find the harmonic mean of the numbers 3, 5, 6, 6, 7, 10 and 12.

Solution:

$$\text{Harmonic mean, } \frac{1}{H} = \frac{1}{N}\sum \frac{1}{X}$$

$$\frac{1}{H} = \frac{1}{7}\left(\frac{1}{3} + \frac{1}{5} + \frac{1}{6} + \frac{1}{6} + \frac{1}{7} + \frac{1}{10} + \frac{1}{12}\right)$$

$$\frac{1}{H} = \frac{1}{7}\left(\frac{140 + 84 + 70 + 70 + 60 + 42 + 35}{420}\right)$$

$$\frac{1}{H} = \frac{501}{2940}$$

$$\Rightarrow H = \frac{2940}{501} = 5.87$$

COMPARISM OF MEASURE OF CENTRAL TENDENCY

We try to emphasize on the merit and demerit of Mean , Median and Mode.

Mean

Merit: - The mean is the sufficient statistics
The mean of any data is unique.
Demerit: - Generally, it is affected by outliers (extreme values).
A non-integral value may be obtained as the mean when the data are discreet.

Median

Merit: - The median of any data is unique.
The median is preferred when the data is discreet.
Demerit: - The median is not a sufficient statistic.
It does not easily learn itself to Algebraic manipulation.

Mode

Merit: - The mode is easily obtained in an un-rolled or raw data set.
It is not affected by outliers.
Demerit: - The mode may not be unique since the set of the data may be bi-modal. It is not a sufficient statistic

ROOT MEAN SQUARE

$$Rms = \sqrt{\frac{\sum x^2}{N}}$$

Example:

The root mean square of the set 1, 3, 4, 5 and 7 is:

Solution:

$$Rms = \sqrt{\frac{\sum x^2}{N}} = \sqrt{\frac{1^2 + 2^2 + 3^2 + 4^2 + 5^2}{5}} = \sqrt{20}$$

$$= 4.47$$

Note: The relationship between arithmetic, Geometric and Harmonic mean is $H \leq G \leq \bar{X}$

MEASURES OF PARTITION

In an ordered set of data, the values are partitioned in various parts in the same manner like;

a) Quartiles
b) Deciles
c) Percentiles.

QUARTILES

These are measured in partitioning the set of data into 4 equal parts (i.e, $\frac{iN}{4}$; $for\ i = 1,2,3$). Lets apply the measures of quartiles on ungrouped and grouped data.

Ungrouped Data

Suppose the N data items are $x_1, x_2, \cdots, x_n$, then the measures includes;

i. First or Lower Quartile: this is denoted by $Q_1 = \frac{1N}{4}$ of the units in the distribution.
ii. Second Quartile: This is denoted by $Q_2 = \frac{2N}{4} = \frac{N}{2}$ (this can also be known as the median of the distribution), that is ,$Q_2 = median$.
iii. Third or Higher Quartile: This is denoted by $Q_3 = \frac{3N}{4}$ of the unit in the distribution.

Example; A sample of 12 paycards of the staff of a branch of a commercial bank revealed the following monthly salaries in (N' 000); $24, 18, 17, 26, 96, 18, 32, 29, 27, 20, 23, 28$. (raw data)

Determine the;
1. First quartile Q_1
2. Second quartile Q_2
3. Third quartile Q_3

Solution

Lets arrange the data given above: $17, 18, 18, 20, 23, 24, 26, 27, 28, 29, 32, 96$. (array data)

When $N = 12$, then $Q_1 = \frac{1N}{4} = \frac{1(12)}{4} = 3$.

Note: the lower quartile Q_1 of the distribution is the 25^{th} percentile of the distribution. Thus, the 3^{rd} position of the sample of paycards in the array data is**18.**

Then, $Q_1 = 3 = N\ 18{,}000$

For $Q_2 = \frac{2N}{4} = \frac{2(12)}{4} = 6$.

Note: the second quartile Q_2 of the distribution is the 50^{th} percentile of the distribution. Thus, the 6^{th} position of the sample of paycards in the array data is **24**.

Then, $Q_2 = 6 = N\ 24{,}000$

For $Q_3 = \frac{3N}{4} = \frac{3(12)}{4} = 9$.

Note: the Third quartile Q_3 of the distribution is the 75^{th} percentile of the distribution. Thus, the 9^{th} position of the sample of paycards in the array data is 28.

Then, $Q_3 = 9 = N\ 28{,}000$

Grouped Data

Suppose the N data items are $x_1, x_2, \cdots, x_n$, then the measures of Quartile of a grouped data are estimated as;

$$Q_i = l_m + \left(\frac{\frac{iN}{4} - Cf_{m-1}}{f_m}\right)C \; ; \qquad for\ i = 1,2,3.$$

Where l_m is the lower class boundary of the quartile class

Cf_{m-1} is the cumulative frequency before the quartile class

f_m is the frequency of the quartile class

C is the class size of the group data.

For $i = 1$, is the lower quartile

$$Q_1 = l_m + \left(\frac{\frac{1N}{4} - Cf_{m-1}}{f_m}\right)C$$

For $i = 2$, is the Second quartile

$$Q_2 = l_m + \left(\frac{\frac{2N}{4} - Cf_{m-1}}{f_m}\right)C$$

For $i = 3$, is the Third quartile

$$Q_3 = l_m + \left(\frac{\frac{3N}{4} - Cf_{m-1}}{f_m}\right)C$$

DECILES

These are measured in partitioning the set of data into 10 equal parts (i.e, $\frac{iN}{10}$; $for\ i = 1,2,3,\cdots,9$). Let's apply the measures of deciles on ungrouped and grouped data.

Ungrouped Data

Suppose the N data items are $x_1, x_2, \cdots, x_n$, then the measures of deciles are denoted as;

$$D_i = \frac{iN}{10} \; ; \; for\ i = 1,2,3,\cdots,9.$$

If $i = 1$, the first decile
$D_1 = \frac{1N}{10}$
If $i = 2$, the Second decile

$$D_2 = \frac{2N}{10} = \frac{N}{5} = 1st\ Quintile$$

If $i = 3$, the Third decile

$$D_3 = \frac{3N}{10}$$

$$\vdots$$

If $i = 5$, the fifth decile

$$D_5 = \frac{5N}{10} = \frac{N}{2} = median$$

Note: the fifth decile of the class os the median of the distribution.

$$\vdots$$

If $i = 9$, the Ninth decile

$$D_9 = \frac{9N}{10}$$

Remember that the percentiles which are multiples of 10 are called **Deciles.**

Grouped Data
Suppose the N data items are $x_1, x_2, \cdots, x_n$, then the measures of Decile of a grouped data are estimated as;

$$D_i = l_m + \left(\frac{\frac{iN}{10} - Cf_{m-1}}{f_m}\right)C \; ; \qquad for\ i = 1,2,3,\cdots,9.$$

Where l_m is the lower class boundary of the Decile class
Cf_{m-1} is the cumulative frequency before the Decile class
f_m is the frequency of the Decile class
C is the class size of the group data.
For $i = 1$, is the First Decile

$$D_1 = l_m + \left(\frac{\frac{1N}{10} - Cf_{m-1}}{f_m}\right)C$$

$$\vdots$$

For $i = 9$, is the Ninth Decile

$$D_9 = l_m + \left(\frac{\frac{9N}{10} - Cf_{m-1}}{f_m}\right)C$$

PERCENTILES

These are measured in partitioning the set of data into 100 equal parts (i.e, $\frac{iN}{100}$; $for\ i = 1,2,3,\cdots,99$). Let's apply the measures of Percentiles on ungrouped and grouped data.

Ungrouped Data
Let say N data items are $x_1, x_2, \cdots, x_n$, then the measures of Percentiles are denoted as;

$$P_i = \frac{iN}{100} \; ; \; for\ i = 1,2,3,\cdots,99.$$

If $i = 1$, is the first Percentile

$$P_1 = \frac{1N}{100}$$

$$\vdots$$

If $i = 50$, *is* the fiftieth Percentile

$$P_{50} = \frac{50N}{100} = \frac{N}{2} = median$$

The fiftieth (50^{th}) Percentile is also known as the median of the distribution.

$$\vdots$$

If $i = 99$, *is* the Ninety Ninth Percentile

$$P_{99} = \frac{99N}{100}$$

Grouped Data
Let say the N data items are $x_1, x_2, \cdots, x_n$, then the measures of percentile of a grouped data are estimated as;

$$P_i = l_m + \left(\frac{\frac{iN}{100} - Cf_{m-1}}{f_m}\right)C \; ; \qquad for\ i = 1,2,3,\cdots,99.$$

Where l_m is the lower class boundary of the Percentile class
Cf_{m-1} is the cumulative frequency before the Percentile class
f_m is the frequency of the Percentile class
C is the class size of the group data.

MEASURE OF VARIATION

It is also called the measure of spread and dispersion. The statistics describes the dispersion in a distribution which indicates how much variation there is among in the distribution. It is important in summarizing data to determine some measure of spread or variability of the data set. Some of such measures are:

- Range (R)
- The Semi-Interquartile range or Quartile Deviation (QD)

- The Mean Absolute Deviation or Mean Deviation (MD)
- Variance (S^2)
- Standard Deviation (S)
- Standard Error (S_E)
- Coefficient of Variation (CV)

Formulae:

Range = largest number – smallest number

$$QD = \frac{Q_3 - Q_1}{2}$$

$$M.D = \frac{\sum|x-\bar{x}|}{N} \text{ or } \frac{\sum f|x-\bar{x}|}{N}$$

$$variance = s^2 = \frac{\sum(x-\bar{x})^2}{N} \text{ or } \frac{\sum f(x-\bar{x})^2}{N}$$

$$standard\ deviation = s$$

$$= \sqrt{\frac{\sum(x-\bar{x})^2}{N}} \text{ or } \sqrt{\frac{\sum f(x-\bar{x})^2}{N}}$$

$$CV = \frac{mean}{standard\ deviation} \times 100$$

Example 1:

Given the set of numbers 12, 6, 7, 3, 15, 10, 18, 5. Find:

(a) The range,
(b) The mean deviation.

Solution:

(a) Range = largest number – smallest number
Range = 18 - 3 = 15
(b) The arithmetic mean is

$$\bar{x} = \frac{12+6+7+3+15+10+18+5}{8}$$

$$= \frac{76}{8} = 9.5$$

The mean deviation is:

$$M.D = \frac{\sum|x-\bar{x}|}{N}$$

$$= \frac{|12-9.5|+|6-9.5|+|7-9.5|+|3-9.5|+|15-9.5|+|10-9.5|+|18-9.5|+|5-9.5|}{8}$$

$$\frac{2.5+3.5+2.5+6.5+5.5+0.5+8.5+4.5}{8}$$

$$\frac{34}{8} = 4.25$$

EXAMPLE 2:

Given the set of data 9, 3, 8, 8, 9, 8, 9, 18. Calculate:

(a) The variance,
(b) The standard deviation,
(c) The coefficient of variation

Solution:

(a) $\bar{x} = \frac{9+3+8+8+9+8+9+18}{8} = \frac{72}{8} = 9$

$$variance = s^2 = \frac{\sum(x-\bar{x})^2}{N}$$

$$= \frac{(9-9)^2+(3-9)^2+(8-9)^2+(8-9)^2+(9-9)^2+(8-9)^2+(9-9)^2}{8}$$

$$= 15$$

(b) Standard deviations $= \sqrt{s^2} = \sqrt{15} = 3.57$

(c) $CV = \frac{mean}{standard\ deviation} \times 100$

$$CV = \frac{9}{3.57} \times 100 = 252.1$$

SKEWNESS

Skewness is the degree of asymmetry, or departure from symmetry of a distribution.

$$skewness = \frac{mean - mode}{standard\ deviation} \qquad (i)$$

To avoid using mode,

$$skewness = \frac{3(mean - median)}{standard\ deviation} \qquad (ii)$$

Equation (i) and (ii) are called Pearson's first and second coefficients of skewness.

Other measures of skewness, defined in terms of quartiles and percentiles are as follows:

$$quartile\ coefficient\ of\ skewness$$
$$= \frac{(Q_3 - Q_2) - (Q_2 - Q_1)}{Q_3 - Q_1}$$
$$= \frac{Q_3 - 2Q_2 + Q_1}{Q_3 - Q_1}$$

$$10 - 90\ percentile\ coefficient\ of\ skewness$$
$$= \frac{(P_{90} - P_{50}) - (P_{50} - P_{10})}{P_{90} - P_{10}}$$
$$= \frac{P_{90} - 2P_{50} + P_{10}}{P_{90} - P_{10}}$$

KURTOSIS

Kurtosis is the degree of peakedness of a distribution.

REGRESSION

Regression is usually used to describe the nature of the relationship between one variable (the *dependent or response* variable denoted by y) and the other variable (the independent or explanatory variable denoted by x). The knowledge of the relationship between the two variables will enable us predict and control events.

The aim of the regression analysis is to learn if *y does depend on x* or to determine the *nature of the regression line*.

To determine if there is a relationship between two variables we are to plot the paired observation of X (independent variable) and Y (dependent variable) on a scatter diagram. The nature of the scatter diagram gives the clue on the nature of the relationship whether there is a linear, inverse, curve linear relationship or no relationship at all.

Recall from linear equation in Further Mathematics that the form of a straight line is **y = mx + c** where m= slope and c = intercept. Similarly, the estimated Linear Regression equation is given by: $Y = a + bX$

Where a and b are numerical constant that estimate α and β respectively.

The Linear Regression Model: The two variables x and y are linearly related if the relationship can be expressed in the following linear model:

$Y_i = \alpha + \beta x_i + e_i$

Where α = regression constant or intercept, β = regression coefficient or gradient and e_i is a random variable with zero mean and variance δ^2.

NOTE:

The estimation of the regression equation can be made by using the free hand method (eye-fitted method) or the method of least squares.

FREE HAND METHOD

In this method, we first plot the scatter diagram of the set of the set of observation (X_i, Y_i) under consideration. Then an average line which is fitted on the set of points. One fits a line in his opinion is the best line to all the points in the scatter diagram. Then the equation of the fitted line equation is determined and is given by:

$$Y = a + bX$$

Where a = intercept on the y-axis and b = slope given by

$b = \frac{y_2 - y_1}{x_2 - x_1}$ using any point (X_1, Y_1) and (X_2, Y_2) on the fitted line.

THE LEAST SQUARE METHOD

The least square method eliminates the human judgment inherent in the free hand method. This method gives us only the line of best fit.

Here $b = \frac{n\sum XY - \sum X \sum Y}{n\sum X^2 - (\sum X)^2}$ and $a = \bar{Y} - b\bar{X}$

$\bar{Y} = \frac{\sum Y_i}{n}$ and $\bar{X} = \frac{\sum X_i}{n}$

$Y = a + bX$ is the required regression equation.

EXAMPLE 1:

The following table gives the advertising expenditure, X_i in #'000 and the resultant sales, Y_i in #'million by a marketing company.

X	85	70	84	85	82	90	92	95	85	86
Y	3	2	4	5	6	7	6	7	5	6

a) Obtain a regression line of Y on X.
b) Determine the value of sale for a given expenditure of #98,000.00 on

SOLUTION:

S/N	X	Y	XY	X^2
1	85	3	255	7225
2	70	2	140	4900
3	84	4	336	7056
4	85	5	425	7225
5	82	6	492	6724
6	90	7	630	8100
7	92	6	552	8464
8	95	7	665	9025
9	85	5	425	7225
10	86	6	516	7396
TOTAL	$\sum X = 854$	$\sum Y = 51$	$\sum XY = 4436$	$\sum X^2 = 73340$

a) The estimated regression equation is given by
Y = a + bX

Where $b = \frac{n\sum XY - \sum X \sum Y}{n\sum X^2 - (\sum X)^2} = \frac{10(4436) - 854(51)}{10(73340) - (854)^2} = 0.1974$

$\bar{Y} = \frac{51}{10} = 5.1$ and $\bar{X} = \frac{854}{10} = 85.4$

$a = \bar{Y} - b\bar{X} = 5.1 - 85.4(0.1974) = -11.7580$

Therefore Y = -11.7580 + 0.1974X is the equation of the estimated regression line.

b) For an expenditure of #98,000.00 on advertisement, the sale is predicted as
Y = -11.7580 + 0.1974X = -11.7580 + 0.1974(98)
= 7.5872 = #7,587,200.00

EXAMPLE 2:

The linear relationship between average body weight, x and food consumption, y for hens from each of 10 strains is estimated to be $y = a + bx$. Using the following data to obtain the values of a and b and hence fit the regression equation. Find y when x is 5.5.

X	4.6	5.1	4.8	4.4	5.9	4.7	5.1	5.2	4.9	5.1
Y	87.1	93.1	89.8	91.4	99.5	92.1	95.5	99.3	93.4	94.4

SOLUTION:

S/N	X	Y	XY	X^2	Y^2
1	4.6	87.1	400.66	21.16	7586.41
2	5.1	93.1	474.81	26.01	8667.61
3	4.8	89.8	431.04	23.04	8064.04
4	4.4	91.4	402.16	19.36	8353.96
5	5.9	99.5	587.05	34.81	9900.25
6	4.7	92.1	432.87	22.09	8482.41
7	5.1	95.5	487.05	26.01	9120.25
8	5.2	99.3	516.36	27.04	9860.49
9	4.9	93.4	457.66	24.01	8723.56
10	5.1	94.4	481.44	26.01	8911.36
	$\sum X=49.8$	$\sum Y=935.6$	$\sum XY=4671.1$	$\sum X^2=249.54$	$\sum Y^2=87670.34$

The estimated regression equation is given by:

Y = a + bX

Where $b = \frac{n\sum XY - \sum X \sum Y}{n\sum X^2 - (\sum X)^2} = \frac{10(4671.10) - 49.80(935.60)}{10(249.54) - (49.80)^2} = \frac{848.844}{15.36} = 7.69$kg of feed per kg of bird

$\bar{Y} = \frac{935.60}{10} = 93.56$ and $\bar{X} = \frac{49.80}{10} = 4.98$

$a = \bar{Y} - b\bar{X} = 93.56 - 4.98(7.69) = 55.26$

Hence, Y = 55.26 + 7.69X.

When X = 5.5,

Y = 55.26 + 7.69(5.5)

Y = 55.26 + 42.295 = 97.555.

CORRELATION

Correlation helps to obtain a measure of the strength of the relationship between X and Y. This can be achieved using the appropriate correlation model. Unlike the regression model, the correlation model requires X and Y to be both random variables. It is important to note that under the correlation model, X and Y vary together and have a joint normal distribution.

THE CORRELATION COEFFICIENT

The correlation coefficient is a measure of the strength of the linear relationship or association between the two variables X and Y. here r is the Correlation Coefficient can be calculated using the

- ✓ Product – moment method
- ✓ Spearman's

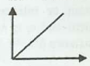
Product – Moment Method

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

Properties of Correlation Coefficient

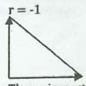
I. The range of the coefficient lies between $-1 \leq r \leq 1$

II. $\|r\| = 1$ which implies r = 1



There is a strong positive linear relationship between x and y / perfect correlation.

III. r = -1



There is a strong negative/ perfect inverse linear relationship between x and y.

IV. r = 0
No relationship between the two variables (x,y).

EXAMPLE:
Compute the correlation coefficient, r in example above

Solution
n = 10, $\sum X=49.8$, $\sum Y=935.6$ $\sum XY=4671.1$
$\sum Y^2=87670.34$, $\sum X^2=249.54$

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}} = \frac{10(4671.1) - 49.8x935.6}{\sqrt{[10(249.54) - (49.8)^2][10(87670.34) - (935.6)^2]}}$$

$$= \frac{118.12}{\sqrt{(15.36)(1356.04)}} = \frac{118.12}{144.3218} = 0.8184$$

r = 0.8184, which is a fairly high positive linear relationship.

COEFFICIENT OF DETERMINATION

The coefficient of determination is given by

$r^2 \times 100 = (0.8184)^2 \times 100 = 81.84\%$.

SPEARMAN'S RANK CORRELATION COEFFICIENT

The correlation coefficient can also be determined for situations when actual measurements cannot be easily obtained or impossible to obtain. We compute correlation coefficient in this case using the Spearman's Rank Correlation coefficient formula given as:

$$r = 1 - \frac{6\sum d^2}{n(n^2-1)}$$

EXAMPLE:

The following table gives the ranking of 7 students in class test in mathematics and statistics.

Student	A	B	C	D	E	F	G
Rank in Maths	3	5	6	7	4	2	1
Rank in Statistics	2	5	7	6	4	1	3

Compute the rank coefficient

SOLUTION:

Student	RANK IN MATHS	RANK IN STATISTICS	D	d^2
A	3	2	1	1
B	5	5	0	0
C	6	7	-1	1
D	7	6	1	1
E	4	4	0	0
F	2	1	1	1
G	1	3	-2	4
				$\sum d^2 = 8$

$$r = 1 - \frac{6\sum d^2}{n(n^2-1)} = 1 - \frac{6(8)}{7(7^2-1)} = 0.86$$

This indicates a high positive correlation between performance in Mathematics and performance in Statistics.

$r^2 x 100 = 0.86^2 \times 100 = 74\%$. Therefore in about 74% of the time performance in statistics is linearly related to performance in mathematics.

EXAMPLE 2:

Use the following data to compute the rank correlation coefficient.

X	65	50	55	65	55	70	65	75
Y	85	74	76	74	80	82	85	88

SOLUTION:

X	Y	Rank of X	Rank of Y	d	d^2
65	85	4	2.5	1.5	2.25
50	74	8	7.5	0.5	0.25
55	76	6.5	6	0.5	0.25
65	74	4	7.5	-3.5	12.25
55	80	6.5	5	1.5	2.25
70	82	2	4	-2	4
65	85	4	2.5	1.5	2.25
75	88	1	1	0	0
					$\sum d^2=23.5$

$$r = 1 - \frac{6\sum d^2}{n(n^2-1)} = 1 - \frac{6(23.5)}{8(8^2-1)} = 0.72$$

$r^2 x 100 = 0.72^2 \times 100 = 51.84\%$. Which indicate that it is only 51.84% of the time when an increase in X results to a corresponding increase in y.

PROBABILITY

Probability measures the degree of uncertainty.

$$\text{Probability} = \frac{\text{no of favourable cases (n)}}{\text{total no of cases/possible events}}$$

Definitions/Things to Note:

(1) Sample space: A sample space is defined as the set of all possible outcomes of a random experiment. Denoted by S.

For example:

i. If a coin is tossed once there are two possible outcomes, "Head" (H) or "Tail" (T) the sample space is given by

S = {H;T}

ii. If the coin is tossed two times,

S = {HH, HT, TH, TT}

iii. If the coin is tossed three times,

S = {HHH, HHT, HTH, THH, THT, TTH, HTT, TTT}

Note: 2^n gives the no. of outcomes in the sample space

iv. Throw a dice S = {1, 2, 3, 4, 5, 6}

v. Throw 2 dice,

	1	2	3	4	5	6
1	1,1	1,2	1,3	1,4	1,5	1,6
2	2,1	2,2	2,3	2,4	2,5	2,6
3	3,1	3,2	3,3	3,4	3,5	3,6
4	4,1	4,2	4,3	4,4	4,5	4,6
5	5,1	5,2	5,3	5,4	5,5	5,6
6	6,1	6,2	6,3	6,4	6,5	6,6

S = 36

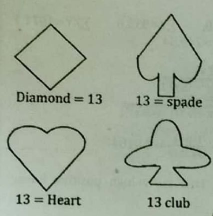
(2) Trial and events:

Event: An event is a subset of a sample space denoted by E

$P_r(E) = \frac{nE}{nS}$

Trials: *Performing a random experiment is called a trial. Tossing a coin or throwing a die is a trial and the turning up of head or tail is an event.*

(3) Cards: *A pack of card consists of four suits i.e Spades, Hearts, Diamonds and clubs, each consists of 13 cards, nine cards numbered 2,3,4,.....,10, an Ace, a king, a Queen and a Jack.*
There are 26 red and 26 black which forms a total of 52 cards.



Total no. of cards = 52.
13 + 13 + 13 + 13 = 52

PROPERTIES OF PROBABILITY

(i) $0 \leq P(E) \leq 1$

i.e the probability of any event E is non negative and take a value from zero to one

(ii) $P_r(\emptyset) = 0$

(iii) $P_r(S) = 1$

(iv) If all possible outcomes $E_1, E_2, ..., E_n$ of a random experiment are mutually exclusive and collectively exhaustive, then the sum of their probability is equal to 1. i.e

$P_r(E_1) + P_r(E_2) + ... + P_r(E_n) = 1$

(v) The sum probability of success and probability of failure is 1

$P_r(E) + P_r(E) = 1$

$p + q = 1$

MUTUALLY EXCLUSIVE EVENT

Here the occurrence of any event E_1 excludes (does not depend) the occurrence of the (E_2). Here
$\Pr\{E_1 \cap E_2\} = 0$
for mutually exclusive events,
$\Pr\{E_1 \cup E_2\} = \Pr(E_1) + \Pr\{E_1\} + \Pr\{E_2\}$.
Otherwise, (dependent events)
$\Pr\{E_1 \cup E_2\} = \Pr\{E_1\} + \Pr\{E_2\} - \Pr\{E_1 \cap E_2\}$

EXAMPLE 1:

If E_1 is the event "drawing an ace from a deck of cards" and E_2 is the event "drawing a king". Find the probability of drawing an Ace of a king.

SOLUTION:

$P_r\{\text{drawing an Ace}\} = \Pr\{E_1\} = \frac{4}{52} = \frac{1}{13}$

$P_r\{\text{drawing a king}\} = P_r\{E_2\} = \frac{4}{52} = \frac{1}{13}$

Since the occurrence of E_1 does not depend on E_2 (mutually exclusive events)

$\Pr\{E_1 \cup E_2\} = \Pr(E_1) + \Pr(E_2)$

$= \frac{1}{13} + \frac{1}{13} = \frac{2}{13}$

EXAMPLE 2:

If E_1 is the event "drawing an Ace from a deck of cards and E_2 is the event "drawing a spade". Find the probability of drawing either an Ace of a spade or both.

SOLUTION:

$P_r(\text{Ace}) = P_r(E_1) = \frac{4}{52}$
$P_r(\text{King}) = P_r(E_2) = \frac{13}{52}$
$\Pr\{E_1 \cap E_2\} = P_r$ (have an Ace and a spade) $= \frac{1}{52}$
Here E_1 and E_2 are not mutually exclusive event since Ace of spade can be drawn from a deck of card.

$P_r(E_1 \cup E_2) = P_r(E_1) + P_r(E_2) - P_r(E_1 \cap E_2) =$

$$\frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}$$

A♣	2	3	4	5	6	7	8	9	10	I	Q	K
A♦	2	3	4	5	6	7	8	9	10	J	Q	K
A♥	2	3	4	5	6	7	8	9	10	J	Q	K
A♠	2	3	4	5	6	7	8	9	10	J	Q	K

FURTHER EXAMPLES

EXAMPLE 3:

A ball is drawn at random from a hot containing 6red balls, 4white balls, and 5blue balls. Determine the probability that the ball drawn is (a) red (b) white (c) blue (d) not red(e) red or white.

SOLUTION:

Let R,W and B denote the events of drawing a red ball, white ball and blue balls respectively. Then

(a) $P_r\{R\} = \frac{\textit{choosing a red ball}}{\textit{total way of choosing ball}} = \frac{6}{6+4+5} = \frac{6}{15}$

(b) $P_r\{W\} = \frac{4}{6+4+5} = \frac{4}{15}$

(c) $P_r\{B\} = \frac{5}{6+4+5} = \frac{5}{15} = 1/3$

(d) $P(\text{Not red}) = P_r(\check{R}) = 1 - P_r(R)$

$= 1 - \frac{2}{5} = \frac{3}{5}$

(e) $P_r(R+W) = \frac{\textit{ways of choosing a red or white ball}}{\textit{total ways of choosing a ball}}$

$= \frac{6+4}{6+4+5} = \frac{10}{15} = \frac{2}{3}$

Alternatively

$P_r(R+W) = P(B) = 1 - \frac{1}{3} = \frac{2}{3}$

CONDITIONAL PROBABILITY

If E_1 and E_2 are two events, the probability that E_2 occurs given that E_1 has occurred is denoted by $P_r\{E_2/E_1\}$, and called the conditional probability of E_2 given that E_1 has occurred.
If E_1 and E_2 are independent event i.e the occurrence of E_1 does not depend on E_2 and vice versa then $P_r\{E_2/E_1\} = P_r\{E_2\}$, otherwise they are dependent event.
Note: $\Pr\{E_1 \cap E_2\} = P_r\{E_1\}.P_r\{E_2/E_1\}$ for dependent events

$\Pr\{E_1 \cap E_2\} = P_r\{E_1\}.P_r\{E_2\}$ for independent events

For three events, E_1, E_2, and E_3

$P_r\{E_1 \cap E_2 \cap E_3\} = P_r\{E_1\} P_r\{E_2/E_1\}.P_r\{E_3/E_1E_2\}$

for dependent event which means:

The probability of occurrence E_1,E_2 and E_3 is equal to (the probability of E_1) x (the probability of E_2 given that E_1 has occurred) x (the probability of E_3 given that both E_1 and E_2 have occurred).

$P_r\{E_1E_2E_3\} = P_r\{E_1\} P_r\{E_2\} P_r\{E_3\}$

for independent event.

EXAMPLE 1:

Let E_1 and E_2 be the events "heads on fifth toss" and heads on sixth toss" of a coin respectively. The probability of heads on both fifth and sixth tosses is?

SOLUTION:

E_1 and E_2 are independent events

$\therefore \Pr\{E_1 \cap E_2\} = p_r(E_1).\ pr(E_2) = (^1/_2)\,(^1/_2) = (^1/_4)$

EXAMPLE 2:

If a box contains 3white and 2black balls, where the balls are not replaced after being drawn. Find the probability that both balls drawn are black.

SOLUTION:

Let E_1 be the event "first ball drawn is black and E_2 the event "second ball drawn is black". Here E_1 and E_2 are dependent events since the balls drawn are not replaced.

$P_r(E_1) = \frac{2}{3+2} = \frac{2}{5}$

$P_r(E_2/E_1)$ = the probability that the second ball drawn is black, given that the first ball drawn was black

$P_r\{E_2/E_1\} = \frac{1}{(3+1)} = \frac{1}{4}$

Therefore, the probability that both ball drawn are black is

$\Pr\{E_1 \cap E_2\} = P_r\{E_1\}.P_r\{E_1\}.P_r\{E_2/E_1\} = \frac{2}{5}.\frac{1}{4} = \frac{2}{20} = \frac{1}{10}$

BAYE'S THEOREM

If the event $A_1, A_2, \ldots, A_n$ form a partition of the same space S and if B is any other event of S such that it can occur only if one of the A_i occurs, hence for any i

$$P\left(^{A_i}/_B\right) = \frac{\Pr(A_i)\Pr(^B/_{A_i})}{\sum_{i=1}^{n}\Pr(A_i)\Pr(^B/_{A_i})}$$

EXAMPLE:

In a certain factory that manufactures lion bulbs, machines A_1, A_2, A_3, Manufacture 40%, 35%, 25% respectively of the total production. The percentage of defective bulbs are 2, 4 and 5. A bulb is selected at random from a days production and is found to be defective. What is the probability:

(a) That it is manufactured by machine A_1
(b) That it is manufactured by machine A_2
(c) Of getting a defective bulb in a day production?

SOLUTION:

Let $P(A_1)$ denote the probability that a bulb is manufactured by machine A_1 and let $P(D/A_1)$ denote the probability that a defective bulb comes from machine A_1

$$P(A_1) = \frac{40}{100},\quad P\left(^D/_{A_1}\right) = \frac{2}{100}$$

$$P(A_2) = \frac{35}{100},\quad P\left(^D/_{A_2}\right) = \frac{4}{100}$$

$$P(A_3) = \frac{25}{100},\quad P\left(^D/_{A_1}\right) = \frac{5}{100}$$

(a) The probability that a defective bulb is manufactured by machine A_1 is

$$P\left(^{A_1}/_D\right) = \frac{P(A_1)P\left(^D/_{A_1}\right)}{P(A_1)P\left(^D/_{A_1}\right) + P(A_2)P\left(^D/_{A_2}\right) + P(A_3)P\left(^D/_{A_3}\right)}$$

$$= \frac{\frac{40}{100}\times\frac{2}{100}}{\left(\frac{40}{100}\times\frac{2}{100}\right)+\left(\frac{35}{100}\times\frac{4}{100}\right)+\left(\frac{25}{100}\times\frac{5}{100}\right)} = \frac{80}{80+140+125} = \frac{80}{345} = \frac{16}{69}$$

(b) The probability that a defective bulb is manufactured by machine A_2 is

$$P\left(^{A_2}/_D\right) = \frac{P(A_2)P\left(^D/_{A_2}\right)}{P(A_1)P\left(^D/_{A_1}\right) + P(A_2)P\left(^D/_{A_2}\right) + P(A_3)P\left(^D/_{A_3}\right)}$$

$$= \frac{\frac{35}{100}\times\frac{4}{100}}{\left(\frac{40}{100}\times\frac{2}{100}\right)+\left(\frac{35}{100}\times\frac{4}{100}\right)+\left(\frac{25}{100}\times\frac{5}{100}\right)} = \frac{35\times 4}{80+140+125} = \frac{140}{345} = \frac{28}{69}$$

(c) The probability of getting a defective bulb

$$P(D) = P(A_1)P\left(^D/_{A_1}\right) + P(A_2)P\left(^D/_{A_2}\right) + P(A_3)P\left(^D/_{A_3}\right)$$

$$\left(\frac{40}{100}\times\frac{2}{100}\right)+\left(\frac{35}{100}\times\frac{4}{100}\right)+\left(\frac{25}{100}\times\frac{5}{100}\right)$$

$$\frac{80}{10000}+\frac{140}{10000}+\frac{125}{10000} = \frac{69}{2000}$$

Permutation

$nP_r = \frac{n!}{(n-r)!}$

Note: Whenever you see the word arrangement is implies permutation.

EXAMPLE 1:

The number of permutation of letters in the word STATISTICS is?

SOLUTION:

$\frac{10!}{3!3!1!2!1!} = 50{,}400$

Since there are 3s', 3t's, 1a, 2i's, and 1c.

EXAMPLE 2:

In how many ways can 5 differently coloured marbles be arranged in a row?

SOLUTION:

Number of arrangement of 5 marbles in a row =

$5\times 4\times 3\times 2\times 1 = 5! = 120$

Note: Number of arrangement of n objects in a row = n(n-1)(n-2)......1= n! ways.

EXAMPLE 3:

In how many ways can 10 people be seated on a bench if only 4 seats are available?

SOLUTION

$10P_4$ ways $= \frac{10!}{(10-4)!} = \frac{10\times 9\times 8\times 7\times 6!}{6!}$

= 5040ways

Combinations

$nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$

Note: STIRLINGS Approximation to n!

When n is large we make use of

$n! \cong \sqrt{2\pi n}\,n^n e^{-n}$

Where e=2.71828.

EXAMPLE 1:

The number of combinations of the letters a, b and c taking two at a time is?

SOLUTION:

$$\binom{3}{2} = 3C_2 = \frac{3\times 2\times 1}{2!\,(3-2)!} = \frac{3\times 2!}{2!} = 3$$

EXAMPLE 2:

In how many ways can a committee of 5 people be chosen out of 9 people?

SOLUTION:

$$\binom{9}{5} = 9C_5 = \frac{9!}{5!\,(9-5)!} = \frac{9!}{5!\,4!} =$$

$$\frac{9\times 3\times 7\times 6\times 5!}{5!\,4!} = \frac{9\times 8\times 7\times 6}{4!} = 126$$

EXAMPLE 3:

Out of 5 mathematics and 7 physicists, a committee consisting of 2 mathematics and 3 physicists to be formed. In how many ways can this be done if :

(a) Any mathematics and any physicists can be included

(b) One particular physicists must be on the committee.

SOLUTION:

(a) Two mathematicians out of 5 can be selected in $\binom{5}{2}$ ways and 3 physicists out of 7 can be selected in $\binom{7}{3}$ways. The total number of possible selection is

$\binom{5}{2}.\binom{7}{3} = 5C_2 \times 7C_3 = 10 \times 35 = 350$

(b) Two mathematicians out of 5 can be selected in $\binom{5}{2}$ ways and 2 additional physicists out of 6 can be selected in $\binom{6}{2}$ ways since one particular physicist must be in the committee.

The total number of possible selection is $\binom{5}{2}.\binom{6}{2} = 5C_2 \times 6C_2 = 10 \times 15 = 150$

Common Probability Distributions

BINOMIAL PROBABILITY DISTRIBUTION

The n repetitions of a Bernoulli experiment give the Binomial distributions with probability given as:

$P_r(x{:}n) = nC_x p^x q^{n-x}; x = 0,1,2,\dots n$

Where P = success, q = failure

$p + q = 1; q = 1 - p.$

The Binomial distribution is a discrete probability distribution with: mean = $\bar{x}$ = np and variance = npq.

EXAMPLE:

Experience has shown that 25% of the staff of a financial house are likely to redraw their accounts before the next pay day. Out of a random sample of 8 employees of the financial house, what is the probability that:

(a) None overdraws his account before the next pay day?

(b) Not more than one employee overdraws his account before he next pay day?

(c) Not less than two employees overdraw their accounts before the next payday?

SOLUTION:

Using binomial distribution let X be the number that will overdraw their accounts before the next pay day:

X = {0, 1, 2, 3, 4, 5, 6, 7, 8}

$P = 25\% = \frac{25}{100} = 0.25$

Q = 1-p = 1 -0.25 = 0.75

(a) P_r(none overdraws) $= 8C_0\,(0.25)^0(0.75)^{8-0} = 0.1001$
(b) P_r(not more than one overdraws) $= P_r(0) + P_r(1) = 8C_0\,(0.25)^0(0.75)^8 + 8C_1\,(0.25)^1\,(0.75)^{8-1} = 0.1001 + 0.267 = 0.3671$
(c) P(not less than 2 employees) = $P_r(2)+P_r(3)+P_r(4)+\dots+P_r(8)$

$= 1-[P_r(0) + P(1)] = 1-[0.3671] = 0.6329$

THE POISSON DISTRIBUTION

The poisson distribution approximates the binomial distribution when n is large and P is small. If this conditions are met then:

$P_r(X; nP) = \frac{(nP)^X e^{-nP}}{X!}; x = 0,1,2,\dots$

EXAMPLE:

On the average 1 house in every 1000 houses in a large village has fire during the year. If 3,500 house are in that village what is the probability that not more than one house will have fire during the year?

SOLUTION:

Let X be the number of houses that will have fire during the year

Using poisson distribution ; $\frac{(nP)^X e^{-nP}}{X!}$

P = 1/1000 = 0.001

$n = 3500$

$P_r(x = 0) + P_r(x = 1) =$

$\frac{(3500 \times 0.001)^0 e^{-3500 \times 0.001}}{0!} + \frac{(3500 \times 0.001)^1 e^{-3500 \times 0.001}}{1!}$

= 0.1359

Note: Mean of poisson distribution μ= np, standard deviation = $\sqrt{npq}$

THE STANDARD NORMAL RANDOM VARIABLE, Z

Here $Z = \frac{x-\mu}{\sigma}$, where μ = mean

σ = std dev.

EXAMPLE

If the normal random variable, X has mean 120 and variance 25, determine the probability that:

(a) The random variables x takes values between 120 and 135

(b) x is less than 130

(c) x is more than 120

SOLUTION

(a) we are to determine the probability that x lies between 120 and 135. Thus

$P_r(120 \le x \le 135) =$.

$P_r[\frac{120-128}{5} \le \frac{X-128}{5} \le \frac{135-128}{5}]$

$= P_r(-1.6 \le Z \le 1.4)$

From the standard normal table we find that 1.4=0.4192 and -1.6 = 0.4452

The total area is 0.4192 + 0.4452 = 0.8644

$P_r(-1.6 \le Z \le 1.4) = 0.8644$

(b) $P_r(x<130) = P_r[\frac{X-128}{5} < \frac{130-128}{5}]$

$= P_r(z < 0.4)$

$P_r(Z < 0.4) = 0.5+0.1554 = 0.6654$

Note: since Z<0.4 we add 0.5 to the value obtained from the standard normal table.

(c) $P_r(x>120) = P_r[\frac{X-123}{5} > \frac{120-128}{5}]$

$= P_r(Z > -1.6) = 0.4452 + 0.5 = 0.9452$

NORMAL APPROXIMATION TO BINOMIAL

When n is large and P is close to 0.5 the normal provides a good approximation to Binomial.

Here mean = μ = nP and variance = σ^2 = npq, $\sigma = \sqrt{variance}$

Recall,

$Z = \frac{X-u}{\sigma} = \frac{X-np}{\sqrt{npq}}$

EXAMPLE:

The chance is 65 in 100 that anyone of 300 beneficiaries of the students' loan scheme will not default in repayment. What are the probabilities that:

(a) 205 or more recipients would not default in repayment

(b) Less than 205 recipients would not default in repayment

(c)Between 188 and 214 recipients would not default in repayment.

Solution

Here $P = \frac{65}{100} = 0.65, q = 1 - p = 1 - 0.65 = 0.35, n = 300$

$np = 300 \times 0.65 = 195$

$\sqrt{npq} = \sqrt{300 \times 0.65 \times 0.35} = 8.26$

(a) $P_r(x \ge 205) = P_r[\frac{X-195}{8.26} \ge \frac{205-195}{8.26}]$

$P_r(z \ge 1.21)$

$= 0.5 - 0.3869 = 0.1131$

(b) $P_r(x < 205) = P_r[\frac{x-195}{826} < \frac{205-195}{826}] =$

$P_r(Z < 1.21) = 0.5+0.3869 =$

0.8869

(c) $P_r(188 < X < 214) =$

$P_r(\frac{188-195}{8.26} < \frac{X-195}{8.26} < \frac{214-195}{8.26}) =$

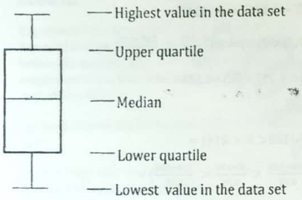
$P_r(-0.85 < Z < 2.30) =$

$0.3023 + 0.4893 = 0.7916$

BOX AND WHISKER PLOTS

The box and whisker plots sometimes called simply a box plot. They were devised by John Tukey in 1977 and offer a simple and effective way of presenting and comparing data set which do not follow an underlying distribution (Non-parametric data). Often box and whisker plots are used to present ordinal (ranked) data.

A typical box and whisker plot is shown below and is annotated to show you what each of the parts represent.



HOW TO CONSTRUCT A BOX AND WHISKEL PLOT

Given a set of data 5, 2, 16, 9, 13, 7, 10

We first arrange the data set in order from least to greatest

2, 5, 7, 9, 10, 13, 16

We obtain the following

(i) Median = 9 since it is placed right at the middle of the data set

(ii) Lower quartile: Here you will need all the data before the median or 9

2 5 7 9 10 13 16

Data before Median

Since the value in the median of the set 2 , 5, 7 is 5, it therefore means the lower quartile is 5.

Then, to get the upper quartile, you will need all the data after the median or 9.

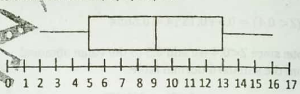
2 5 7 9 10 13 16

Data set after Median

Here we see that the value at the middle of the set 10, 13, 16 is 13, which means the upper quartile is 13.

We now have all the data to construct the box and whisker plot for the above data.

Finally, now make a number line and graph above the number line 2, 5, 9, 13, 16



ONE-WAY ANALYSIS OF VARIANCE (ANOVA) OR A COMPLETELY RANDOMIZED DESIGN

This is the simplest experiment in which there is only one factor observed at P levels. The main objectives of the one-way anova is usually to indicate whether P – treatments means differ.

Model: $Y_{ij} = \mu + t_i + e_{ij}; \begin{matrix} i = 1,2 \dots p \\ j = 1,2 \dots n \end{matrix}$

Y_{ij} = observed value of the ith treatment in the jth replication i.e (ij)th observation.

$\mu = Overall\ mean$

$ti = ith\ treatment\ effect$

$e_{ij} = random\ error$

Assumptions

(1) Normality (2) Independence

(3) Constant variance

Hypothesis

$H_0: t_1 = t_2 = t_2 \dots = t_p = 0$

$H_1: t_i \neq 0\ for\ at\ least\ one\ i$

i) One way analysis of variance table

source of Variation	degree of Freedom	sum of Squares	means Squares	F_0
Treatment	P – 1	$SS_{treatment}$	$MS_{treatment}$	$F_0 = \frac{MS_{treatment}}{MS_{error}}$
Error	P(n – 1) N – P	SS_{error}	MS_{error}	
Total	N – 1	SS_{total}		

Where

$SS_{total} = \sum_i \sum_j Y_{ij}^2 - \frac{Y_{..}^2}{N}$

$SS_{treatment} = \frac{1}{n}\sum_{i=1}^{P} Y_{ij}^2 - \frac{Y_{..}^2}{N}$

$SS_{error} = SS_{total} - SS_{treatment}$

$MS_{treatment} = \frac{SS_{treatment}}{P-1}$

$MS_{error} = \frac{SS_{error}}{P(n-1)}$ or $\frac{SS_{error}}{N-P}$

Example:

A completely randomized design (CRD) was conducted to compare the % Nitrogen of five varieties of corn. The result of the experiment is given on the table below:

Varieties	Replication 1	2	3	4	5	6	y_i
1	33	37	60	64	46	50	290
2	50	56	79	77	72	58	392
3	72	64	80	88	72	80	456
4	77	73	79	87	81	77	474
5	67	73	84	80	69	83	456

Determine at 5% level of significance whether there exist differences in the result given.

Note: $N = n \times p = 6 \times 5$

Solution:

State the model

$yij = \mu + t_i + e_{ij}; \begin{matrix} i = 1,2 \dots 5 \\ j = 1,2 \dots 6 \end{matrix}$

State the hypothesis

$H_0: t_i = 0; i = 1,2 \dots 5$

$H_1: t_i \neq 0\ for\ at\ least\ one\ i$

$N = 30\ (No\ of\ observations)$

$Y = \sum_{i=1}^{5}\sum_{j=1}^{6} Y_{ij}^2 = 2068$

[$Summing\ up\ all\ the\ Y_i$]

$SS_{total} = \sum_{i=1}^{5}\sum_{j=1}^{6} Y_{ij}^2 - \frac{Y_{..}^2}{N}$

$= 33^2 + 37^2 + 60^2 + \dots 83^2 - \frac{(2068)^2}{30}$

$= 148614 - 142554.13 = 6059.87$

$SS_{treatment}$ or $SS_{Varieties} = \frac{1}{n}\sum_{i=1}^{5} Y_{i.}^2 - \frac{Y^2}{N}$

$= \frac{1}{6}[290^2 + 392^2 + 456^2 + \dots + 456^2] - \frac{(2068)^2}{30}$

$= 3831.20$

$SS_{error} = SS_{total} - SS_{Varieties}$

$= 6059.87 - 3831.20$

$= 2228.67$

$MS_{error} = \frac{SS_{error}}{N-P} = \frac{2228.67}{30-5} = 89.15$

$MS_{Varieties} = \frac{SS_{Varieties}}{P-1} = \frac{3831.2}{5-1} = 957.80$

$F_0 = \frac{MS_{Varieties}}{MS_{error}} = \frac{957.80}{89.15} = 10.74$

ANOVA TABLE

Source of variation	Degree of freedom	Sum of squares	Mean squares	F_0
Varieties	P – 1 = 5 – 1 = 4	3831.20	957.80	10.74
Error	P(n – 1) =5(6 – 1) = 25	2228.67	89.15	
Total	N – P= 30 – 5 = 25	6059.87		

At 5% level of significance $= F_{p-1,N-p}^{\alpha}$

$= F^{0.05}{}_{4,25} = 2.76$

Since $F_{cal} = 10.74 > F_{tabulated} = 2.76$, we reject the null hypothesis and conclude that there are differences in the treatments

TWO-WAY ANALYSIS OF VARIANCE

The generalization of the mathematical model for one factor experiment leads us to assume for two factor experiment that the model is

$Y_{ij} = N + t_i + b_j + e_{ij}; \begin{matrix} i = 1 \dots . p \\ j = 1 \dots . b \end{matrix}$

where

$Y_{ij} = (ij)th\ observation$

$\mu = Grand\ mean$

$t_i = ith\ treatment\ effect$

$b_j = ith\ block\ effect$

$e_{ij} = ith\ random\ error\ term$

$$e_{ij} \sim NID\ (0, \sigma^2)$$

Hypothesis

$H_0: t_i = 0, i = 1,2 \dots p$

$H_1: t_i \neq 0. for\ at\ least\ one\ i$

$H_0: b_j = 0, j = 1,2 \dots b$

$H_1: b_j \neq 0, for\ at\ least\ one\ j$

Note: The assumption for one way and two way ANOVA are the same

Computational formula S for the two-way ANOVA

$SS_{total} = \sum_i \sum_j Y_{ij}^2 - \frac{Y_{..}^2}{N=b \times p}$

$SS_{treatment} = \frac{1}{b} \sum_i Y_{i.}^2 - \frac{Y_{..}^2}{N}$

$SS_{block} = \frac{1}{p} \sum_j Y_{.j}^2 - \frac{Y_{..}^2}{N}$

$SS_{error} = SS_{total} - SS_{treatment} - SS_{block}$

$MS_{treatment} = \frac{SS_{treatment}}{p-1}$

$MS_{block} = \frac{SS_{block}}{b-1}$

$MS_{error} = \frac{SS_{error}}{(p-1)(b-1)}$

$F_0\ (treatment)\ \frac{MS_{treatment}}{MS_{error}}$

$F_0\ (block)\ \frac{MS_{block}}{MS_{error}}$

Example:

Fabric sample

Varieties	1	2	3	4	5	$y_{i.}$
1	1.3	1.6	0.5	1.2	1.1	5.7
2	2.2	2.4	0.4	2.0	18	8.8
3	1.8	1.7	0.6	1.5	1.3	6.9
4	3.9	4.4	2.0	4.1	3.4	17.8
$y_{.j}$	9.2	10.1	3.3	8.5	7.6	39.2

Test for differences in treatment means at $\propto = 0.01$

Solution:

Model: $Y_{ij} = \mu + t_i + b_j + e_{ij};\ \begin{matrix} i = 1,2,\dots 4 \\ j = 1,2,\dots 5 \end{matrix}$

$H_0: t_i = 0, i = 1,2 \dots 4$

$H_1: t_i \neq 0. for\ at\ least\ one\ i$

$SS_{total}: 1.3^2 + 1.6^2 + 0.5^2 + \dots + 3.4^2 - \frac{(39.2)^2}{20}$

$= 25.69$

$SS_{treatment} = \frac{1}{b} \sum_{i=1}^{4} Y_{i.}^2 - \frac{Y_{..}^2}{pb}$

$= \frac{1}{5} [5.7^2 + 8.8^2 + 6.9^2 + 7.8^2] - \frac{(39.2)^2}{20}$

$= 18.04$

$SS_{block} = \frac{1}{p} \sum_{j=1}^{5} Y_{.j}^2 - \frac{Y_{..}^2}{pb}$

$= \frac{1}{4} [9.2^2 + 10.1^2 + 3.5^2 + 8.8^2 + 7.6^2] - \frac{(39.2)^2}{20}$

$= 6.69$

$SS_{error} = SS_{total} - SS_{treatment} - SS_{block}$

$= 25.69 - 6.69 - 18.04$

$= 0.96$

ANOVA TABLE

Source of variation	Degree of freedom	Sum of squares	Mean squares	F_0
Chemical types (treatment)	$P - 1 = 4 - 1 = 3$	18.04	6.01	F_0 (treatment) $= \frac{6.01}{0.08} = 75.13$
Fabric samples (blocks)	$b - 1 = 5 - 1 = 4$	6.69	1.67	F_0 (block) $= \frac{1.67}{0.08} = 20.88$
Error	$(P-1)(b-1) = (4-1)(5-1) = 12$	0.96	0.08	
Total	$Pb - 1 = 20 - 1 = 19$	25.69		

$$F^{\propto}{}_{(p-1),(p-1)(b-1)} = F^{0.05}{}_{3,12} = 3.49$$

Conclusion

Since $F_{calculated} = 75.13 > F_{tabulated} = 3.49$, we therefore reject the null hypothesis and conclude that the treatment means differ.

PAST QUESTIONS

FEDERAL UNIVERSITY OF TECHNOLOGY, OWERRI
DEPARTMENT OF STATISTICS
2017/2018 HARMATTAN SEMESTER EXAMINATION
STA 211: INTRODUCTION TO STATISTICS AND PROBABILITY
INSTRUCTION: ANSWER ALL QUESTIONS; TIME: 2 HRS 30 Mins

HIBIT I: Given the distribution

Values (X)	35	45	55	65	75	85	95
Frequency (f)	6	20	30	20	10	8	6

e the information contained in EXHIBIT I to answer questions 1 and 2.

Calculate the mean of the distribution. (A) 61.4 (B) 32.5 (C) 30.2 (D) 60.6

Calculate the standard deviation of the distribution. (A) 40.63 (B) 15.51 (C) 10.22 (D) 14.23

The statistic that is not affected by extreme values is: (A) median (B) median (C) mean (D) mode

The degree of the linear relationship between two variables at a time is called ….
(A) Skewness (B) Product moment (C) Regression (D) Correlation

A die is tossed once. If the number is odd, what is the probability that it is less than 4.
(A) 2/3 (B) 1/3 (C) 1/6 (D) 1/2

The well defined rule by which numbers are assigned to each outcome of an experiment is known as … (A) Random variable (B) Probability distribution (C) Bayes Theorem (D) Independent events

A sample statistic that is used to estimate a population parameter is referref to as …..
(A) Estimator (B) Estimating (C) Estimate (D) Estimation

One of the regimes of kurtorsis is: (A) Mesokurtic (B) Leptokurrttic (C) Mesorkurtic (D) Lepttokurtic

The scores of ten students in English and Mathematics is shown as;

English (X)	80	75	75	70	65	65	100	75	80	100
Mathematics (Y)	50	47	45	32	32	47	65	45	52	47

alculate the correlation between English and Mathematics using the Spearman rank correlation method.
(A) 0.63 (B) 0.64 (C) 0.73 (D) 0.74

. The two ways of calculating probabilities of a random variable are:
) Classical and Posteriori (B) Frequency and Observed
) Classical and Priori (D) Frequency and Posteriori

EXHIBIT II

e probability that a man watches Johntee TV show given that his wife does is $3(1-X^2)$ while the
bability that his wife watches Johntee TV is $\frac{2}{3}$. The probability that the couple watches the TV show is
$-X^2$.

e the information that is contained in Exhibit II, to answer questions 11 and 12.

. Obtain the value of X. (A) 1/2 (B) 3/4 (C) 1/4 (D) 2/3

. Obtain the probability that a man watches Johntee TV show given that his wife does.
(A) 1/2 (B) 3/4 (C) 1/4 (D) 2/3

. The first and last observation of a distribution is 10 km and 145 km respectively. The class interval of third class, given that the distribution is grouped into ten distinct classes, is..:
(A) 37.00 – 50.49 (B) 37.00 – 50.50 (C) 23.50 – 36.99 (D) 23.50 – 37.00

. Which of these mathematical statements is true, given that X_i and $\bar{X}$ are the ith value and mean spectively of a distribution. (A) $\sum(X_i - \bar{X}) > 0$ (B) $\sum(X_i - \bar{X}) < 0$ (C) $\sum(X_i - \bar{X}) \neq 0$ (D) $\sum(X_i - \bar{X}) = 0$

. Obtain the probability of obtaining at least four girls in a family of five Children.
(A) 0.17 (B)0.15 (C) 0.18 (D) 0.19

. Find the probability of obtaining a sum of 6 in a toss of two fair dice.
(A) 0.17 (B) 0.13 (C) 0.14 (D) 0.83

. A Typist makes an average of four errors per page. What is the probability that she makes at least two rors on the next page: (A) 0.90 (B) 0.91 (C) 0.80 (D) 0.81

EXHIBIT III

The inter quartile range of a distribution, whose upper and lower quartiles are $2X^2+1$ and $8X-9$ respectively, is 12. **Use this Information contained in Exhibit III to answer Questions 18, 19 and 20.**

18. Determine the upper quartile of the distribution. (A) 19 (B) 20 (C) 18 (D) 14
19. Determine the lower quartile of the distribution. (A) 6 (B) 8 (C) 2 (D) 7
20. Determine the 75th Percentile of the distribution. (A) 19 (B) 20 (C) 18 (D) 14
21. The errors that occur in any hypothesis test about a parameter are ….. errors:
    (A) Types IA and IIB (B) Types C and D (C) Types A and B (D) Types I and II
22. The probability of rejecting a false hypothesis is:
    (A) Power of performance (B) Power of testability performance (C) Power of test (D) Power of hypothesis
23. The test in which the alternative hypothesis is non-directional is … test:
    (A) two tailed (B) Type I (C) Right tailed (D) Left tailed

EXHIBIT IV

In a Statistics Examination, the mean grade is 43% with a variance of 6.25. The grades are approximately normally distributed, all students with grades 40 to 47 received grade D and 20 students received grade D. **Given that $\phi(1.6)=0.9452$ and $\phi(-1.20)=0.1151$, Use this Information contained in Exhibit IV to answer Questions 24 and 25.**

24. Obtain the probability of students with grades 40 to 47. (A) 0.12 (B) 1.0 (C) 0.95 (D) 0.83
25. Determine the number of students that took the Examination. (A) 167 (B) 20 (C) 24 (D) 21
26. The act of taking data measurement on a variable of interest using measurement tool is?
    (A) Scale Measure (B) Scaling (C) Data Measurement (D) Scale
27. If the moment coefficient of kurtosis is > 0, it means the distribution is ------ $x = 2+\sqrt{5}$, $2-\sqrt{5}$
    (A) mesokurtic (B) flatkurtic (C) platykurtic (D) leptokurtic

EXHIBIT V

145.5 – 135.5

The class marks of a frequency distribution are 128, 137, 146, 155, 164, 173 and 182
Use the information contained in EXHIBIT V to answer questions 28, 29 and 30.

28. The class size is (A) 10 (B) 11 (C) 8 (D) 9 128 – 137
29. The class boundaries of the second class is 146 – 155
    (A) (132.5, 141.5) (B) (137, 146) (C) (133, 141) (D) (134.5, 141.5)
30. The class limits of the last class are (C) (178, 182) (D) (178, 186) (A) (173, 183) (B) (169, 177)

EXHIBIT VI

The foilowing computations were obtained from the number of plants (X) in a Farm and it's yield (Y).

$\bar{X}=12.88$, $var(X)=47.36$, $\sum_{i=1}^{n} X_i^2=1705$, $\sum_{i=1}^{n} Y_i=1992$, $\sum_{i=1}^{n} Y_i^2=585808$ *and* $\sum_{i=1}^{n} X_i Y_i=31330$.

Use this Information contained in Exhibit VI to answer Questions 31, 32, 33 and 34.

31. Obtain the value of n. (A) 9 (B) 10 (C) 7 (D) 8
32. The gradient of the line of best fit for the data is …. (A) -14.01 (B) 15.02 (C) 14.01 (D) 15.01
33. The intercept of the line of best fit for the data is … (A) -45.57 (B) 55.67 (C) 45.67 (D) 45.57
34. The coefficient of correlation, Using the product moment correlation method, between X and Y is ….
    (A) 0.96 (B) 0.97 (C) 0.94 (D) 0.95
35. The Methods of Point Estimation are ….. and ……. Estimation.
    (A) Point and Intennal (B) Point and Confidence (C) Point and Internal (D) Point and Interval

EXHIBIT VII

A random variable X follows the binomial distribution with parameters 5 and 0.4.
Use the information contained in EXHIBIT VII to answer questions 36, 37 and 38.

36. Find the probability that $X=2$ is (A) 0.45 (B) 0.44 (C) 0.35 (D) 0.34
37. Find the probability that $X<2$ is (A) 0.34 (B) 0.26 (C) 0.68 (D) 0.91
38. Find the probability that $X\geq2$ is (A) 0.32 (B) 0.67 (C) 0.34 (D) 0.66

133 – 141 4, 4

EXHIBIT VIII

A fair coin is tossed three times
Use the information contained in EXHIBIT VIII to answer questions 39, 40 and 41.

39. Find the probability of obtaining exactly three heads (A) 3/8 (B) 5/8 (C) 1/8 (D) 2/8
40. Find the probability of obtaining at least three heads. (A) 3/8 (B) 1 (C) 1/4 (D) 1/8
41. Find the probability of obtaining at most three heads is (A) 1/4 (B) ½ (C) 1/16 (D) 1/8
42. Each class of a class interval contains ---------------values.
    (A) homogeneous (B) compacted (C) opposite (D) heterogeneous
43. One of this options is not part of a good statistical Table.
    (A) Foot note (B) Content title (C) Table label (D) Foot source
44. Table 1 stands for -----. (A) First Table seen in Page one (B) First Table for data one (C) First Table in Chapter one D) First Table in the work
45. Non-numeric data are ------ in nature. (A) quantitative (B) distinct (C) Qualitative (D) calculative
46. Information employed for a study which had been collected by another investigator is ---- data.
    (A) Tertiary (B) Supplementary (C) Primary (D) secondary
47. A study that generates information about an enquiry from part of the targeted population is ----.
    (A) Observation (B) Sample survey (C) Census (D) Part population survey
48. The quality scale adopted for scaling kind of information is called --------.
    (A) Interval (B) Ratio (C) Nominal (D) Ordinal
49. Tabular arrangement of data where frequency is associated with corresponding class interval is called --- distribution. (A) Interval (B) Class interval (C) Data (D) Frequency
50. The least value that belongs to a class interval is called-----.
    (A) Lower class value (B) Smallest class value (C) Least class limit (D) Lower class limit
51. The range of a distribution is six times its class interval size. Find the number of data values observed for the distribution. (A) 36 (B) 37 (C) 34 (D) 35
52. Find the number of classes required for proper grouping of sixty set of observations
    (A) 8 (B) 9 (C) 6 (D) 7
53. The lower class limit equals the number of classes required for grouping a set of distribution. If the upper class limit is twice the number of classes and the range equals thrice the lower class limit, find the range of the distribution. (A) 9 (B) 1 (C) 3 (D) 6
54. A Table giving all possible values of a discrete random variable with their associated probabilities is called (A) Cumulative probability density (B) Cumulative density (C) Probability density (D) Probability distribution
55. A Normal distribution that has zero mean and unit variance is called …. distribution.
    (A) Standard Normal (B) Gaussian (C) Binomial (D) Normal
56. If the variable Z follows standard normal distribution then $P(1.50\leq Z\leq1.83)$ is
    (A) 0.5124 (B) 0.8996 (C) 0.0332 (D) 0.0228
57. The parameters of the binomial distribution are (A) n, p (B) p, q (C) n, p, q (D) n, q
58. One advantage of the mean is that it utilizes information on all items in a series.
    (A) Often time (B) Seldomly true (C) True (D) False
59. One limitations of the mean is that (A) It is affected by extreme values (B) it can be calculated for qualitative data (C) it can be calculated when there are open class (D) it can be combined for several classes
60. The probability distribution in which both the mean and variance are equal is …. distribution.
    (A) Normal (B) Uniform (C) Binomial (D) Poisson
61. When is a Bar chart preferred to a Pie chart
    (A) When there is need to emphasize a significant aspect of the data (B) When number of class interval is low (C) When number of classes is small (D) When number of classes is high
62. Which of the following is not an advantage of the Component Bar chart (A) It provides a bar chart of each common component of the categories (B) It provides a within category comparison of contributions of the components (C) It provides all the information contained in the data (D) It provides a bar of the first component of the categories

63. Which of the following is not an advantage of Multiple Bar Chart: (A) It provides for within category comparison (B) It provides for between categories comparison of the contributions of the first component (C) It enjoys all the advantage s of the simple bar chart (D) It has at least two bar charts on the same diagram

EXHIBIT IX

The weight (measured in KG) of fifty Paw-Paw heads harvested at FUTO farms LTD were classified as follows:

Weight interval	0.1 – 1.49	1.50- 2.89	2.90 – 4. 29	4.30 – 5.69	5.70 – 7.09	7.10 – 8.49	8.50 – 9.89	9.90 – 11.29
frequency	12	6	$2X_1+3$	8	8	X_1	X_1	1

Use the information contained in EXHIBIT IX to answer questions 64 and 65.

64. Find the frequency of the third class. (A) 6 (B) 21 (C) 10 (D) 9

65. Calculate the Harmonic mean of the distribution. (A) 2.09 Kg (B) 2.09 (C) 2.08 Kg (D) 2.08

66. Calculate the Geometric mean of the distribution. (A) 3.09 (B) 3.09 Kg (C) 3.07 (D) 3.07 Kg

67. The representation of the entire data points on the cartesian plane is called …
(A) Histogram (B) Data plane (C) Scattergram (D) Area plot

68. The value of the upper and lower class boundaries of a class are (X^2+2) and $(4X+1)$ respectively. If the class mark is 1.5, find the value of the lower class boundary.
(A) -16 (B) -15 (C) 4 (D) 2

69. If A, B, C are any three events, the probability of the union of the three events is:
(A) P(A) P(B) P(C) - $P(A\cap B\cap C)$ (B) P(A) + P(B) + P(C) + $P(A\cap B)$ + $P(A\cap C)$ + $P(B\cap C)$ - $P(A\cap B\cap C)$
(C) P(A) + P(B) + P(C) (D) P(A) + P(B) + P(C) – $P(A\cap B)$ - $P(A\cap C)$ - $P(B\cap C)$ + $P(A\cap B\cap C)$

70. If A, B, C are mutually exclusive events, the probability of the union of the three events is:
(A) P(A) P(B) P(C) - $P(A\cap B\cap C)$ (B) P(A) + P(B) + P(C) + $P(A\cap B)$ + $P(A\cap C)$ + $P(B\cap C)$ - $P(A\cap B\cap C)$
(C) P(A) + P(B) + P(C) (D) P(A) + P(B) + P(C) – $P(A\cap B)$ - $P(A\cap C)$ - $P(B\cap C)$ + $P(A\cap B\cap C)$

$\frac{1}{n}\left(\frac{1}{12} + \frac{1}{6} \ldots\right)$

English (x)	Math: (y)	d	d^2
80	50	3.5	12.25
75	47	6	36
75	45	6	36
70	32	8	64
65	32	10.5	110.25
65	47	10.5	110.25
100	65	1.5	2.25
75	43	6	36
80	52	3.5	12.25
100	47	1.5	

$1.5^2 + 2$

$6\Sigma d^2$

Values (x)	35	45	55	65	75	85	95
Frequency (f)	6	20	30	20	10	8	6

Mean, $\bar{x} = \frac{\Sigma fx}{\Sigma f}$

$\Sigma fx = 6(35) + 20(45) + \cdots + 8(85) + 6(95)$
$= 6060$

$\Sigma f = 6 + 20 + \cdots + 8 + 6$
$= 100$

$\Rightarrow \bar{x} = \frac{6060}{100} = 60.6$ (D)

2) $\sigma = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n}} = \sqrt{\frac{\Sigma f(x-\bar{x})^2}{\Sigma f}}$

$= \sqrt{\frac{\Sigma fx^2 - \Sigma f(\bar{x})^2}{\Sigma f}}$

where $\Sigma fx^2 = 6(35^2) + 20(45^2) + \cdots + 8(85^2) + 6(95^2)$

$= 391300$

$\Rightarrow \sigma = \sqrt{\frac{391300 - 100(60.6)^2}{100}}$

$= \sqrt{\frac{391300 - 367236}{100}} = \sqrt{240.64}$

$= 15.51$ (B)

3) D

4) D

5) $\Omega = 1, 2, 3, 4, 5, 6$ (sample space)

Let E be the event of obtaining an odd number:

Then, $E = 1, 3, 5 \Rightarrow n(E) = 3$

$P(E) = \frac{3}{6} = \frac{1}{2}$ (D)

6) (A) Random variable.

7) (A)

(P91)

8) (A)

9)

x	80	75	75	70	65	65	100	75	80	100
y	50	47	45	32	32	47	65	45	52	47

x	y	R_x	R_y	d	d^2
80	50	3.5	3	0.5	0.25
75	47	6	5	1	1
75	45	6	7.5	-1.5	2.25
70	32	8	9.5	-1.5	2.25
65	32	9.5	9.5	0	0
65	47	9.5	5	4.5	20.25
100	65	1.5	1	0.5	0.25
75	45	6	7.5	-1.5	2.25
80	52	3.5	2	1.5	2.25
100	47	1.5	5	-3.5	12.25
					43.75

$d^2 = (R_x - R_y)^2$

$r = 1 - \frac{6\Sigma d^2}{n(n^2-1)} = 1 - \frac{6(43.75)}{10(100-1)}$

$= 1 - \frac{262.5}{990} = 1 - 0.2652$

$= 0.7348 \simeq 0.73$ (C)

10) (A)

11) Let M be the event that the man watches the TV show; Let W be the event that the wife watches the TV show.

$P(M/W) = 3(1-x^2)$

$P(W) = 2/3$; $P(M\cap W) = \frac{3}{4} - x^2$.

From conditional probability:

$P(M/W) = \frac{P(M\cap W)}{P(W)}$

$\Rightarrow 3(1-x^2) = \frac{\frac{3}{4} - x^2}{2/3}$

Cross-multiplying; we obtain

$\frac{2}{3}\left[3(1-x^2)\right] = \frac{3}{4} - x^2$

$2(1-x^2) = \frac{3}{4} - x^2$

$2-2x^2 = \frac{3}{4} - x^2$ (multiply thru by 4)

$\Rightarrow 8-8x^2 = 3-4x^2$

$8x^2 - 4x^2 = 8-3$

$4x^2 = 5$

$x^2 = \frac{5}{4} \Rightarrow x = \frac{\sqrt{5}}{2}$

$x = \frac{\sqrt{5}}{2}$ (No option is correct)

(12) $P(M|W) = 3\left(1-\frac{5}{4}\right) = \frac{-3}{4}$ (no option)

Since Prob here is negative, we infer that the problem is faulty.

(13) $X_{min} = 10Km$; $X_{max} = 145Km$

Since $K = 10$; but $K = 1+3.222\log n$.

and class size $= \frac{Range}{K} = \frac{145-10}{10} = 13.5$

Starting from 10, the upper class of the last class must be 145.

10 – ?
23.5 – ?
37.0 – ?
50.5 – ?
64.0 – ?
77.5 – ?
91.0 – ?
104.5 – ?
118.00 – ?
131.5 — 145

Now, $145 - 131.5 = 13.5$, same as the class size, thus,

$37.0 + 13.5 = 50.5$; reduce 50.5 by 0.01

ie, 37.00 — **50.49** (A)

0.01 allows each class to vary.

(14) (D)

Proof: Given $\sum(x_i - \bar{x}) = 0$

$\Rightarrow \sum x_i - \sum\bar{x}$; but $\sum\bar{x} = n\bar{x}$

$\because \bar{x} = \frac{\sum x}{n} \Rightarrow \sum x = n\bar{x}$

$\therefore \sum x_i - \sum\bar{x} = n\bar{x} - n\bar{x} = 0$

(15) $P = \frac{1}{2}$, Since it's either a girl or a boy

$n = 5$

$\therefore P(X=x) = {}^nC_x\, p^x q^{n-x}$

$P(X \geq 4) = P(X=4) + P(X=5)$

$P(X=4) = {}^5C_4\left(\frac{1}{2}\right)^4\left(\frac{1}{2}\right)^{5-4} = 5\left(\frac{1}{32}\right) = \frac{5}{32}$

$P(X=5) = {}^5C_5\left(\frac{1}{2}\right)^5\left(\frac{1}{2}\right)^0 = \frac{1}{32}$

$\therefore P(X \geq 4) = \frac{5}{32} + \frac{1}{32} = \frac{6}{32} = 0.1875$

$\simeq 0.19$ (D)

(16) Ω

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

$n(6) = 5$; $N = 36$

$\therefore P(6) = 5/36 = 0.1389 \simeq 0.14$ (C)

(17) $\lambda = \mu = 4$ errors per page.

$P(X=x) = \frac{\lambda^x e^{-\lambda}}{x!}$; $x = 0, 1, 2, \ldots$

$P(X \geq 2) = 1 - P(X<2) = 1 - \left[P(X=0) + P(X=1)\right]$

$P(X=0) = \frac{4^0 e^{-4}}{0!} = 0.0183$

$P(X=1) = \frac{4^1 e^{-4}}{1!} = 0.0733$

$\Rightarrow P(X=0) + P(X=1) = 0.0183 + 0.0733 = 0.0916$

$P(X \geq 2) = 1 - 0.0916 = 0.9084$

$\simeq 0.91$ (B)

) $Q_3 = 2x^2+1$; $Q_1 = 8x-9$

and $Q_3 - Q_1 = 12$

$\Rightarrow 2x^2+1-(8x-9) = 12$

$2x^2+1-8x+9-12 = 0$

$2x^2-8x-2 = 0$

$\Rightarrow x^2-4x-1 = 0$

but, $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

$x = \frac{4 \pm \sqrt{16+4}}{2} = \frac{4 \pm 2\sqrt{5}}{2}$

$x = 2+\sqrt{5}$ or $x = 2-\sqrt{5}$

$\Rightarrow Q_3 = 2(2+\sqrt{5})^2+1 = 19+8\sqrt{5} = \mathbf{36.89}$

(no option)

(19) $Q_1 = 8(2+\sqrt{5}) - 9 = 7+8\sqrt{5} = 24.89$.

(no option).

See that $Q_3 - Q_1 = \underline{\underline{12}}$

(20) 75th percentile is the same as the third quartile, hence, (no option is correct)

1) D

2) C

23) A

(24) $\mu = 43\%$; $\sigma^2 = 6.25 \Rightarrow \sigma = \sqrt{6.25} = 2.5$

$40 - 47 = D$

$n(D) = 20$

$Z = \frac{X-\mu}{\sigma}$

$P[40 \leq X \leq 47] = P\left[\frac{40-43}{2.5} \leq Z \leq \frac{47-43}{2.5}\right]$

$= P[-1.2 \leq Z \leq 1.6] = \phi(1.6) - \phi(-1.2)$

$= 0.9452 - 0.1151 = 0.8301$

$\simeq 0.83$ (D)

(25) $N = \frac{n}{P} = \frac{20}{0.8301} = 24.09$

$\simeq 24$ (C)

(26) (C)

(27) (D)

(28) Given

128, 137, 146, 155, 164, 173 and 182 ;

Class ~~mar~~ size $= 137-128 = 146-137 = \cdots = 182-173$

$= 9$ (D)

(29) Given the class marks 137.

the class interval is, by substracting and adding 4,: 133 – 141; adding and subtracting 0.5, we obtain:

(132.5, 141.5) (A)

(30) Given 182, adding and subtracting 4 we obtain:

(178, 186) (D)

note, for questions 29 and 30, class size is 9, but difference between the Uc & Lc of a class is one less than the class size, ie, 8, and $\frac{8}{2} = 4$.

That's how the '4' came about.

Uc = Upper class Limit, not boundary

Lc = Lower class Limit, not boundary.

(31) $\bar{x} = 12.8$, $V(x) = 47.36$; $\sum_{i=1}^{n} x_i^2 = 1705$, $\sum_{i=1}^{n} Y_i = 1992$, $\sum_{i=1}^{n} Y_i^2 = 585808$; $\sum_{i=1}^{n} x_i Y_i = 31330$

but, $V(x) = \frac{\sum x^2 - n\bar{x}^2}{n}$

$\Rightarrow 47.36 = \frac{1705 - n(12.88)^2}{n}$

$47.36n = 1705 - 165.8944n$

$47.36n + 165.8944n = 1705.00$

$213.2544n = 1705$

$n = \frac{1705}{213.2544} = 7.9951 \simeq 8$

$n = 8$ (D)

(32) gradient, $b = \frac{\sum xy - n\bar{x}\bar{y}}{\sum x^2 - n\bar{x}^2}$

$\sum xy - n\bar{x}\bar{y} = 31330 - 8(12.88)\left(\frac{1992}{8}\right)$

$= 31330 - 25656.96$

$= 5673.04$

$\sum x^2 - n\bar{x}^2 = 1705 - 8(12.88^2) = 1705 - 1327.1552$

$= 377.8448$

$\Rightarrow b = \frac{5673.04}{377.8448} = 15.0142$

$b \simeq 15.01$ (D)

(33) the intercept, $a = \bar{y} - b\bar{x}$

$a = \frac{1992}{8} - 15.0142(12.88)$

$= 249 - 193.3829 = 55.6171$

$a \simeq 55.62$

Using $b = 15.01$;

$a = 249 - 193.3288$

$= 55.67$ (B)

(34) $\rho = \frac{\sum xy - n\bar{y}\bar{x}}{\sqrt{(\sum x^2 - n\bar{x}^2)(\sum y^2 - n\bar{y}^2)}}$

$\sum xy - n\bar{y}\bar{x} = 5673.04$

$\sum x^2 - n\bar{x}^2 = 377.8448$

$\sum y^2 - n\bar{y}^2 = 585808 - 8(249^2) = 89800$

$\Rightarrow \rho = \frac{5673.04}{\sqrt{377.8448(89800)}} = \frac{5673.04}{\sqrt{33930463.0}}$

$\rho = \frac{5673.04}{5824.9861} = 0.9739 \simeq 0.$

(B)

(35) The methods of estimation are Point and Interval estimation.
But the methods of POINT estimation are "Method of Moment", "Method of Least Square", "Method of Maximum Likelihood".
If the question was for method of estimation (D) would be correct, else, no option is correct.

(36) $X \sim b(5, 0.4)$

$\Rightarrow n = 5,\ p = 0.4,\ q = 1 - p = 1 - 0.4 = 0.6$

$P(X = x) = {}^5C_x (0.4)^x (0.6)^{5-x}$; $x = 0, 1, 2, \ldots, 5$

ie, $P(X = x) = {}^nC_x p^x q^{n-x}$.

then, $P(X = 2) = {}^5C_2 (0.4)^2 (0.6)^3$

$= 10 \times 0.16 \times 0.216$

$= 0.3456 \simeq 0.35$ (C)

(37) $P(X < 2) = P(X = 0) + P(X = 1)$

$P(X = 0) = {}^5C_0 (0.4)^0 (0.6)^5 = 0.07776$

$P(X = 1) = {}^5C_1 (0.4)^1 (0.6)^4 = 0.2592$

$P(X = 0) + P(X = 1) = 0.0778 + 0.2592 = 0.337$

but $0.337 \simeq 0.34$

$\Rightarrow P(X < 2) = 0.34$ (A)

(38) $P(X \geq 2) = 1 - P(X < 2) = 1 - 0.34 = 0.66$ (D)

(39) $\Omega = \{HHH, HHT, HTT, HTH, THH, THT, TTH, TTT\}$

$P(H = 3) = \frac{1}{8}$ (C)

(40) $P(H \geq 3) = \frac{1}{8}$ (D)

(41) $P(H \leq 3) = P(H=0) + P(H=1) + P(H=2) + P(H=3)$

$= \frac{1}{8} + \frac{3}{8} + \frac{3}{8} + \frac{1}{8} = 1$

(no option)

(42) (B)

(43) (D)

(44) (D)

(45) (C)

(46) (D)

(47) B

(48) B

(49) D

(50) D

(51) Let R = range, C = class size, K = number of classes.

From definition, $C = \frac{R}{K}$

but $R = 6C$

$\Rightarrow C = \frac{6C}{K} \Rightarrow KC = 6C \Rightarrow K = 6$

but $K = 1 + 3.222 \log n$

$\Rightarrow 6 = 1 + 3.222 \log n$

$3.222 \log n = 6 - 1 = 5$

$\log n = \frac{5}{3.222} = 1.5518$

$\Rightarrow n = \text{Antilog}(1.5518) = 10^{1.5518}$

$= 35.6287 \simeq 36$ (A)

(52) $K = 1 + 3.222 \log n$; but $n = 60$

$\Rightarrow K = 1 + 3.222 \log 60$

$= 1 + 3.222(1.7782)$

$= 1 + 5.7294 = 6.7294$

$K \simeq 7$ (D)

(53) $L_c = K$; $U_c = 2K$; $R = 3K = 3L_c$.

C = class size

$C = \frac{R}{K}$

$C \simeq (2K - K) + 1$

ie, $C = (U - L) + 1$

$\Rightarrow 2K - K + 1 = K + 1 = \frac{3K}{K}$

$\Rightarrow K + 1 = 3 \Rightarrow K = 3 - 1 = 2$

but $R = 3K \Rightarrow R = 3(2) = 6$ (D)

(54) (D)

(55) (A)

(56) Given $P(1.50 \leq Z \leq 1.83)$

$= \phi(1.83) - \phi(1.50)$

$= 0.9664 - 0.9332$

$= 0.0332$ (C)

(57) (A) ie, $X \sim b(n, p)$

(58) (C)

(59) (A)

(60) (D)

(61) (C)

(62) (D)

(63) (B)

(64)

weight interval	freq.	class mark	$f(\frac{1}{x})$
0.1 - 1.49	12	0.795	15.0943
1.50 - 2.89	6	2.195	2.7335
2.90 - 4.29	$2x_1+3=9$	3.595	2.5035
4.30 - 5.69	8	4.995	1.6016
5.70 - 7.09	8	6.395	1.2510
7.10 - 8.49	$x_1=3$	7.795	0.3849
8.50 - 9.89	$x_1=3$	9.195	0.326
9.90 - 11.29	1	10.595	0.0944
	50		23.9892

Total freq = 50

i.e. $50 = 12+6+(2x_1+3)+8+8+x_1+x_1+1$

$50 = 38 + 4x_1$

$\Rightarrow 4x_1 = 50-38 = 12$

$x_1 = \frac{12}{4} = 3$

$\Rightarrow 2x_1+3 = 2(3)+3 = 6+3$

$= 9$ (D)

(65) $HM = \frac{\Sigma f}{\Sigma \frac{f}{x_i}}$

$\Rightarrow HM = \frac{50}{23.9892} = 2.0843$ kg

$\simeq 2.08$ kg (C)

(66) $GM = \sqrt[\Sigma f]{X_1^{f_1} \times X_2^{f_2} \times \dots \times X_n^{f_n}}$

$\Rightarrow \sqrt[50]{0.795^{12} \times 2.195^{6} \times \dots \times 10.595^{1}}$

$= \sqrt[50]{3.023539141 \times 10^{24}}$

$= 3.0875$ kg

$\simeq 3.09$ kg (B)

(67) (C)

(68) $U_b = x^2+2; \quad L_b = 4x+1$

$\bar{x} = 1.5$

but $\bar{x} = \frac{U_b + L_b}{2}$

$\Rightarrow 1.5 = \frac{(x^2+2)+(4x+1)}{2}$

$3 = x^2+2+4x+1$

$x^2+4x+3-3 = 0$

$x^2+4x = 0$

$x(x+4) = 0$

$x = 0$ or $x = -4$

when $x=0$, $L_b = 4(0)+1 = 1$

when $x=-4$, $L_b = 4(-4)+1 = -16+1 = -1$

$\therefore L_b = -15$ (B)

(69) (D)

(70) (C)

where, L_b = lower class boundary, not li

U_b = Upper class boundary, not limit.

FEDERAL UNIVERSITY OF TECHNOLOGY, OWERRI
DEPARTMENT OF STATISTICS
2016/2017 RAIN SEMESTER EXAMINATION
STA 211: INTRODUCTION TO STATISTICS AND PROBABILITY
INSTRUCTION: ANSWER ALL QUESTIONS; TIME: 3 HOURS

(1) The term used to describe the degree of asymmetry of a given distribution is -------?
(A) skewness (B) kurtosis (C) correlation (D) regression

(2) When a distribution is positively skewed, it means that …….. ?
(A) mean = median (B) mean < median (C) mean > median (D) mean –median = 0

(3) The term used to describe the degree of peakedness of a distribution usually taken relative to the normal distribution is ---------? (A) skewness (B) kurtosis (C) correlation (D) moment

(4) A data set is normally distributed with mean 100 and variance 25. Find $P(X<90)$?
(A) 0.0228 (B) 0.0105 (C) 0.2381 (D) 0.1680

(5) Assume X has a binomial distribution with n= 4 and p= 0.41. Find $P(X \geq 2)$?
(A) 0.542 (B) 0.351 (C) 0.163 (D) 0.028

(6) A die is tossed once. Let X be the event of getting at most 2. What is the probability of event X?
(A) 1/2 (B) 1/3 (C) 1/4 (D) 1/5

(7) The mean number of accidents per week at a certain T-junction is 3. What is the probability that in any given week 4 accidents will occur at this T-junction?
(A) 0.861 (B) 0.921 (C) 0.168 (D) 0.621

(8) It is known that 2 out of every 10 bolts produced by a Machine is defective. In a pack of six bolts, find the probability that at least 2 bolts are defective? (A) 0.26 (B) 0.02 (C) 0.39 (D) 0.65

(9) The probability that a student guesses the answer to a question correctly is 0.25. What is the probability that He guesses 2 correctly out of 5 questions? (A) 0.26 (B) 0.25 (C) 0.75 (D) 0.65

Exhibit I: Using the information contained in Exhibit I, answer Questions 10 through 16.
The weights (in kg) of 13 students were recorded as follows:
60, 55, 47, 49, 78, 50, 51, 46, 55, 61, 52, 40, 43

(10) Determine the harmonic mean of the weights of the students?
(A) 51.43 Kg (B) 50.44 Kg (C) 50.44 (D) 51.43

(11) Determine the Geometric mean of the weights of the students?
(A) 51.98 Kg (B) 52.10 Kg (C) 52.10 (D) 51.98

(12) Determine the second quartile of the weights of the students? (A) 47 (B) 47.5 (C) 50 (D) 51

(13) Determine the third quartile of ~~the weights of the students?~~ (A) 55 (B) 55.5 (C) 57.5 (D) 58.5

(14) Determine the seventh decile of the weights of the students? (A) 55 (B) 55.5 (C) 57.5 (D) 58.5

(15) Determine the 55th percentile of the weights of the students? (A) 47.7 (B) 47.5 (C) 50.7 (D) 51.7

(16) Determine the modal mark of the weights of the students? (A) 50 (B) 51 (C) 55 (D) 60

(17) One of these statistics is not displayed using the Box and Whisker Plot?
(A) Median (B) Quartiles (C) Range (D) Mean

(18) One of these sequences is true when a distribution is skewed to the left?
(A) mode > mean > median (B) mean > mode > median (C) mode > median > mean (D) median > mean > mode

(19) A distribution is said to be symmetric if ……..?
(A) mode > mean > median (B) mean > mode > median (C) mode > median > mean (D) median = mean = mode

(20) One of these statistics is not influenced by wild number?
(A) Mean (B) Median (C) Mode (D) Variance

(21) In a regression study, the plot of the data points in the Cartesian plane is called …….?
(A) Data Plot (B) Line Bolt (C) Regression Plane (D) Scattergram

(22) On the average, 3 vehicles passes through FUTO gate every five minutes. What is the probability that atleast four vehicles will pass through FUTO gate in any interval five minutes?
(A) 0.25 (B) 0.15 (C) 0.35 (D) 0.26

(23) Student's scores in a Mathematics Examination are normally distributed with mean 75% and standard deviation 10%. Find the probability that a student selected at random scores less than 60%?
(A) 0.05 (B) 0.06 (C) 0.07 (D) 0.94

(24) In a pie chat, the sector for Education is 720. If the total allocation to the development plan is N5m, the actual amount allocated to Education is: (A) N 0.72m (B) N 2.1m (C) N 1.1m (D) N 1m

(25) Two events A and B are said to be mutually exclusive if?
(A) If they cannot occur at the same time (B) They always occur together (C) The occurrence of A has no effect on B (D) P(AnB) = P(A).P(B)

(26) Two events A and B are said to be Independent if …….?
(A) P(AnB) = P(A).P(B) (B) P(B/A) = P(A) (C) P(A/B) = P(B) (D) P(AnB) = P(A) + P(B)

(27) Evaluate the probability of either event A or B occurring given that P(A) = 2/3, P(B) = ¼ and $P(A \cap B) = 4/5$.
(A) 19/60 (B) 7/60 (C) 23/60 (D) 9/60

Exhibit II: Using the information contained in Exhibit II, answer Questions 28, 29 and 30.

A Box contains 300 Beads of which 5% are defective; a second Box contains 110 Beads of which 7% are defective while the other three Boxes contain 380 Beads with 4% defective each. It is known that the Boxes are identical in appearance.

(28) What is the probability of obtaining a defective Bead? (A) 0.50 (B) 0.05 (C) 0.01 (D) 0.07

(29) What is the probability of obtaining a defective Bead given that it is contained in Box 3?
(A) 0.17 (B) 0.09 (C) 0.07 (D) 0.20

(30) Determine the total number of non-defective Beads that it is contained in Box 2?
(A) 102 (B) 103 (C) 101 (D) 95

Exhibit III: Using the information contained in Exhibit III, answer Questions 31 and 32.

The probability that a Man drinks a given brand of alcoholic drink is $\frac{2}{3}$ and the probability that his wife drinks the same brand of alcoholic drink is $\frac{1}{9}$.

(31) Find the probability that the couple drink the same brand of alcoholic drink?
(A) 0.07 (B) 0.08 (C) 0.67 (D) 0.11

(32) Find the probability that the Wife drinks given that the Man drinks the same brand of alcoholic drink?
(A) 0.07 (B) 0.08 (C) 0.67 (D) 0.11

(33) Suppose the heights of 100 students in STA 211 class are normally distributed with mean 1.57 and variance 0.04. What is the proportion of students whose heights are between 1.47 and1.75?
(A) 0.5074 (B) 0.3159 (C) 0.1915 (D) 0.90

(34) Frequency polygon is obtained when frequency is plotted on -----------?
(A) class mark (B) class size (C) class boundaries (D) class interval

Exhibit IV: Using the information contained in Exhibit IV, answer Questions 35 Through 41

The distribution shown in Table 1 represents the distance (Km) covered by some selected students from their Home to School. The distance (Km) covered is recorded in one decimal point and summarized in Table 1.

Table 1

Serial Number	Distance (Km)	Frequency
1	10-14	3
2	15-19	12
3	20-24	17
4	25-29	9
5	30-34	4
6	35-39	1

(35) Determine the upper class boundary of the third class? (A) 20.50 (B) 24.50 (C) 22.00 (D) 21.99

(36) Determine the lower class boundary of the fifth class? (A) 29.50 (B) 34.55 (C) 34.50 (D) 32.00

(37) Determine the class mark of the second class? (A) 10.00 (B) 19.50 (C) 15.50 (D) 17.00

(38) Calculate the standard deviation of the distribution? (A) 6.7[illegible] (B)7.85 (C) 5.61 (D) 7.75

(39) Calculate the mean $(\overline{X})$ of the distribution? (A) 22.21 (B) 22.22 (C)22.21 Km (D)22.22 Km

(40) Determine the modal value of the distribution? (A) 22.45 (B) 21.40 (C) 22.40 (D) 21.42

(41) Determine the median value of the distribution? (A) 21.75 (B) 21.85 (C) 23.80 (D) 21.80

Exhibit V: Using the information contained in Exhibit V, answer Questions 42 and 43.

The following computations were recorded from the number of plants (X) collected from a Maize Farm and its corresponding yield (Y). $\overline{Y} = 28.40$, $\text{var}(Y) = 48.40$, $\sum X^2 = 385.00$, $\sum X = 55.00$, $\sum Y^2 = 8500.00$ and $\sum XY = 1727.00$.

(42) Find the value of n ? (A) 7 (B) 8 (C) 9 (D) 10

(43) Using the product moment correlation method, the coefficient of linear correlation between X and Y is ……? (A) 0.85 (B) 0.87 (C) 0.86 (D) 0.84

(44) Ten set of observations each for X and Y were examined by a student. It was recorded that $\sum(\text{Rank}(X) - \text{Rank}(Y))^2 = 99.00$. Calculate the coefficient of linear correlation between X and Y?
(A) 0.40 (B) 0.60 (C) 0.50 (D) 0.39

Exhibit VI: Using the information contained in Exhibit VI, answer Questions 45, 46 and 47.

Two fair Dice were thrown and the sample space generated in an experiment performed by a STA 211 student.

(45) What is the probability that sum less than 2 turns? (A) 0.00 (B) 0.10 (C) 0.50 (D) 0.03

(46) What is the probability that sum equals 7 turns? (A) 0.17 (B) 0.03 (C) 0.50 (D) 0.08

(47) What is the probability that at least sum equals 7 turns? (A) 0.17 (B) 0.05 (C) 0.08 (D) 0.58

Exhibit VII: Using the information contained in Exhibit VII, answer Questions 48 and 49.

The mean of a distribution is five times its modal mark X. The distribution has a standard deviation of 0.20 with a skewness of $5X^2 + 20$.

(48) find the modal mark X? (A) 2 (B) 5 (C) 25 (D) 10

(49) find the mean of the distribution? (A) 2 (B) 5 (C) 25 (D) 10

(50) The mean of a distribution is thrice its median value. If the skewness of the distribution is 0.75 and the standard deviation is 4, find the median value of the distribution?
(A) 0.2 (B) 0.5 (C) 0.7 (D) 0.02

(51) Information generated from Government agencies are example of -------- data?
(A) Primary (B) Published (C) Secondary (D) Tertiary

(52) One of these is not among the main methods of data collection?
(A) Interviewer (B) Census (C) Sample survey (D) Observation

(53) A data measurement scale which can be used for any kind of data measurements is ------ scale?
(A) Ratio (B) Interval (C) Ordinal (D) Nominal

(54) The overlap class interval is most suitable for ------ data grouping?
(A) Continuous (B) Discrete (C) Decimal (D) Regular

(55) The least value X of a distribution is 6. If the mid-range of the distribution is twice the least value X, obtain the maximum value of the distribution? (A) 4 (B) 24 (C) 30 (D) 34

(56) In simple linear regression analysis, there ………?
(A) is only one independent variable in the model. (B) could be several linear independent variables in the model. (C) is only one non-linear term in the model. (D) is at least one nonlinear term in the model

(57) The mean of the squared deviations of sample values from the mean is?
(A) Standard deviation (B) Sample variance (C) Population variance (D) Mean deviation

(58) The values of the variance and standard deviation are?
(A) Never negative (B) Always positive (C) Never zero (D) Both positive negative

(59) The highest value of a distribution is five times its least value X. if the mid-range of the distribution is 12, obtain the value of X? (A) 4 (B) 6 (C) 8 (D) 9

(60) The number of classes required to partition a given distribution on students score in a test is 9. Determine the total number of students that participated in the test?
(A) 105 (B) 205 (C) 305 (D) 405

(61) The mid range of a class is twice the class interval size. If there are X^2 number of classes required for proper grouping of the class, determine the value of X^2?
(A) 2 (B) 4 (C) 6 (D) 8

(62) The number of books read by students in the last 12 months were recorded by a teacher and displayed as shown in the Stem and Leaf chart :

Stem	Leaf
0	6 7
1	0 2 2 5 9
2	1 3 5

The student's responses are ....?
(A) 6, 7, 0, 2, 2, 5, 9, 1, 3, 5 (B) 31, 12, 1, 52, 70, 91, 60, 21, 51, 21 (C) 12, 23, 19, 6, 10, 7, 15, 25, 21, 12 (D) 0, 7, 0, 2, 0, 5, 9, 1, 3, 5

Exhibit VIII: Using the information contained in Exhibit VIII, answer Questions 63 and 64.
The upper class limit of a class is five times its lower limit X. If the class mark of the class is 32.

(63) Determine the upper class limit of the class? (A) 80.0 (B) 94.0 (C) 53.5 (D) 32.0

(64) Determine the lower class limit of the class? (A) 16.0 (B) 18.8 (C) 10.7 (D) 6.4

(65) The diagram which shows the relationship between two or more set of quantities is ....?
(A) Double bar chart (B) Ogive (C) Line graph (D) Pictograph

(66) The diagram which conveys the idea about the shape of the density of the random variables in its original form is? (A) Stem and leaf plot (B) Bar chart (C) Ogive (D) Line graph

(67) One of these conditions is true about the relationship between the Arithmetic mean ($\overline{X}$), Harmonic mean ($\overline{X}_H$) and Geometric mean ($\overline{X}_G$), when the set of observations of a distribution have common values? (A) $\overline{X}=\overline{X}_G \geq \overline{X}_H$ (B) $\overline{X} \leq \overline{X}_G = \overline{X}_H$ (C) $\overline{X}=\overline{X}_G=\overline{X}_H$ (D) $\overline{X} \geq \overline{X}_G \geq \overline{X}_H$

(68) The line chosen in a simple linear regression study such that the sums of squares of errors are minimized is called?
(A) Scatter line (B) Line of best fit (C) Regression line (D) Least squares line

Exhibit IX: Using the information contained in Exhibit IX, answer Questions 69 and 70
The regression coefficient in a simple linear regression between the dependent variable (Y) and independent variable (X) is one-fifths of the intercept. A random sample of eight observations yielded $\sum X = 464.4$ and $\sum Y = 46.2$.

(69) Calculate the regression coefficient of the simple linear regression?
(A) 0.09 (B) 0.10 (C) 0.18 (D) 0.20

(70) Calculate the intercept of the simple linear regression? (A) 0.45 (B) 0.50 (C) 0.90 (D) 1.00

STA211 SOLUTION (EXAM) 2016/2017

1 A
2 C
3 B
4 Mean (μ) = 100, variance (σ^2) = 25
To find $P(x < 90)$
Using the z-distribution formula
$z = \frac{x-\mu}{\sigma} = \frac{90-100}{5} = -\frac{10}{5} = -2$
$P(z<-2) = 0.0228$ (using the negative z-table) A.

5 $P(x \geq 2) = 1 - P(x<2)$, $n=4$, $P=0.41$
Using a binomial distribution
$1 - \sum_{x=0}^{1} b(x:4, 0.41)$
$= 1 - {}^4C_0(0.41)^0(1-0.41)^{4-0} + {}^4C_1(0.41)^1(0.59)^{4-1}$
$= 1 - {}^4C_0(0.41)^0(0.59)^4 + {}^4C_1(0.41)^1(0.59)^3$
$= 1 - (0.1212) + 4(0.0842)$
$= 1 - [0.1212 + 0.3368]$
$= 1 - 0.4580 = 0.542$ A

6 $Pr(\text{At most } 2) = Pr(1) + Pr(2)$
$= \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$ B

7 Using Poisson distribution
$P(x;\mu) = \frac{e^{-\mu}(\mu^x)}{x!}$ where $x=4$, $\mu=3$
$P(4;3) = \frac{(2.71828^{-3})(3^4)}{4!} = 0.168$ C

8 Here $\mu = 0.2$, $x=3$
Using Poisson distribution
$1 - Pr(x \leq 3)$
$= 1 - e^{-0.2}\left[\frac{0.2^0}{0!} + \frac{0.2^1}{1!}\right]$
$= 1 - e^{-0.2}[1 + 0.2] = 1 - 0.8187(1.2)$
$= 0.017 \simeq 0.02$ B.

9 Using binomial distribution where
$P = 0.25$, $n=5$ and $x=2$
$b(x;n,p) = {}^nC_x p^x(1-p)^{n-x}$
$= {}^5C_2(0.25)^2(0.75)^{5-2}$
$= 10 \times 0.0625 \times 0.421875$
$= 0.26$ A.

10 Harmonic mean $= \frac{n}{\sum_i^n \left(\frac{1}{x_i}\right)}$
$= \frac{13}{\left[\frac{1}{60}+\frac{1}{55}+\frac{1}{47}+\frac{1}{49}+\frac{1}{78}+\frac{1}{50}+\frac{1}{51}+\frac{1}{46}+\frac{1}{55}+\frac{1}{61}+\frac{1}{52}+\frac{1}{40}+\frac{1}{43}\right]}$
$= 51.43$ Kg A.

11 Geometric mean $= \left(\prod^n x_i\right)^{1/n}$
$= [60 \times 55 \times 47 \times 49 \times 78 \times 50 \times 51 \times 46 \times 55 \times 61 \times 52 \times 40 \times 43]^{1/13}$
$= 52.105$ B.

12 Second quartile $Q_2 = \frac{i(n+1)}{4}$
Arrange the values in ascending order which is
40, 43, 46, 47, 49, 50, 51, 52, 55, 55, 60, 61, 78
$Q_2 = \frac{2(13+1)}{4} = \frac{2(14)}{4} = 7$th value
The 7th value is 51 D.

13 Using the quartile equation in the above question 12.
$Q_3 = \frac{3(13+1)}{4} = \frac{3(14)}{4} = \frac{42}{4} = 10.5$th value
Which is between 10th and 11th
$Q_3 = 55 + 0.5(60-55)$
$= 55 + 0.5(5) = 55 + 2.5 = 57.5$ C.

14 Decile $= \frac{j(n+1)}{10}$
7th decile $= \frac{7(13+1)}{10} = \frac{7(14)}{10} = \frac{98}{10} = 9.8$th value
It lies between the 9th and 10th value
$D_7 = 55 + 0.8(55-55)$
$= 55 + 0.8(0) = 55$ A.

15 Percentile $= \frac{K(n+1)}{100}$
55th Percentile (P_{55}) $= \frac{55(13+1)}{100}$
$= \frac{55(14)}{100} = \frac{770}{100} = 7.7$th value
Between 7th & 8th value
$P_{55} = 51 + 0.7(52-51)$
$51 + 0.7 = 51.7$ D.

16 Modal mark is the number with highest frequency which is 55 (C)

17 D
18 C
19 D
20 C
21 D
22 Using the Poisson distribution
$1 - Pr(x \leq 4)$
$= 1 - e^{-0.6}\left[\frac{0.6^0}{0!} + \frac{0.6^1}{1!} + \frac{0.6^2}{2!} + \frac{0.6^3}{3!}\right]$
$= 1 - e^{-0.6}[1 + 0.6 + 0.18 + 0.036]$
$= 1 - 0.99664 = 0.0034$

(23) Using z-distribution

$z = \frac{x-\mu}{\sigma}$ where $\begin{cases} x = 0.6 \\ \mu = 0.75 \\ \sigma = 0.1 \end{cases}$

$z = \frac{0.6-0.75}{0.1} = -1.5$

$P(z<-1.5) = 0.07$ (C)

(24) Total degree of a pie chart is 360°

$72^\circ = \frac{72}{360} = \frac{1}{5}$

$\frac{1}{5} \times 5m = ₦1m$ (D)

(25) A

(26) A

(27) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= \frac{2}{3} + \frac{1}{4} - \frac{4}{5} = \frac{40+15-48}{60} = \frac{7}{60}$ (B)

(28) $P(D/B_1) = 0.05$; $P(D, B_2) = 0.07$

$P(D, B_3) = 0.04$, $P(D, B_4) = 0.04$, $P(D, B_5) = 0.04$

By Baye's theorem; $P(B_i/D) = \frac{P(B_i)P(D/B_i)}{\Sigma P(B_i)P(D/B_i)} = \frac{P(B_i \cap D)}{P(D)}$

$P(D) = \Sigma P(B_i)P(D/B_i) = \frac{1}{5}(0.05) + \frac{1}{5}(0.07) + \frac{3}{5}(0.04)$

$0.01 + 0.014 + 0.024 = 0.048 \simeq 0.05$ B.

(29) $P(D/B_3) = \frac{P(D \cap B_3)}{P(B_3)}$ Conditional Probability

$P(B_3/D) = \frac{P(D/B_3) \cdot P(B_3)}{P(D)} = \frac{\frac{1}{5}(0.04)}{0.05}$

$= \frac{0.008}{0.048} = 0.17$ A.

(30) If 7% are defective, then 93% are non defective, but $n = 10$

$\left(\frac{93}{100} \times 110\right)$ are non-defective

$= 102$ beads A.

(31) $P(A \cap B) = P(A) \cdot P(B)$

$= \frac{2}{3} \times \frac{1}{9} = \frac{2}{27} = 0.07$ (A).

(32) $P(B/A) = \frac{P(A \cap B)}{P(A)} = \frac{0.07}{0.67} = 0.11$ D.

(33) $P(1.47 < z < 1.75)$

$= P\left(\frac{1.47-1.57}{0.04}\right) < \frac{x-1.57}{0.04} < P\left(\frac{1.75-1.57}{0.04}\right)$

$= P(-0.5 < z < 0.9) = P(z < 0.9) - P(z < -0.5)$

$= 0.8159 - 0.3085 = 0.5074$ (A)

(34) D

class	Freq	Class boundary	class mark	Cummulative freq
10 – 14	3	9.5 – 14.5	12	3
15 – 19	12	14.5 – 19.5	17	15
20 – 24	17	19.5 – 24.5	22	32
25 – 29	9	24.5 – 29.5	27	41
30 – 34	4	29.5 – 34.5	32	45
35 – 39	1	34.5 – 39.5	37	46

(35) B

(36) C

(37) D

Class Interval	class Mark(x)	Freq(F)	FX	X^2	FX^2
10 – 14	12	3	36	144	432
15 – 19	17	12	204	289	3468
20 – 24	22	17	374	484	8228
25 – 29	27	9	243	729	6561
30 – 34	32	4	128	1024	4096
35 – 39	37	1	37	1369	1396

(38) standard deviation $= \sqrt{\sigma^2}$

$\sigma^2 = \frac{\Sigma fx^2 - \frac{(\Sigma fx)^2}{N}}{N} = \frac{24154 - \frac{(1022)^2}{46}}{46}$

$= \frac{24154 - 22706.174}{46} = \frac{1447.826}{46}$

$\sigma^2 = 31.4745$

$\sigma = \sqrt{31.4745} = 5.61$ (C)

(39) Mean$(\mu) = \frac{\Sigma fx}{\Sigma f} = \frac{1022}{46} = 22.22$ (B)

(40) Modal Value $= L + \frac{f_m - f_{m-1}}{(f_m - f_{m-1}) + (f_m - f_{m+1})} \times C$

$= 19.5 + \frac{17-12}{(17-12)+(17-9)} \times 5$

$= 19.5 + \frac{5}{5+8} \times 5$

$19.5 - \frac{25}{13} = 19.5 - 1.92$

$= 21.42$ (D)

(41) Median Value $= L + \frac{\left(\frac{n}{2} - Cf_{m-1}\right)}{f_m} \times C$

$= 19.5 + \frac{\left(\frac{46}{2} - 15\right)}{17} \times 5$

$= 19.5 + \frac{(23-15)}{17} \times 5$

$19.5 + 2.35 = 21.85$ (B)

(42) To get n. We use

$\sigma^2 = \frac{\Sigma y^2 - n\bar{y}^2}{n} \Rightarrow \sigma^2 n = \Sigma y^2 - n\bar{y}^2$

$(48.40)n = 8500 - n(806.56)$

$806.56n + 48.40n = 8500$

$854.96n = 8500$

$n = 9.94$

$n \simeq \underline{\underline{10}}$

(43) $r = \frac{n\Sigma xy - \Sigma x \Sigma y}{\sqrt{[n\Sigma x^2 - (\Sigma x)^2][n\Sigma y^2 - (\Sigma y)^2]}}$

$r = \frac{(10 \times 1727) - (55)(284)}{\sqrt{[(10 \times 385) - (55)^2][10 \times 8500 - (284)^2]}}$

$r = \frac{17270 - 15620}{\sqrt{(3850-3025)(85000-80650)}}$

$r = \frac{1650}{\sqrt{825 \times 4344}} = \frac{1650}{\sqrt{3583800}}$

$= \frac{1650}{1893.093} = 0.87$ B.

(44) Using Spearman's rank formula where $n = 10$, $d^2 = 99$

$r = 1 - \frac{6\Sigma d^2}{n(n^2-1)} = 1 - \frac{6(99)}{10[(10)^2-1]}$

$= 1 - \frac{6(99)}{10(100-1)} = 1 - \frac{6(99)}{10(99)}$

$= 1 - 0.6 = 0.4$ (A)

(45)

	1	2	3	4	5	6
1	1,1	1,2	1,3	1,4	1,5	1,6
2	2,1	2,2	2,3	2,4	2,5	2,6
3	3,1	3,2	3,3	3,4	3,5	3,6
4	4,1	4,2	4,3	4,4	4,5	4,6
5	5,1	5,2	5,3	5,4	5,5	5,6
6	6,1	6,2	6,3	6,4	6,5	6,6

(A)

(46) $\frac{6}{36} = \frac{1}{6} = 0.1667 \simeq 0.17$ (A)

(47) $\frac{21}{36} = \frac{7}{12} = 0.583 \simeq 0.58$ (D)

(48) Skewness $= \frac{\text{Mean} - \text{Mode}}{\text{Standard Deviation}}$

$5x^2 + 20 = \frac{5x - x}{0.2}$

$(5x^2+20)0.2 = 4x$

$x^2 + 4 = 4x$

$x^2 - 4x + 4 = 0$

$x^2 - 2x - 2x + 4 = 0$

$x(x-2) - 2(x-2) = 0$

$(x-2)(x-2) = 0$

$x = 2$ twice A.

(49) Mean $= 5(x)$

$= 5(2) = 10$ D.

(50) Skewness $= \frac{3(\text{Mean} - \text{Median})}{\text{standard deviation}}$

$0.75 = \frac{3(3x - x)}{4}$

$3 = 3(2x)$

$1 = 2x \Rightarrow x = \frac{1}{2} = 0.5$ B.

(51) C

(52) B

(53) D

(54) B

(55) Midrange $= \frac{\text{Highest Value} - \text{lowest Value}}{2}$

$12 = \frac{x-6}{2} \Rightarrow 24 = x - 6$

$x = 24 + 6 \Rightarrow x = 30$ (C)

(56) B

(57) A

(58) A

(59) Midrange $= \frac{\text{Highest Value} - \text{lowest Value}}{2}$

$12 = \frac{5x - x}{2} \Rightarrow 24 = 4x \Rightarrow x = 6$ (B).

(60) $2^k \geq n \Rightarrow 2^9 \geq 512 \Rightarrow n = 512$

(61) Midrange, $R = xy$

Midrange $= 2y$

$2y = xy$

$x = 2$

$\Rightarrow x^2 = 4$ (B)

(62) C

(63) Class Mark $= \frac{UCL + LCL}{2}$

$32 = \frac{5x + x}{2} \Rightarrow 64 = 6x$

$x = \frac{64}{6} = 10.67 \simeq 10.7$ (LCL)

$UCL = 5(x) = 5(10.7)$

$= 53.5$

(64) C

(65) C

(66) B

(67) D

(68) B Using the regression formula where a is intercept and b is the regression coefficient

$a = \frac{\Sigma y - b\Sigma x}{n}$

$a = \frac{46.2 - \frac{a}{5}(464.4)}{8}$

$8a = 46.2 - 92.88a$

$8a + 92.88a = 46.2$

$100.88a = 46.2 \Rightarrow a = \frac{46.2}{100.88} = 0.457$

$a \simeq 0.5$

Since $b = \frac{a}{5} \Rightarrow b = \frac{0.5}{5} = 0.1$

(69) B

(70) B

FEDERAL UNIVERSITY OF TECHNOLOGY, OWERRI
DEPARTMENT OF STATISTICS
2015/2016 HARMATTAN SEMESTER EXAMINATION
STA 211: INTRODUCTION TO STATISTICS AND PROBABILITY I FOR SMAT
INSTRUCTION: ANSWER ALL QUESTIONS
TIME: 3 HOURS

1. In a single toss of eight fair coins, find the probability of getting at least three heads?
(A) 0.67 (B) 0.96 (C) 0.86 (D) 0.76
2. A computer typist makes six errors per page on the average. What is the probability that she makes at most two errors on the next page? (A) 0.04 (B) 0.05 (C) 0.5 (D) 0.06

EXHIBIT I

Let M and N be two events associated with an experiment. Suppose $P(M) = 0.6$, $P(N) = x, P(M \cup N) = 0.8$. Using this information, answer questions 3 and 4.

3. For what choice of x will the two events be independent? (A) 0.40 (B) 0.50 (C) 0.60 (D) 0.70
4. For what choice of x will the two events be mutually exclusive? (A) 0.20 (B) 0.25 (C) 0.75 (D) 0.70
5. The set of all possible distinct outcomes of an experiment is called?
(A) Event (B) Elementary Event (C) Sample Space (D) Probability Sample Space
6. An experiment is said to be a random experiment if? (A) The set of all possible outcomes of the experiment is known before the trials is performed (B) Experiment is performed under identical conditions (C) A and B (D) It is performed without any stringent conditions.
7. The line drawn to *minimize* the deviation between the data points and the fitted line is ?
(A) Regressor line (B) Normal Line (C) Line of best fit (D) Normal Line with best selected error
8. The process through which an appropriate Statistic is used to approximate corresponding Parameter is called?
(A) Inference Assertion (B) Estimate (C) Estimation (D) Statistics
9. The function of the random variables that does not depend on the unknown Parameter is called?
(A) Statistics (B) Statistic (C) Estimation (D) None Listed
10. One of these methods is a method of point estimation? (A) Method of Squares (B) Method of Optimum Squares (C) Method of Least Squares (D) Method of Maments

EXHIBIT II

The weight of Paw-Paw heads packed in a bag was measured separately (in grams) and recorded by eight students as follows: 79,120, 49, 68, 75, 110, 99, 112. Using this information, answer questions 11, 12, 13 and 14.

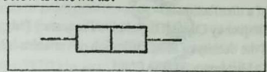
11. Calculate the geometric mean of the distribution on the weight of paw-paw heads as recorded by the students?
(A) 75.63 grams (B) 75.63 (C) 85.63 (D) 85.63 grams
12. Calculate the upper quartile of the distribution on the weight of paw-paw heads as recorded by the students?
(A) 110.0 (B) 110.5 (C) 111.0 (D) 111.5
13. Calculate the lower quartile of the distribution on the weight of paw-paw heads as recorded by the students?
(A) 73.05 (B) 73.15 (C) 73.25 (D) 73.75
14. Evaluate the standard deviation on the weight of paw-paw heads? (A) 13.37 (B) 13.87 (C) 23.37 (D) 23.87
15. Which of these measures is resistant to wild values? (A) Mean (B) Mode (C) Median (D) Range

EXHIBIT III

The data below represents the age (years) of seven students selected randomly from STA 211 class: 19, 21, 26, 20, 28, 20, 22. Using this information, answer questions 16, 17, 18 and 19.

16. Calculate the harmonic mean of the age of the students? (A) 21.88 years (B) 20.88 years (C) 21.88 (D) 20.88
17. Calculate the upper quartile of the distribution on the age of the students? (A) 26 (B) 27 (C) 28 (D) 25
18. Calculate the lower quartile of the distribution on the age of the students?
(A) 20.0 (B) 20.10 (C) 20.50 (D) 21.0

19. Evaluate the variance of the age of the students? (A) 9.92 (B) 9.92 years (C) 11.58 years (D) 11.58
20. A value of a data that is unusual from the bulk of the data is?
(A) Even Number (B) Wild Number (C) Odd Number (D) None of the above
21. The figure below is known as?



(A) Four Number Summary plot (B) Five Number Summary plot (C) Six Number Summary plot (D) A and B
22. Probability of an event can be described in two ways namely?
(A) Classical and Priori (B) Frequency and Optimized (c) Posteriori and Frequency (D) Frequency and Priori
23. Two dice are thrown together, what is the probability that a sum of 4 or 9 turns?
(A) 0.08 (B) 0.12 (C) 0.25 (D) 0.16
24. The probability that Obi watches a television programme is 0.40 and the probability that his sister watch such programme is 0.75. The probability that both Obi and the Sister watched the TV programme is 0.30. What is the probability that Obi's sister watched the TV programme given that Obi does?
(A) 0.75 (B) 0.25 (C) 0.40 (D) 0.60
25. Evaluate the probability of getting at least three girls in a family of five Children?(A) 0.5(B) 0.4 (C) 0.3 (D)0.2
26. In a single toss of eight fair coins, find the probability of getting fewer than six heads?
(A) 0.76 (B) 0.86 (C) 0.96 (D) 0.67
27. The two basic types of data are; (A) Primary and Second hand (B) Primary and Numeric (C) Primary and Secondary (D) Numeric and Non-numeric
28. One of these types of data is a second hand data used by investigators;
(A) Primary (B) Secondary (C) Numeric (D) Non-Numeric
29. All except one is an example of Non-Numeric data. (a) Race (B) Sex (C) Geographical Regions (D) Waist Size
30. The two basic methods for the classification of the sources of data are; (A) Magazine and Research Reports (B) Primary and Secondary (C) Published and Unpublished (D) Printed and Virgin
31. One of the following is not a method of data collection. (A) Survey (B) Published (C) Census (D) Experiment
32. One of these methods of data collection attempt to understand "Cause and Effect" relationships and investigator is not able to control the variables of interest. (A) Questionnaire (B) Survey (C) Observation (D) Experiment
33. The act of taking data measurement on a variable of interest using measurement tool is?
(A) Data Measurement (B) Scale (C) Scale Measure (D) Scaling
34. One of these is a scale used for data measurement; (A) Norminnal (B) Ordinal (C) Ordinnal (D) Norminna
35. A table showing the summary of the proportion of the total number of items of a set of data falling into the different classes is called? (A) Cumulative frequency distribution (B) Frequency distribution (C) Relative frequency distribution (D) None of the above
36. The first and last observation on the distance of seventy students from their home to School is 10 km and 145 km respectively. Evaluate the class interval of the third class when the data is properly grouped using frequency distribution? (A) 48.58 – 67.87 (B) 29.29 – 48.58 (C) 39.29 – 58.58 (D) 48.58 – 67.86
37. The two types of frequency distribution are? (A) Discrete and Ungrouped (B) Discrete and Relative grouped (C) Ungrouped and Continuous (D) Continuous and Grouped
38. The Chart used for the representation of quantitative data is? (A) Multiple Bar Chart (B) Component Bar Chart (C) Frequency Polygon (D) Simple Bar Chart
39. Which of these bar diagram is associated with these properties; I. Height of bars are proportional to the volume of data II. Bars are drawn to scale III. Bars consist of vertical bars of equal width IV. It is used for qualitative data representation(A) Multiple Bar Chart (B) Component Bar Chart (C) Histogram (D) Simple Bar Chart

40. Which of these bar diagram provides within category comparison of the contributions of the various components to the overall volume? (A) Multiple Bar Chart (B) Component Bar Chart (C) Histogram (D) Simple Bar Chart

41. A chart used for representing accumulated frequencies of a distribution is?
(A) Ogive (B) Cumulative Frequency Polygon (C) Frequency Curve (D) Relative Frequency Polygon

42. The bar diagram that conveys an idea about the shape of the density of the distribution of variables is?
(A) Multiple Bar Chart (B) Component Bar Chart (C) Histogram (D) Pie Chart

43. One of these is not a feature of the Stem and Leaf Plot? (A) Each observation considered consists of two parts (B) It retains the original information as the data themselves (C) Outliers are easy to detect (D) The plot does not display the shape of the data.

44. The sum and standard deviation of a set of sixty observations on the scores of students in a given Statistics Course test are 300% and 0.0072 respectively. Find the sum of squares $\left(\sum X^2\right)$ of the ungrouped scores of students in the test to two decimal places. (A) 1500.00 (B) 1498.42 (C) 300.01 (D) 1511.42

45. If X_i and $\overline{X}$ are the ith value and the mean of a set of observations respectively. Then, one of these mathematical statement is true? (A) $\sum(X_i-\overline{X})\neq 0$ (B) $\sum(X_i-\overline{X})=0$ (C) $\sum(X_i-\overline{X})>0$
(D) $\sum(X_i-\overline{X})<0$

46. One of these charts at a glance conveys the idea about the shape of the density of a set of observations?
(A) Multiple Bar Chart (B) Component Bar Chart (C) Histogram (D) Simple Bar Chart

EXHIBIT IV

The number of a particular bus and the mileage covered by such Bus in a week were recorded by the Management of Imo Transport Company and classified as follows;

Mileage Interval	Frequency
120.10 – 139.49	3
139.50 – 158.89	7
158.90 – 178.29	11
178.30 – 197.69	8
197.70 – 217.09	5

Using Exhibit IV, answer questions 47, 48, 49, 50 and 51.

47. Evaluate the upper class boundary of the second class (that is, 139.50 – 158.89)?
(A) 158.899 (B) 158.895 (C) 158.891 (D) 158.890

48. Evaluate the average mileage covered by the bus in a week? (A) 151.45 (B) 159.82 (C) 161.45 (D) 171.45

49. Evaluate the most frequent mileage covered by the bus in a week?
(A) 179.98 (B) 161.98 (C) 169.98 (D) 171.98

50. Find the median of the distribution? (A) 171.24 (B) 161.24 (C) 151.24 (D) 158.24

51. Evaluate the class size of the distribution? (A) 19.390 (B) 19.400 (C) 19.395 (D) 19.405

52. One of these mathematical statement is true about the relationship between the Arithmetic mean $(\overline{X})$, Harmonic mean $(\overline{X}_H)$ and Geometric mean $(\overline{X}_G)$.
(A) $\overline{X}_H>\overline{X}>\overline{X}_G$ (B) $\overline{X}_H>\overline{X}_G>\overline{X}$ (C) $\overline{X}>\overline{X}_G>\overline{X}_H$ (D) $\overline{X}>\overline{X}_H>\overline{X}_G$

EXHIBIT V

The following computations were recorded from the number of plants in a Cassava Plot (X) and it's yield (Y).

$\overline{Y}=12.83$, $var(X)=47.36$, $\sum_{i=1}^{n}X_i^2=1705$, $\sum_{i=1}^{n}Y_i=1992$, $\sum_{i=1}^{n}Y_i^2=585808$ *and* $\sum_{i=1}^{n}X_iY_i=31330$. Use this information to answer Questions 53, 54, 55 and 56.

53. Find the value of n? (A) 6 (B) 7 (C) 8 (D) 9

54. The gradient of the line of best fit for the data is ? (A) 14.01 (B) 15.01 (C) -14.01 (D) 15.02

55. The intercept of the line of best fit for the data is ? (A) 55.67 (B) 45.57 (C) -45.57 (D) 35.67

56. Using the product moment correlation method, the coefficient of correlation between X and Y is ?
(A) 0.94 (B) 0.95 (C) 0.96 (D) 0.97

57. The methods of point estimation are?
(A) Point and Internal (B) Point and Interval (C) Point and Internal (D) Point and Confidence

58. The mean and variance of the CGPA of a random sample of forty students in year three at the Department of Statistics FUTO in 2015/2016 Session are 4.34 and 0.067 respectively. Evaluate the 99% confidence interval for approximating the average CGPA of the Population of students in year three at FUTO?
(A) (4.10, 4.31) (B) (4.18, 4.30) (C) (4.19, 4.39) (D) (4.19, 4.41)

EXHIBIT VI

Consider the One Way Classification ANOVA Table, evaluate and complete the missing value in the Table.

Source of Variation	Degrees of Freedom	Sum of Squares (SS)	Mean Squares (MS)	F ratio
Treatment		19.74	6.58	
Error	32		4.23	-
Total			-	-

Using the information above, answer questions 59, 60, 61, 62 and 63.

59. The sum of squares due to total variation is? (A) 148.41 (B) 158.41 (C) 157.41 (D) 159.41

60. The sum of squares due to error is? (A) 129.57 (B) 139.67 (C) 138.67 (D) 129.67

61. The degrees of freedom due to treatment? (A) 3 (B) 4 (C) 5 (D) 6

62. The degrees of freedom due to total variation? (A) 33 (B) 34 (C) 35 (D) 36

63. The value of the F ratio is? (A) 1.56 (B) 1.46 (C) 1.65 (D) 1.64

EXHIBIT VII

Given below is the probability table for two events (A and B), use the information to answer questions 64 and 65.

event	A	B	Total
A	0.4	0.3	0.7
B	0.3	0.6	0.9
Total	0.7	0.9	-

64. If A and B are independent events, find $P(A\cap B)$? (A) 0.50 (B) 0.60 (C) 0.33 (D) 0.63

65. If A and B are not independent events, $P(A/B)$? (A) 0.30 (B) 0.60 (C) 0.70 (D) 0.90

66. The objective of the completely randomized Design is? (A) Partition a system in a recognized form (B) Partition a system into existing form among the means (C) Discover differences among several means of a given population (D) Detect difference in the mean of a population

67. One of these properties is not a property of a good estimate?
(A) Unbiasedness (B) Efficiency (C) Invertibility (D) Consistency

68. Is it possible that a Man 12 meters tall who walks across a river whose depth is 10.5 meters could be drown in the river? (A) Yes (B) No (C) uncertain (D) A and B

EXHIBIT VIII

The lower class limit of a class is two-third that of the upper class limit. The class mark of the class is 2.25. Using Exhibit VIII, answer questions 69 and 70.

69. Find the upper class limit of the class. (A) 2.24 (B) 2.74 (C) 2.70 (D) 2.80

70. Find the lower class boundary of the class. (A) 1.70 (B) 1.80 (C) 1.75 (D) 1.85

GOOD LUCK! GOOD LUCK!! GOOD LUCK!!!

STA 211 SOLUTION 2015/2016

① The probability of getting at least three heads is given by

$$\frac{\sum_{x=3}^{8} {}^nC_x}{2^n} \quad \text{Where } n=8$$

Thus, we have;

$$\frac{{}^8C_3+{}^8C_4+{}^8C_5+{}^8C_6+{}^8C_7+{}^8C_8}{2^8}$$

$$= \frac{56+70+56+28+8+1}{256} = \frac{217}{256} = 0.86 \text{ (C)}$$

② Using the poisson distribution with $x=0,1,2$ and $\lambda=4$, we have

$P(X;\lambda) = P(0,1,2;4)$

$$\therefore P = \frac{e^{-\lambda}\lambda^x}{x!} = \frac{e^{-6}6^0}{0!} + \frac{e^{-6}6^1}{1!} + \frac{e^{-6}6^2}{2!}$$

$= 0.01487 + 0.0446 + 0.002478$

$\cong 0.06$ (D)

③ $P(M)=0.6$, $P(N)=x$, $P(M\cup N)=0.8$

for independent events

$P(M\cap N) = P(M)\cdot P(N)$

Also

$P(M\cup N) = P(M)+P(N) - P(M\cap N)$

$P(M\cup N) = P(M)+P(N) - P(M)\cdot P(N)$

$0.8 = 0.6 + x - 0.6x$

$0.8 - 0.6 = x - 0.6x$

$0.2 = x(1-0.6)$

$0.2 = 0.4x \Rightarrow x = \frac{0.2}{0.4}$

$x = 0.5$ (B)

④ For mutually exclusive events

$P(M\cup N) = P(M) + P(N)$

$0.8 = 0.6 + x$

$x = 0.8 - 0.6 = 0.2$ (A)

⑤ C

⑥ B

⑦ C

⑧ C

⑨ B

⑩ C

⑪ Geometric mean $= X_G$

$$= \sqrt[n]{x_1 \cdot x_2 \cdots x_n} = \left(\prod_{i=1}^{n} x_i\right)^{1/n}$$

$= (79 \times 120 \times 49 \times 68 \times 75 \times 110 \times 99 \times 112)^{\frac{1}{8}}$

$= (2.8895 \times 10^{15})^{0.125}$

$= 85.625g \simeq 85.63$ grams (D)

⑫ Upper quartile $= Q_3$

$Q_3 =$ Value of $\frac{3(n+1)}{4}$th item

Here $n = 8$, so

Value of $\frac{3(8+1)}{4}$th item

$= \frac{27}{4} = 6.75$th item

By arranging the items in ascending order, we have;

49, 68, 75, 79, 99, 110, 112, 120

We will make use of the 6th and 7th items, Which are 110 and 112 respectively so, the 6.75th item is

$= 6$th item $+ 0.75$(7th item $-$ 6th item)

$= 110 + 0.75(112-110)$

$= 110 + 1.5 = 111.5$ (D)

⑬ Lower Quartile, Q_1 ; $n=8$

$Q_1 =$ value of $\frac{(n+1)}{4}$th item

$=$ value of $\frac{(8+1)}{4}$th item $= 2.25$th item

The 2nd and 3rd item will be used in this case.

$= 2$nd item $+ 0.25$(3rd item $-$ 2nd item)

$= 68 + 0.25(75-68)$

$= 68 + 1.75 = 69.75$

⑭ Standard deviation (S.D) $= \sqrt{Var}$

$$Var = \sum_{i=1}^{N} \frac{(x_i-\bar{x})^2}{N} = \sigma^2$$

$$\bar{x} = \frac{49+68+75+79+99+110+112+120}{8}$$

$$= \frac{712}{8} = 89$$

29

$$\sigma^2 = \frac{\left\{\begin{array}{l}(49-89)^2+(68-89)^2+(75-89)^2+(79-89)^2 \\ +(99-89)^2+(110-89)^2+(112-89)^2+(120-89)^2\end{array}\right\}}{8}$$

$$\sigma^2 = \frac{4368}{8} = 546$$

$S.D = \sqrt{var.} = \sqrt{546} = 23.37$ (C)

⑮ (D)

⑯ Re-arranging; 19, 20, 20, 21, 22, 26, 28

Harmonic Mean, $\frac{1}{H} = \frac{1}{N}\sum\frac{1}{x}$; $N=7$

$$\frac{1}{H} = \frac{1}{7}\left\{\frac{1}{19}+\frac{1}{20}+\frac{1}{20}+\frac{1}{21}+\frac{1}{22}+\frac{1}{26}+\frac{1}{28}\right\}$$

$$\frac{1}{H} = \frac{1}{7}\{0.053+0.05+0.05+0.048+0.045+0.038+0.036\}$$

$\frac{1}{H} = \frac{1}{7}\{0.32\} \Rightarrow H = \frac{7}{0.32}$

$H = 21.88$ years (A)

⑰ Upper quartile $= Q_3$

$Q_3 =$ Value of $\frac{3(n+1)}{4}$th item

$= \frac{3(7+1)}{4}$th item $= 6$th item

6th item $= 26 = Q_3$ (A)

⑱ Lower Quartile $= Q_1$

$Q_1 =$ Value of $\frac{(n+1)}{4}$th item

$= \frac{7+1}{4} = 2$nd item

2nd item $= Q_1 = 20$ (A)

⑲ Variance $= \sigma^2 = \sum_{i=1}^{N} \frac{(x_i-\bar{x})^2}{N}$

$$\bar{x} = \frac{19+20+20+21+22+26+28}{7}$$

$$\bar{x} = \frac{156}{7} = 22.29$$

$$\sigma^2 = \frac{\left\{\begin{array}{l}(19-22.29)^2+(20-22.29)^2+(20-22.29)^2 \\ (21-22.29)^2+(22-22.29)^2+(26-22.29)^2+(28-22.29)^2\end{array}\right\}}{7}$$

$$\sigma^2 = \frac{\left\{\begin{array}{l}10.8241+1.6641+13.7641+5.2441 \\ +32.6041+5.2441+0.0841\end{array}\right\}}{7}$$

$= 69.4287 - 0.92$ (A)

⑳ B

㉑ B

㉒ A

㉓

	1	2	3	4	5	6
1	1,1	1,2	1,3	1,4	1,5	1,6
2	2,1	2,2	2,3	2,4	2,5	2,6
3	3,1	3,2	3,3	3,4	3,5	3,6
4	4,1	4,2	4,3	4,4	4,5	4,6
5	5,1	5,2	5,3	5,4	5,5	5,6
6	6,1	6,2	6,3	6,4	6,5	6,6

Total outcome $= 36$

Probability of 4 turns $= \frac{3}{36}$

Probability of 9 turns $= \frac{4}{36}$

Probability that a sum of 4 or 9 turns

$= \frac{3}{36} + \frac{4}{36} = \frac{7}{36} = 0.19$

㉔ Probability that Obi watches a T.V Programme $= P(A) = 0.4$

Probability that Obi's sister watches a T.V Programme $= P(B) = 0.75$

$P(A\cap B) = 0.30$

$$P(B/A) = \frac{P(A\cap B)}{P(A)} = \frac{0.3}{0.4} = 0.75 \text{ (A)}$$

㉕ Probability of getting at least three girls in a family of five children is

$$\frac{\sum_{x=3}^{5} {}^nC_x}{2^n} \quad \text{where } n=5$$

Thus we have

$$\frac{{}^5C_3+{}^5C_4+{}^5C_5}{2^5} = \frac{10+5+1}{32}$$

$$= \frac{16}{32} = 0.5 \text{ (A)}$$

㉖ $$\frac{\sum_{x=0}^{5} {}^nC_x}{2^n} \quad \text{where } n=8$$

$$\frac{{}^8C_0+{}^8C_1+{}^8C_2+{}^8C_3+{}^8C_4+{}^8C_5}{2^8}$$

$$= \frac{1+8+28+56+70+56}{256} = \frac{217}{256} = 0.86 \text{ (B)}$$

27) C
28) B
29) D
30) C
31) B
32) D
33) D
34) B
35) B

36) Class interval $= \frac{\text{Range}}{K}$

Range $= 145 - 10 = 135$

$K = 1 + 3.22\log_{10} n = 1 + 3.22\log_{10} 70$

$= 1 + 5.94 = 6.94 \simeq 7$

$\therefore$ Class interval $= \frac{135}{7} = 19.29$

Classes	class interval
1st	10 − 29.29
2nd	29.29 − 48.58
3rd	48.58 − 67.87 − 3rd class
4th	67.87 − 87.16
5th	87.16 − 106.45

3rd class is 48.58 − 67.87 (A)

37) D
38) D
39) D
40) B
41) B
42) B
43) D

44) $\sigma^2 = \sqrt{\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2} \Rightarrow \sigma^2 = \frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2$

$\Sigma x = 300,\ \sigma = 0.0072,\ n = 60$

$(0.0072)^2 = \frac{\Sigma x^2}{60} - \left(\frac{300}{60}\right)^2 \Rightarrow 0.000052 = \frac{\Sigma x^2}{60} - 25$

$0.000052 + 25 = \frac{\Sigma x^2}{60} \Rightarrow 25.000052 = \frac{\Sigma x^2}{60}$

$\Sigma x^2 = 25.000052 \times 60 = 1500$ (A)

45) B
46) B

Class interval	f	class boundary	x	fx	Cf
120.10 − 139.49	3	120.095 − 139.495	129.795	389.385	3
139.50 − 158.89	7	139.495 − 158.895	149.195	1044.365	10
158.90 − 178.29	11	158.895 − 178.295	168.595	1854.545	21
178.30 − 197.69	8	178.295 − 197.695	187.995	1503.96	29
197.70 − 217.09	5	197.695 − 217.095	207.395	1036.975	34
	34			5829.23	

47) Upper class boundary of the second class $= 158.895$ (B)

48) Mean $= \frac{\Sigma fx}{\Sigma f} = \frac{5829.23}{34} = 171.45$ (D)

49) Mode $= l_i + \left[\frac{D_1}{D_1 + D_2}\right] \times c$

$D_1 = f - f_{-1} = 11 - 7 = 4$

$D_2 = f - f_1 = 11 - 8$

$l_i = 158.895$; $c = 19.4$

Mode $= 158.895 + \left[\frac{4}{4+3}\right] \times 19.4$

$= 169.98$ (C)

50) Median $= l_i + \left[\frac{N/2 - C_{f-1}}{f_m}\right] \times c$

$l_i = 158.895,\ N = 34,\ C_{f-1} = 10,\ f_m = 21$

$c = 19.4$

Median $= 158.895 + \left[\frac{\frac{34}{2} - 10}{11}\right] \times 19.4$

$= 158.895 + \left[\frac{17 - 10}{11}\right] \times 19.4 = 171.24$ (A)

51) B
52) B

53) Using the formula

$$\sigma^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n}$$

or

$$\sigma^2 = \frac{\Sigma x_i^2 - n\bar{x}^2}{n}$$

$$47.36 = \frac{1705 - n(12.88)^2}{n}$$

$47.36n = 1705 - 165.8544n$

$47.36n + 165.8544n = 1705$

$213.2144n = 1705$

$n = \frac{1705}{213.2144}$

$\therefore n = 7.9966$

$n \simeq 8$ (C)

54) Using the formula

$$b = \frac{n\Sigma xy - \Sigma x \Sigma y}{n\Sigma x^2 - (\Sigma x)^2}$$

but $\Sigma x = n\bar{x} = 8 \times 12.88 = 103.04$

$$b = \frac{8(31330) - (103.04)(1992)}{8(1705) - (103.04)^2} = \frac{250640 - 205225.68}{13640 - 10617.2416}$$

$= \frac{45414.32}{3022.7584} \Rightarrow b = 15.02$ (D) (gradient of the line)

55) $Y = a + bX$

$a = \bar{y} - b\bar{x}$

$\bar{y} = \frac{1992}{8} = 249$; $\bar{x} = 12.88$

$a = 249 - (15.02)(12.88)$

$= 249 - 193.4576$

$= 55.54$ (Intercept)

(A) closest option

56) $$r = \frac{n\Sigma xy - \Sigma x \Sigma y}{\sqrt{[n\Sigma x^2 - (\Sigma x)^2][n\Sigma y^2 - (\Sigma y)^2]}}$$

$$r = \frac{8(31330) - (103.04)(1992)}{\sqrt{[8(1705) - (103.04)^2][8(585808) - (1992)^2]}}$$

$$r = \frac{250640 - 205255.68}{\sqrt{(3{,}022.7584)(718400)}} = \frac{45414.32}{46599.89}$$

$= 0.97 \Rightarrow$ (D)

57) B

58) Since the size of the data is greater than 30 (i.e $n > 30$), we will use the Z-statistic

$\mu = \bar{x} \pm Z_{\alpha}\frac{\sigma}{\sqrt{n}}$

Where $\bar{x} = 4.34$, $\sigma = 0.067$, $\alpha = (1 - 0.99) = 0.01$

$\therefore \mu = 4.34 \pm Z_{0.01}\left(\frac{0.067}{\sqrt{40}}\right)$

$= 4.34 \pm (3.62)\frac{0.067}{\sqrt{40}}$

$= 4.34 \pm 0.03835$

$\mu = (4.30, 4.38)$

Mean square (MS) $= \frac{\text{Sum of squares (SS)}}{\text{Degree of freedom (Df)}}$

F ratio $= \frac{\text{Mean Square treatment}}{\text{Mean square Error}}$

Source of variation	Degree of freedom	Sum of squares	Mean square	F ratio
Treatment	3	19.74	6.58	1.65
Error	32	135.36	4.23	
Total	35	155.1		

59) 155.1
60) 135.36
61) A
62) C
63) C

64) $P(A \cap B) = P(A) \cdot P(B) = 0.7 \times 0.9 = 0.63$ (D)

65) $P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.63}{0.9} = 0.70$ (C)

66) C 67) C 68) C

69) Let the lower class limit be y and upper class limit be x. Then,

$y = \frac{2}{3}x$

Class Mark $= \frac{\text{upper class limit + lower class limit}}{2}$

$$2.25 = \frac{x + \frac{2}{3}x}{2}$$

$4.5 = x + \frac{2}{3}x$

$4.5 = \frac{5x}{3}$

$x = \frac{4.5 \times 3}{5} = 2.7$

Upper class limit, $x = 2.7$ (C)

70) lower class limit $y = \frac{2}{3}x$

$= \frac{2}{3} \times 2.7 = 1.8$

class limit	class mark	class boundary
1.8 − 2.7	2.25	1.75 − 2.75
2.8 − 3.7		

lower class boundary $= 1.75$ (C)

FEDERAL UNIVERSITY OF TECHNOLOGY, OWERRI

DEPARTMENT OF STATISTICS

2014/2015 SESSION HARMATTAN SEMESTER EXAM

TIME: 3HRS COURSE CODE: STA 211

1). A normal distribution of data has mean of 100 and variance of 25. Find P(X < 90) (A) 0.0228 (B) 0.0105 (C) 0.2381 (D) 0.1680 (E) none of the above

2). For a continuous random variable, the probability of a single value of X is always (A) 0 (B) 1.0 (C) between 0 and 1 (D) either 0 or 1 (E) none of the above

3). Given that the sum of the multiple of each observation (x_i) and its associated frequency (f_i) is 207.75. Evaluate the total observations required such that the mean of the set of observations is 4.155? (A) 50 (B) 55 (C) 60 (D) 65 (E) 70

4). One of these is not a characteristic of the simple bar chart; (A) The bars are separated from each other by equal gaps (B) The area of the bars constructed is proportional to the value of items or attributes been represented (C) All bars stand on the same base line (D) It is a diagrammatic representation used for, comparisons of attributes with respect to a single diagram (E) Each bar is drawn to scale to represent each of the attribute considered

5). The class mark of a class is 49.99 and its upper class limit is 54.61. Evaluate the lower class boundary of the class? (A) 44.365(B) 44.465 (C) 45.365 (D) 45.465 (E) 46.465

6). The mean and standard deviation of a binomial distribution with n=25 and p=0.20 are respectively (A) 4 and 3 (B) 5 and 4 (C) 6 and 4(D) 5 and 2 (E) none of the above

7). The modal value of a distribution can be estimated graphically using-----. (A) Frequency Polygon (B) Ogive (C) Bar Chart (D) Histogram (E) all of the above

8). The number of classes required for even grouping of a set of observations is 7 while the length of each class is 6. Determine the range of the set of observations? (A) 32 (B) 38 (C) 42 (D) 48 (E) 39

9). The standard deviation of ten ungrouped observations ($x_i, i = 1,2,...,10$) is 0.0312 and its sum ($\sum_{i=1}^{10} x_i$) is 310. Find the sum of the square of each observation ($\sum_{i=1}^{10} x_i^2$)? (A) 5.75 (B) 8.81 (C) 9.61 (D) 9.98 (E) 8.61

The following table gives the frequency distribution of time (in minutes) that 90 students spent waiting in line to use ATM. Use the information to answer questions 10-12

Waiting time (minutes)	frequency
0 – 6	5
7 – 13	27
14 – 20	30
21 – 27	20
28 – 34	8

10). The class width is -------- (A) 6 (B) 5 (C) 7 (D) 8 (E) none of the above

11). The lower class boundary of the second class is____ (A) 6.5 (B) 7.0 (C) 7.5 (D) 13.5 (E) none of the above

12). The class mark of the third class is ___ (A) 16.5 (B) 17 (C) 17.5 (D) 20.5 (E) none of the above

13). To graph a Box and Whisker plot, which of the following is NOT necessary (A) minimum (B) median (C) first quartile (D) maximum (E) none of the above

The information below shows the summary for Height (x) vs. Femur Length (y) measurements (both in inches) for 10 men: Use the information to answer question 14 to 17

$$\sum_{i=1}^{10} X_i = 681.8, \quad \sum_{i=1}^{10} Y_i = 418.8, \quad \sum_{i=1}^{10} X_i^2 = 46520.4,$$
$$\sum_{i=1}^{10} Y_i^2 = 17622.76, \quad \sum_{i=1}^{10} X_i Y_i = 28589.09$$

14). The least square regression line Y on X is (A) $\hat{y} = -1 + 26.36x$ (B) $\hat{y} = 26.36 - x$ (C) $\hat{y} = -26.36 + x$ (D) None of the above (E) All above

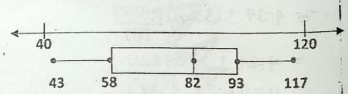
15). Suppose a crime scene investigator digs up the femur of a man and finds that it is 38.5 inches long. Based on our regression line for the height vs. femur length data, what would we estimate the man's height to have been? (A) 54.80 (B) 34.80 (C) 64.80 (D)none of the above (E) All of the above

16). Calculate the coefficient of correlation for the data set of height vs femur length for 10 men (A) r = 0.651 (B) r = 0.821 (C) r = 1.651 (D) r = -0.651 (E) none of the above

17). Interpret the product moment correlation coefficient for question 3 above (A) Strong Positive Correlation (B) Strong Negative Correlation (C) Weak Positive Correlation (D) No correlation (E) none of the above

18). The height, in feet, of lecture halls in Federal University of Technology, Owerri are given below. 35, 42, 42.5, 60, 60, 70, 76, 78, 81, 100 The data can be displayed through ____ so that it is divided into quarters. (A) Stem and leaf chart (B) Histogram (C) Pie chat (D) Box and Whisker plot (E) None

The figure below show Box and Whisker plot of a data set; use the figure to answer question 19 – 22



19). The upper extreme is (A) 120 (B) 93 (C) 40 (D) None of the above (E) All of the above

20). The median is (A) 43 (B) 82 (C) 93 (D) 58 (E) None

21). The lower quartile is (A) 43 (B) 82 (C) 93 (D) 58 (E) 0

22). The upper quartile is (A) 43 (B) 82 (C) 93 (D) 58 (E) 0

23). If cov(x,y) = 0 then (A) x and y are correlated (B) x and y are uncorrelated (C) none (D) x and y are linearly related

24). Ogives for more than type and less than type distribution intersect at (A) mean (B) median (C) mode (D) origin

25). Sum of the deviations about mean is (A) Zero (B) minimum (C) maximum (D) one (E) none of the above

26). The graph obtain by plotting the frequencies of the classes against the corresponding midpoints is called (A). Ogive (B) Bar Chart (C) Histogram (D) Frequency polygon

27). The highest and the lowest value of a set of data are 96 and 25 respectively. If the data are to be arranged in 9 groups, the range and class size are? (A) 71,8 (B) 72,8 (C) 71,7 (D) 71,9 (E) -71,8

28). The middle value of an ordered series is called (A). 2ndquartile(B) 5thdecile((C) 50th percentile(D) all the above (E). none of the above

29). What percentage of values lies between 5th and 25th percentiles? (A) 5% (B) 20% (C) 30% (D) 75% (E) 20%

30). Which of the following is not a measure of central tendency? (A) Mode (B) Variance (C) Median (D) Mean

31). Given the following data set: 12 32 45 14 24 31. The total deviation from the mean for the data values is (A) 0 (B) 26.3 (C) 29.5 (D) 12 (E) none of the above

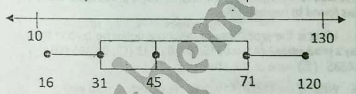
Use the following table to answer question 32 – 33 given that the mean is 5.5

Scores	3	4	5	6	7	8	9
No. of Students	6	x-2	X	8	6	4	2

32). The Value for X is ? (A) (1)0 (B) 8 (C). 7 (D). 9 (E) 11

33). The mode is: (A) 3 (B) 4 (C) 5 (D) 6 (E). 8

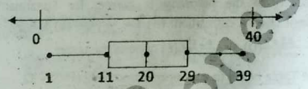
The prices of the watches at a store are displayed in the Box and Whisker plot below. Use it to answer question 34 – 35



34). If all of the watches under ₦31 are on clearance, then about what fraction on the watches are on clearance? (A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{1}{4}$ (D) none of the above

35). If all of the watches from ₦ 31 to ₦ 71 are on sale, then about what fraction of the watches are on a sales? (A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{1}{4}$ (D) none of the above

36). Choose the set of data that is displayed in the box and whisker plot



a) 1, 30, 39, 12, 13, 20, 11, 22, 29
b) 12, 28, 13, 10, 1, 39, 30, 20, 22
c) 40, 25, 9, 11, 29, 27, 1, 14, 20
d) None of the above

37). A survey that includes every element in the population is called? (A) Sample Survey (B) Census (C) Experimentation (D) Population (E). Estimation

38). Class intervals of the type 30-39,40-49,50-59 represents (A) inclusive type (B) exclusive type (C) open-end type (D) none.

39). Correlation analysis deals with the problem of establishing a linear relationship existing between two variables, It is a statistical tool which help to predict one variable from the other variable? A. True B. False C. May Be True D. All of the above E. None of the Above

40). Geometric mean of two numbers $\left(\frac{1}{6}\right)$ and $\left(\frac{4}{25}\right)$ is (A) $\left(\frac{1}{10}\right)$ (B) $\left(\frac{1}{100}\right)$ (C) 10 (D) 100 (E). none of the above

41). Evaluate the probability of either event A or B occurring given that P(A) = 2/3, P(B) = ¼, and P(An(B) = 4/5 (A) 19/60 (B) 7/60 (C) 23/60 (D) 9/60

42). Probability always lie between: (A) -1 & 1 (B) 0 & 1 (C) -1 and 0 (D) All of the above

A teacher asked 10 of her students how many books they had read in the last 12 months. Their answers were as represented in the below Stem and Leaf chart. Use the information to answer questions 43 – 45.

Stem	Leaf
0	6 7
1	0 2 2 5 9
2	1 3 5

43). Their answers are: (A) 6, 7, 0, 2, 2, 5, 9, 1, 3, 5 (B) 31, 12, 1, 52, 70, 91, 60, 21, 51, 21 (C) 12, 23, 19, 6, 10, 7, 15, 25, 21, 12 (D) None of the above

44). On the average the students read: (A) 10 books (B) 3 books (C) 19 books (D) 15 books (E) none of the above

45). The Standard deviation from the mean books read is (A) 10.3 (B) 6.7 (C) 0 (D) 44.9 (E) none of the above

46). If the standard deviation of 18, 18, 19, 17, 19.5, 19, 17, 18.5, 18, 17 is 0.9. What is the standard deviation of 18-k, 18-k, 19-k, 17-k, 19.5-k, 19-k, 17-k, 18.5-k, 18-k, 17-k is: (A) 0.9 (B) 0.9-k (C) 0.9k (D) 0.9/k (E) none of the above

47). Which of the following is correct for Spearman's Coefficient of rank correlation: (A) $1-\frac{6\sum d^2}{n(n^2-1)}$ (B) $1+\frac{6\sum d^2}{n(n^2-1)}$ (C) $1-\frac{6\sum d^2}{n(n^2+1)}$ (D) $1+\frac{6\sum d^2}{n(n^2+1)}$ (E) none of the above

Use the Table below to answer question number 48 – 52.

Class interval	f	x	U	Fu	Fu^2	Cum.f
60.0 – 64.9	4	62.45	-2	-8	16	4
65.0 – 69.9	5	67.45	-1	-5	5	9
70.0 – 74.9	11	72.45	0	0	0	20
75.0 – 79.	11	77.45	1	11	11	31
80.0 – 84.9	5	82.45	2	10	20.	36
85.0 – 89.9	4	87.45	3	12	36	40
Total	40		3	19	88	

48). Find the first quartile (Q_1). (A) 69.96 (B). 69.06 (C.) 69.16. (D). 69.69 (E) none of the above

49). Find the third quartile (Q_3). (A) 78.09 (B). 79.05 (C) 77.05 (D). 76.05 (E) none of the above

50). Find the mean. (A). 72.83 (B). 73.83 (C). 74.83 (D). 75.83 (E) none of the above

51). Find the median. (A). 72.5 (B). 73.5 (C). 74.5 (D). 75.5

52). Find the variance. (A). 49.360 (B). 49.370 (C). 49.380 (D). 49.390 (E) none of the above

53). In regression analysis, the variable that is being predicted is called the _____ variable. (A) independent (B) dependent (C) regression (D) all of the above (E) none of the above

54). Which of the following values cannot be a value of the correlation coefficient? (A) -1 (B) 0.66 (C) -1.01 (D) 0 (E) -0.78

55). If the computed value of the correlation coefficient is 0.71, then the slope of the least squares line will be (A) positive (B) negative (C) zero (D) all of the above (E) none of the above

56). The coefficient of variation can be used to compare the _____of data sets. (A) centralness (B) variability (C) uniqueness (D) all of the above (E) none of the above

57). If a measurement in a data set is below the mean value for that set, then the deviation from the mean will be (A) positive (B) negative (C) zero (D) all of the above (E) none of the above

58). Data that are measured over an interval are called (A) discrete data (B) continuous data (C) interval data (D) all of the above (E) none of the above

59). The shape of the frequency distribution and the relative frequency distribution will always be (A) the same (B) different (C) skewed (D) all of the above (E) none of the above

60). A Box contains 400 beads of which 4% are defective: a second Box contains 120 beads of which 6% are defective while the third Box contains 500 beads with 5% defectives. If the three Boxes are identical in appearance, find the probability of selecting a defective bead given that it is contained in Box 2? (A) 0.40 (B) 0.45 (C) 0.46 (D) 0.50 (E) 0.55

61). Which measure of dispersion ensures highest degree of reliability? (A) Range (B) Mean deviation (C) Quartile deviation (D) Standard deviation

62). The probability that students pass STA 211 examination is 3/4. What is the probability that one student pass and one student fail the examination? (A). 1/16 (B). 2/16 (C). 3/16 (D). 4/16 (E) none of the above

Use this question to answer number 63 and 64. Two students of mathematics went for a written interview. Student A has 25% chance of passing and student B has 45% chance of passing. Both students have 18% chances of passing.

63). If B has passed, what is the probability that A will also pass the examination? (A). 1/5 (B). 2/5 (C). 3/5 (D). 4/5

64). What is the probability that neither A nor B pass the examination? (A). 0.28 (B). 0.38 (C). 0.48 (D). 0.58

Use this question to answer number 65 and 66. In a certain factory that manufacture Lion bulb, machine A_1, A_2 and A_3 manufacture 40%, 35% and 25% respectively of the total production. The percentages of defective bulbs are 2, 4 and 5. A bulb is drawn at random from a day's production and it was found to be defective.

65). What is the probability of getting a defective bulb in the day's production? (A). 0.0345 (B). 0.0243 (C). 0.0465 (D). 0.0335 (E) none of the above

66). What is the probability that the defective bulb was manufactured by machine A_2? (A). 0.6041 (B). 0.5626 (C). 0.3257 (D) 0.4058 (E) none of the above

67). The sum of the relative frequencies for all the classes of a data set will always equal (A) The sample size (B) the number of classes (C) One (D) any value greater than one

68) Respective Probabilities of Certainty and Impossibility are: (A) 0 & 1 (B) 1 & 0 (C) -1 & 1 (D) -1 & 0

69) A subset of a population is called a (A) census (B) sample (C) small (D) population (E) none of the above

70) Skewness of a Normal distribution is (A)3 (B)1(C) 0 (D) 2

STA 211 SOLUTION 2014/2015

1. $Z = \frac{X-\mu}{\sigma}$

where $\mu = 100$, $\sigma^2 = 25$, $\sigma = \sqrt{25} = 5$

Then $P(X<90) = P\left[\frac{X-100}{5} < \frac{90-100}{5}\right]$

$P(Z<-2.0)$

$= 0.5+(-0.47725)$

$= 0.5-0.47725 = \underline{0.0228}$ A

2.

3. $\bar{x} = \frac{\sum fx}{\sum f} \Rightarrow \bar{x} = \frac{\sum fx}{N}$

$N = \frac{\sum fx}{\bar{x}} = \frac{207.75}{4.155} = \underline{50}$ A

4. D

5. Class mark $= \frac{UCL + LCL}{2}$

Where UCL = Upper class limit
LCL = Lower class limit

$49.99 = \frac{54.61 + LCL}{2}$

$54.61 + LCL = 2(49.99)$

$54.61 + LCL = 99.98$

$LCL = 99.98 - 54.61$

$LCL = \underline{45.37}$ C

6. For a binomial distribution

$\mu = np$, $\sigma = \sqrt{npq}$

$n = 25$, $p = 0.2$, $q = 1-p = 1-0.2 = 0.8$

$\mu = 25 \times 0.2 = 5$

$\sigma = \sqrt{25 \times 0.2 \times 0.8} = 2$ D

7. D

8. Class size $= \frac{\text{Range}}{\text{no of classes}}$

$\Rightarrow 6 = \frac{\text{Range}}{7} \Rightarrow \text{Range} = 6 \times 7 = \underline{42}$ C

9. $\sigma = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$

$n = 10$ observations

$\sum x = 310$ and $\sigma = 0.0312$

$\sigma^2 = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2$

$(0.0312)^2 = \frac{\sum x^2}{10} - \left(\frac{310}{10}\right)^2$

$(0.0312)^2 = \frac{\sum x^2}{10} - 31^2$

$(0.0312)^2 + 31^2 = \frac{\sum x^2}{10}$

$0.00097344 + 961 = \frac{\sum x^2}{10}$

$961.00097344 = \frac{\sum x^2}{10}$

$\sum x^2 = 961.0097344 \times 10$

$\sum x^2 = 9610.097344$

or 9.61×10^3 C

10. Class width = Difference between upper (or lower) class limits of consecutive classes

$\therefore$ class width $= 7 - 0 = 7$ C

11. A

12. Class mark $= \frac{UCL + LCL}{2}$

$= \frac{20+14}{2} = 17$ B

13. E

14. The least square regression line, y on x is $\hat{y} = a + bx$

where $b = \frac{n\sum XY - \sum X \sum Y}{n\sum X^2 - (\sum X)^2}$

$a = \bar{y} - b\bar{x}$

$b = \frac{10(28589.09) - (681.8)(418.8)}{10(46520.4) - (681.8)^2}$

$b = \frac{353.06}{352.76} = 1.0008$

Then $a = \bar{y} - b\bar{x}$

$\bar{y} = \frac{\sum Y}{n} = \frac{418.8}{10} = 41.88$

$\bar{x} = \frac{\sum X}{n} = \frac{681.8}{10} = 68.18$

$a = 41.88 - (1.0008 \times 68.18)$
$= -26.36$

The least square regression line of y on x will be;
$\hat{y} = -26.36 + x$ C

(15) Based on the regression line
$\hat{y} = -26.36 + x$
let $\hat{y}$ = height and x = femur length
then $y = -26.36 + 38.5$
$y = 12.14$ D

(16) $r = \frac{n\sum XY - \sum X \sum Y}{\sqrt{[n(\sum x^2) - (\sum x)^2][n(\sum y^2) - (\sum y)^2]}}$

$r = \frac{10(28587.09) - (618.8)(418.8)}{\sqrt{[10(46520.4) - (618.8)^2][10(17622.76) - (418.8)^2]}}$

$r = \frac{353.06}{542.46} = 0.651$ A

(17) A
(18) D
(19) A
(20) B
(21) D
(22) C
(23) B
(24) D
(25) A
(26) D
(27) Range = 96 − 25 = 71
Class size = $\frac{71}{9} = 8$ A
(28) D
(29) % values = $\left(\frac{25N}{100} - \frac{5N}{100}\right) \times 100$
$\frac{20N}{100} \times 100 = 20N\%$
Let N = unknown value = 1
$\Rightarrow 20(1)\% = 20$ B
(30) B

(31) Let $\bar{x} = \frac{\sum X}{n}$
$\bar{x} = \frac{12+32+45+14+24+31}{6} = \frac{158}{6}$
$= 26.33$
$\sum(x - \bar{x}) = (12 - 26.33) + (32 - 26.33) + (45 - 26.33) + (14 - 26.33) + (24 - 26.33) + (31 - 26.33) = 0.02 \simeq 0$ A

(32) $\frac{3(6)+4(x-2)+5(x)+6(8)+7(6)+8(4)+9(2)}{6+(x-2)+x+8+6+4+2} = 5.5$
$\frac{18+4x-8+5x+48+42+32+18}{6+x-2+x+8+6+4+2} = 5.5$
$\frac{150+9x}{24+2x} = 5.5$
$5.5(24+2x) = 150 + 9x$
$132 + 11x = 150 + 9x$
$11x - 9x = 150 - 132$
$2x = 18 \Rightarrow x = \frac{18}{2} = 9$ D

(33) Mode = 5 (Highest freq. 9) ⇒ C

(34) If all of the watches under #31 are on clearance falls at Q_1
Then $Q_1 = \frac{1}{4}$ C

(35) If all watches from #31 to #71 which is Q_1 and Q_3
Interquartile range = $Q_3 - Q_1$
$= \frac{3}{4} - \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$ A

(36) 1, 30, 39, 12, 13, 20, 11, 22, 29 and 12, 28, 13, 10, 1, 39, 30, 20, 22 are both displayed in the box and whisker plot.
The correct option is either A or B
(37) B
(38) A
(39) B
(40) $G = \left(\frac{1}{6} \times \frac{4}{25}\right)^{1/2} = \left(\frac{4}{150}\right)^{1/2}$
$= 0.16$ E

(41) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
where $P(A) = \frac{2}{3}$, $P(B) = \frac{1}{4}$, $P(A \cap B) = \frac{4}{5}$
$\therefore P(A \cup B) = \frac{2}{3} + \frac{1}{4} - \frac{4}{5}$
$= \frac{40 + 15 - 48}{60} = \frac{7}{60}$ B

(42) B

(43) Their answers:
1st row = 6, 7
2nd row = 10, 12, 12, 15, 19
3rd row = 21, 23, 25
∴ we have; 6, 7, 10, 12, 12, 15, 19, 21, 23, 25
C

(44) $\bar{x} = \frac{6+7+10+12+12+15+19+21+23+25}{10}$
$\bar{x} = \frac{150}{10} = 15$ Books D

(45) $\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{x})^2}$
$\sqrt{\frac{(6-15)^2+(7-15)^2+(10-15)^2+(12-15)^2+(12-15)^2+(15-15)^2+(19-15)^2+(21-15)^2+(23-15)^2+(25-15)^2}{10}}$
$= \sqrt{\frac{81+64+25+9+9+0+16+36+64+100}{10}}$
$\sigma = \sqrt{\frac{404}{10}} = \sqrt{40.4} = 6.4$ B

(46) A
(47) A
(48) $Q_1 = l_i + \left[\frac{\frac{N}{4} - C_{f-1}}{f_{Q_1}}\right] \times C$
$= 70 + \left[\frac{\frac{40}{4} - 9}{11}\right] \times 4.9$
$= 70 + \left[\frac{10-9}{11}\right] \times 4.9 = 70 + \frac{4.9}{11}$
$= 70 + 0.445 = 70.445$ E

(49) $Q_3 = l_i + \left[\frac{\frac{3N}{4} - C_{f-1}}{f_{Q_3}}\right] \times C$
$75 + \left[\frac{\frac{3(40)}{4} - 20}{11}\right] \times 4.9$
$75 + \left[\frac{30 - 20}{11}\right] \times 4.9$
$75 + \frac{49}{11} = 75 + 4.45$
$= 79.45$ B

(50) $\frac{\sum fx}{\sum f} = \frac{4(62.45)+5(67.45)+11(72.45)+11(77.45)+5(82.45)+4(87.45)}{40}$
$= \frac{249.8+337.25+796.95+851.95+412.25+349.8}{40}$
$= \frac{2998}{40} = 74.95$ E

(51) Median $= l_i + \left[\frac{\frac{N}{2} - C_{f-1}}{f_{med}}\right] \times C$
$70 + \left[\frac{\frac{40}{2} - 9}{11}\right] \times 4.9 = 70 + \left[\frac{20-9}{11}\right] \times 4.9$
$= 70 + 4.9 = 74.9$ C

(52) $\sigma^2 = \frac{\sum f(x - \bar{x})^2}{\sum f}$
$= \frac{[4(62.45-74.95)^2 + 5(67.45-74.95)^2 + 11(72.45-74.95)^2 + 11(77.45-74.95)^2 + 5(82.45-74.95)^2 + 4(87.45-74.95)^2]}{40}$
$= \frac{[4(156.25) + 5(56.25) + 11(6.25) + 11(6.25) + 5(56.25) + 4(156.25)]}{40}$
$= \frac{625 + 281.25 + 68.75 + 68.75 + 281.25 + 625}{40}$
$= \frac{1950}{40} = 48.75$ E

(53) B
(54) C $-1 \leq r \leq 1$
(55) A
(56) B
(57) A
(58) A
(59) A

(60) let $X=400, Y=120, Z=500$

Total $= 400+120+500=1020$

$P(X)=\frac{400}{1020}$; $P(Y)=\frac{120}{1020}$; $P(Z)=\frac{500}{1020}$

Let event E be that the selected bead is defective

$P(E/X)=4\%=\frac{4}{100}=0.04$

$P(E/Y)=6\%=\frac{6}{100}=0.06$

$P(E/Z)=5\%=\frac{5}{100}=0.05$

P(defective bead)

$=P(X)\cdot P(E/X)+P(Y)\cdot P(E/Y)+P(Z)\cdot P(E/Z)$

$\frac{400}{1020}\times 0.04+\frac{120}{1020}\times 0.06+\frac{500}{1020}\times 0.05$

$0.0157+0.007+0.0245$

$=0.047$ C

(61) D

(62) $P(\text{Pass})\times P(\text{fail})$

$\frac{3}{4}\times\frac{1}{4}=\frac{3}{16}$ C

(63) $P(A\cap B)=P(A)\cdot P(B)$

$18\%=P(A)\cdot 45\%$

$\frac{18}{100}=P(A)\cdot\frac{45}{100}$

$P(A)=\frac{18}{100}\times\frac{100}{45}=\frac{18}{45}=\frac{2}{5}$ B

(64) $P(\overline{A\cup B})=1-P(A\cup B)$

But $P(A\cup B)=P(A)+P(B)-P(A\cup B)$

$\frac{25}{100}+\frac{45}{100}-\frac{18}{100}$

$0.25+0.45-0.18=0.52$

$P(\overline{A\cup B})=1-P(A\cup B)$

$1-0.52=0.48$ C

(65) $P(A_1)=40\%=\frac{40}{100}=0.4$

$P(A_2)=35\%=\frac{35}{100}=0.35$

$P(A_3)=25\%=\frac{25}{100}=0.25$

let event E be that selected bulb is defective

$P(E/A_1)=\frac{2}{100}=0.02$

$P(E/A_2)=\frac{4}{100}=0.04$

$P(E/A_3)=\frac{5}{100}=0.05$

$P(\text{defective})=P(A_1)\cdot P(E/A_1)+P(A_2)\cdot P(E/A_2)+P(A_3)\cdot P(E/A_3)$

$=(0.4)(0.02)+(0.35)(0.04)+(0.25)(0.05)$

$=0.008+0.014+0.0125$

$=0.0345$ A

(66) $P(A_2/E)=\frac{P(A_2)P(E/A_2)}{P(A_1)P(E/A_1)+P(A_2)\cdot P(E/A_2)+P(A_3)P(E/A_3)}$

$=\frac{0.014}{0.0345}=0.4058$ D

(67) C

(68) B

(69) A

(70) C

FEDERAL UNIVERSITY OF TECHNOLOGY, OWERRI

DEPARTMENT OF STATISTICS

HARMATTAN SEMESTER EXAMINATIONS 2013/2014 SESSION

STA. 211: INTRODUCTION TO STATISTICS AND PROBABILITY-MULTIPLE CHOICE QUESTIONS

INSTRUCTIONS: ANSWER ALL THE QUESTIONS TIME: 2 HRS

Each question is followed by four options lettered A to D. Find out the correct option for each question and shade in pencil on your answer sheet the answer space which bears the same letter as the option you have chosen. Give only one answer to each question.

1. The probability that an event A will happen is 0.5 and an event B will not happen is 0.6. What is the probability that neither A nor B will happen?
   A) 0.3 B) 0.6 C) 0.1 D) 0.9
2. A bag contains 5 red, 3 green and 2 blue balls. Two balls are drawn at random with replacement. What is the probability that none of the balls drawn is blue?
   A) 0.64 B) 0.25 C) 0.15 D) 0.09
3. 20% of the students in a school of Mathematical Sciences are majoring in Pure Mathematics. 15% in Industrial Mathematics. 35% in Computer Science, and 30% in Statistics. The graphical device(s) which can be used to present these data is (are):
   A) a line graph B) only a bar graph C) only a pie chart D) both a bar graph and a pie chart
4. A researcher is gathering data from four geographical areas designated:
   South = 1; North = 2; East = 3; West = 4. The designated geographical regions represent
   A) qualitative data B) quantitative data C) label data D) either quantitative or qualitative data
5. Tickets numbered 1 to 20 are mixed up and then a ticket is drawn at random. What is the probability that the ticket drawn has a number which is a prime number?
   A) 0.4 B) 0.8 C) 0.3 D) 0.6

Exhibit 1-1: Let A and B be two events associated with an experiment. Suppose that P(A) = 0.4 while P(AUB) = 0.7. Let P(B) = p. Use **Exhibit** 1-1 to answer questions 6-7.

6. For what choice of p are A and B mutually exclusive?
   A) 0.4 B) 0.7 C) 0.3 D) 0.0.5
7. For what choice of p are A and B independent?
   A) 0.5 B) 0.3 C) 0.7 D) 0.4

Exhibit 1-2: In a bolt factory, machines, X, Y, and Z manufacture 25, 35, and 40 percent of the total output, respectively. Of their outputs, 5, 4, and 2 percent, respectively, are defective bolts. A bolt is chosen at random and found to be defective. Use Exhibit 1-2 to answer questions 8-9.

8. What is the probability that the bolt is defective?
   A) 0.345 B) 0.362 C) 0.406 D) 0.0.232
9. What is the probability that the defective bolt came from Machine Y?
   A) 0.345 B) 0.362 C) 0.232 D) 0.406

Exhibit 1-3: The following table shows the distribution of 100 workers in FUTO bread factory.

Income group	No. of workers
140-159	9
160-179	20
200-219	33
200-219	25
220-239	11
240-259	2

Use Exhibit 1-3 to solve questions 10 – 12

10. Determine the lower quartile A) 15 B) 175.5 C) 25 D) 177.5
11. Determine the first deciles A) 162.5 B) 177.5 C) 141.7 D) 160.5
12. Determine the first percentiles A) 162.5 B) 177.5 C) 25 D) 142.4
13. One of the statements below is NOT correct? A) $D_5 = P_{50} = Q_2 =$ Median

B) Median deviation = $\frac{1}{n}\sum_{i=1}^{n}\left[x_i - \bar{x}\right]$ C) $Var(X) = E[X(X-1)] + E(X) - [E(X)]^2$

D) Geometric mean = $\sqrt[n]{x_1 + x_2 + ... + x_n}$

14. The degree of peakedness of a distribution is known as
A) Deciles B) Coefficient of variation C) Skewness D) Kurtosis.

Exhibit 1-4: The following are the prices in dollar of a certain FUTO product at forty different stores

60	87	68	86	74	75	64	73
82	67	89	73	75	70	88	67
65	74	68	72	80	74	76	78
78	63	75	71	83	85	72	74
76	77	75	75	71	80	84	77

Use Exhibit 1-4 to solve 15 - 21

15. Determine the approximate number of classes using STURGE method (A) 4 (B) 5 (C) 6 (D) 7
16. Determine the range A) 5 B) 29 C) 20 D) 24
17. Determine the frequency of the fourth class A) 11 B) C) 19 D) 30
18. Determine the cumulative frequency of the fourth class A) 11 B) 29 C) 19 D) 30.
19. The lower class limits and boundary of the fourth class are A) 60 and 75 respectively B) 60 and 74.5 respectively C) 74.5 and 75 respectively D) 75 and 74.5 respectively.
20. The midpoint of a class or class mark for the third class is A) 72 B) 71 C) 73 D) 74
21. Compute the class interval A) 6 B) 5 C) 4 D) 13
22. A graph of a frequency distribution which is obtained by plotting frequency against class interval is known as A) Histogram B) Pie chart C) Bar chart D) frequency table.
23. The feature possessed by the units of the population that changes during the period under consideration is called: A) Constant B) Variable C) Attributes D) Characteristics
24. One of these is not a method of data collection: A) Census B) Geographical Survey C) Observation D) Sample Survey
25. The scale used to categorize information, where the categories are ordered and assigned numbers to preserve the ordering of the categories is called: A) Nominal Scale B) Ratio Scale C) Ordinal Scale D) Not Scale
26. Which of these is the Characteristic of a sample Bar Chart? A) Consists of bars of equal widths B) Bars stand on the same base line C) Heights of bars are proportional to magnitude of value being represented D) All of the above
27. Comparison of two or more characters within each attribute could be done using? A) Component bar chart B) Simple bar chart C) Multiple bar chart D) Histogram
28. Which of these diagrams could be used to assess the extent in which observations spread about their centers? A) Histogram B) Component bar chart C) Stem and Leaf plot D) None of the above
29. Given the data, 25, 18, 30, 8, 10, 5, 10, 35, 40, 45. Obtain the third Quartile (Q_3) of the data set.
A) 35.00 B) 30.01 C) 34.26 D) 36.25
30. Given the data, 25, 18, 30, 8, 10, 5, 10, 35, 40, 45. Calculate the variance of the data set. A) 14.45 B) 208.93 C) 15.45 D) 108.83
31. If the mean of a distribution is greater than its median and its median is greater than its mode then the distribution is said to be .... skewed.
A) Negatively B) Positively C) Normally D) None of the above

32. Statistical inferences on a population are made on the basis of a ________ drawn from the population. A) a population B) inferential statistics C) a census D) a sample
33. The class mark for the class 25-30 is A) 24.5 B) 29.5 C) 4 D) 27.5.
34. An organization of observed data into tabular form in which classes and frequencies are used is called A) a bar chart B) a pie chart C) a frequency distribution D) a frequency polygon.
35. The mode for the following quiz scores: 7, 4, 4, 7, 2, 9, 10, 6, 7, 3, 8, 5 (out of a possible 10 points) recorded by an instructor for 12 students present in a statistics class is
A) 9.5 B) 7 C) 6 D) 3.
36. In the simple linear regression model with x representing the independent variable and y representing the dependent variable, correlation analysis is used to A) find the least-squares regression line B) find the slope of the regression line C) measure the strength of the linear relationship between x and y D) draw a scatter plot
37. In simple linear regression analysis, there A) is only one independent variable in the model B) could be several linear independent variables in the model C) is only one nonlinear term in the model D) is at least one nonlinear term in the model.
38. Given the following information: $\sum x = 24$, $\sum y = 16$, $\sum x^2 = 180$, $\sum y^2 = 90$, $\sum xy = 75$, and $n = 10$. The predicted value $\hat{y}$ of the dependent variable y when $x = 2$ is
A) 1.4804 B) 1.1296 C) 3.752 D) 3.2722.
39. In experiment where the probability of a success is 0.4, if you are interested in the probability of 2 successes out of 7 trials, the correct probability is A) 0.0774 B) 0.1600 C) 0.2613 D) 0.0016.
40. If n is the number of trials and p the probability of success for a binomial experiment, the standard deviation for the resulting binomial distribution is
A) $\sqrt{n(1-p)}$ B) $\sqrt{p(1-p)}$ C) $\sqrt{np}$ D) $\sqrt{np(1-p)}$.
41. If X is a normal random variable with a mean of 15 and a variance of 9, then $P(X < 18)$ is A) 0.7486 B) 0.8413 C) 0.3413 D) 0.1587.
42. The area under the standard normal curve between -2.0 and -1 is
A) 0.0228 B) 0.1359 C) 0.4772 D) 0.3413.
43. Which of the following is a unitless measure of dispersion?
A) Standard deviation B) Mean deviation C) Coefficient of variation D) Range
44. Sum of the deviations about mean is A) Zero B) minimum C) maximum D) one
45. The first quartile divides a frequency distribution in the ratio A) 4 : 1 B) 1 :4 C) 3 : 1 D) 1 : 3
46. Histogram is useful to determine graphically the value of
A) mean B) median C) mode D) all the above
47. Median can be located graphically with the help of
A) Histogram B) Ogives C) Bar diagram D) Scatter diagram
48. Shoe size of most of the people in Owerri is No. 7 which measure of central value does it represent? A).mean B) second quartile C) eighth decile D) mode
49. Which of the following measures is most affected by extreme values
A) Standard deviation B) Quartile deviation C) Mean deviation. D) Range
50. If the minimum value in a set is 9 and its range is 57, the maximum value of the set is
A) 33 B) 66 C) 48 D) 24
51. In a distribution with standard deviation equal to 6, if all observation is multiplied by 2 what will be the result of the standard deviation A) 12 B) 6 C) 18 D) 6
52. In a symmetric distribution A) mean ≠ median ≠ mode B) mean = median = mode C) mean > median > mode D) mean< median < mode
53. Geometric mean of two numbers (1/9) and (4/25) is A) (1/10) B) (1/100) C) 10 D) 0.133
54. If the grouped data has open-end classes, one cannot calculate.
A).median B) mode C) mean D) quartile
55. Ogives for more-than-type and less-than-type distribution intersect at
A) mean B) median C) mode D) origin
56. Raw data means? A) Primary data B) secondary data C) data collected for investigation

D) Well classified data.

57. The sum of the relative frequencies for all classes of a set of data will always equal
A) the sample size B) the number of classes C) one D) any value larger than One

58. If the coefficient of correlation equals 0.61, it indicates that the proportion of the variation in the dependent variable explained by the variation in the independent variable is A) 61% B) 98% C) 37% D) cannot be determined

59. Suppose the correlation between X and Y is - 0.95. Which of the following conclusions is correct?
A) The linear relation between X and Y is strong, Y decreases when X increases.
B) The linear relation between X and Y is weak, Y decreases when X increases.
C) The linear relation between X and Y is strong, Y increases when X increases.
D) The linear relation between X and Y is weak, Y increases when X increases.

60. If events A and B are not dependent, one of the following is not true.
A) $P(A \cup B) = P(A) + P(B)P(A^c)$ B) $P(A \cap B) = P(A)P(B)$ C) $P(A/B) = P(A)$ D) None of the above

61. Let Y be a Binomial random variable with parameters, n and p. Determine n if $p = 0.25$. A) 5 B) 3 C) 4 D) None of the above

62. Let Y be a Poisson random variable with parameter μ. If the probability, $p(0) = 0.04979$, Find the variance of Y. A) 0.04 B) 0.049 C) 3 D) 1

63. Find the probability that a random variable having the standard normal distribution will take on a value between -0.25 and 0.45. A) 0.0987 B) 0.1736 C) 0.2723 D) None of the above.

64. A random variable Y has a normal distribution with mean 4.35 and variance 0.59. Obtain a value of Y that gave a z score of 1.44. A) 0.85 B) 1.44 C) 5.20 D) 5.02

65. If Y is random variable having a normal distribution, what is the probability of getting a value within one standard deviation of the mean. A) 0.6826 B) 0.9544 C) 0.9984 D) 0.99994

66. Records show that the probability is 0.00005 that a car will have a flat tire while crossing a certain bridge. Use the Poisson distribution to approximate the binomial probabilities that, among ~~10,000~~ ~~cars crossing this~~ bridge, exactly two will have a flat tire. A) 0.0758 B) 0.0005 C) 0.0785 D) 0.5

67. A random variable Y takes on the following values 0, 1, 2, 3 with the corresponding probabilities 1/8, 3/8, 3/8, 1/8. Calculate the mean of Y. A) 0.5 B) 1.5 C) 3 D) 2

68. A random variable Y takes on the following values 0, 1, 2, 3 with the corresponding probabilities 1/8, 3/8, 3/8, 1/8. Calculate the variance of Y. A) 0.7 B) 1.57 C) 0.75 D) 2

69. In Owerri, the probabilities are 0.86, 0.35, and 0.29 that a family owns a colour television set, a black-and-white set, or both kinds of sets. What is the probability that a family owns either or both kinds of sets? A) 0.29 B) 0.92 C) 0.86 D) None of the above

70. If we randomly pick two television tubes in succession from a shipment of 240 television tubes of which 15 are defective, what is the probability that they will both be defective?
A) 0.0037 B) 0.0036 C) 0.004 D) None of the above

STA211 SOLUTION 2013/2014

(1) $P(A) + P(B) = 1$

$P(A) = 0.5$, $P(\bar{B}) = 0.6$

$\therefore P(B) + P(\bar{B}) = 1$

$P(B) = 1 - P(\bar{B}) = 1 - 0.6 = 0.4$

$\therefore$ Probability of neither A nor B

$= 1 - P(A \cup B)$

$= 1 - [P(A) + P(B)] = 1 - [0.5 + 0.4]$

$= 1 - 0.9 = 0.1$ C

(2) Total no of balls $= 5 + 3 + 2 = 10$

no of blue balls = 2

$P(\text{Blue}) = \frac{2}{10} = 0.2 = P(B)$

$P(\text{Not Blue}) = P(\bar{B}) = 1 - 0.2 = 0.8$

Probability of the first drawn not blue $= 0.8$

$\therefore$ Probability of first and second drawn not being blue $= P(\bar{B}) \times P(\bar{B})$

$= 0.8 \times 0.8 = 0.64$ A

(3) D

(4) A

(5) Sample space = 20

Prime numbers are numbers divisible by 1 and itself, 1 not inclusive

$\{2, 3, 5, 7, 11, 13, 17, 19\} = 8$

$P(\text{Prime number}) = \frac{8}{20} = 0.4$ A

(6) $P(A) = 0.4$, $P(A \cup B) = 0.7$, $P(B) = P$

for mutually exclusive event,

$P(A \cup B) = P(A) + P(B)$

$P = P(B) = P(A \cup B) - P(A)$

$= 0.7 - 0.4 = 0.3$ C

(7) For Independent event,

$P(A \cap B) = P(A) \cdot P(B)$

Also

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Where $P(B) = P$

$P(A \cup B) = P(A) + P - P(A) \cdot P$

$0.7 = 0.4 + P - 0.4 \times P$

$0.7 - 0.4 = P - 0.4P$

$0.3 = P(1 - 0.4)$

$0.3 = 0.6P$

$P = \frac{0.3}{0.6} \Rightarrow P = 0.5$ A

(8) Let X, Y and Z be the events of total output by machines X, Y and Z respectively; and Let E be the event that the selected machine is defective

$P(X) = \frac{25}{100} = 0.25$, $P(Y) = \frac{35}{100} = 0.35$

$P(Z) = \frac{40}{100} = 0.4$, $P(E/X) = \frac{5}{100} = 0.05$

$P(E/Y) = 0.04$, $P(E/Z) = \frac{2}{100} = 0.02$

$P(\text{Bolt defective}) = P(X) \cdot P(E/X) + P(Y) \cdot P(E/Y) + P(Z) \cdot P(E/Z)$

$= (0.25 \times 0.05) + (0.35 \times 0.04) + (0.4 \times 0.02)$

$= 0.0125 + 0.014 + 0.008$

$= 0.0345$ A

(9) Applying Baye's theorem

$$P(Y/E) = \frac{P(Y \cap E)}{P(X \cap E) + P(Y \cap E) + P(Z \cap E)}$$

$$= \frac{P(E/Y)P(Y)}{P(E/X)P(X) + P(E/Y)P(Y) + P(E/Z)P(Z)}$$

$$= \frac{0.04 \times 0.35}{0.05 \times 0.25 + 0.35 \times 0.04 + 0.4 \times 0.02}$$

$$= \frac{0.014}{0.0345} = 0.406 \quad D$$

(10)

Class boundary	income group	no of workers(f)	Cumm. freq
139.5 − 159.5	140 − 159	9	9
159.5 − 179.5	160 − 179	20	29
179.5 − 199.5	180 − 199	33	62
199.5 − 219.5	200 − 219	25	87
219.5 − 239.5	220 − 239	11	98
239.5 − 259.5	240 − 259	2	100

Class size = Upper class boundary − lower class boundary

$\Rightarrow 159.5 - 139.5 = 20$

N = Total frequency = 100

Q_1 = Lower quartile

$$Q_1 = L_i + \left[\frac{\frac{N}{4} - C_{f-1}}{f_{Q_1}}\right] \times C$$

$$\Rightarrow \frac{N}{4} = \frac{100}{4} = 25$$

25 falls under Cf 29, which is in the second class

$$Q_1 = 159.5 + \left[\frac{25-9}{20}\right] \times 20$$

$$159.5 + 16 = 175.5 \quad B$$

(11) First decile $= \frac{N}{10} = \frac{100}{10} = 10$

10 falls into the second class

$$D_1 = L_i + \left[\frac{\frac{N}{10} - C_{f-1}}{f_{D_1}}\right] \times C$$

$$159.5 + \left[\frac{10-9}{20}\right] \times 20$$

$$159.5 + 1 = 160.5 \quad D$$

(12) First Percentile $= \frac{N}{100} = \frac{100}{100} = 1$

Which falls in the first class

$$P_1 = L_i + \left[\frac{\frac{N}{100} - C_{f-1}}{f_{P_1}}\right] \times C$$

$$139.5 + \left[\frac{1-0}{9}\right] \times 20$$

$$= 141.7 \quad C$$

(13) C

(14) D

(15) From STURGE method

no of classes = $1 + 3.22 \log(n)$

n = Total frequency = 40

no of classes = $1 + 3.22 \log 40$

$= 5.82 \simeq 6 \quad C$

(16) Range = Highest − lowest

$89 - 60 = 29 \quad B$

To determine the frequency of the forth class, we must draw the frequency distribution table

class width or class interval

$$= \frac{\text{range}}{\text{no of classes}} = \frac{29}{6} =$$

$$= 4.8 \simeq 5$$

Class limit/ interval	Tally	Freq	Class mark	Cf
60 − 64	|||	3	62	3
65 − 69	~~||||~~	5	67	8
70 − 74	~~||||~~ ~~||||~~		11	72	19
75 − 79	~~||||~~ ~~||||~~		11	77	30
80 − 84	~~||||~~	5	82	35
85 − 89	~~||||~~	5	87	40

(17) A

(18) D

(19) D

(20) A

(21) Class Interval $= \frac{\text{Range}}{\text{no of classes}} = \frac{29}{6}$

$= 4.8 = 5$

(22) C

(23) B

(24) B

(25) C

(26) D

(27) A

(28) C

(29) Re-arranging

5, 8, 10, 10, 18, 25, 30, 35, 40, 45

$$\frac{3N}{4} = \frac{3}{4} \times 10 = \frac{30}{4} = 7.5\text{th position}$$

We take the 8th position = 35 A

(30) Variance $= \frac{\sum_{i=1}^{n} (x - \bar{x})^2}{n-1}$

$$\bar{x} = \frac{25+18+30+8+10+5+10+35+40+45}{10}$$

$$\bar{x} = \frac{226}{10} = 22.6$$

$$\text{Variance} = \sigma^2 = \frac{\{(25-22.6)^2 + (18-22.6)^2 + (30-22.6)^2 + (8-22.6)^2 + (10-22.6)^2 + (5-22.6)^2 + (10-22.6)^2 + (35-22.6)^2 + (40-22.6)^2 + (45-22.6)^2\}}{10-1}$$

$$\sigma^2 = \frac{\{5.76 + 21.16 + 54.76 + 213.16 + 158.76 + 309.76 + 158.76 + 153.76 + 302.76 + 501.76\}}{9}$$

$$\sigma^2 = \frac{1880.4}{9} = 208.93 \quad B$$

(31) B

(32) D

(33) Class mark $= \frac{\text{lower class limit} + \text{upper class limit}}{2}$

$$= \frac{25+30}{2} = 27.5 \quad D$$

(34) C

(35) B

(36) C

(37) A

(38) The estimated regression equation is given by $y = a + bx$

Where $a = \bar{y} - b\bar{x}$

$\bar{y} = \frac{\Sigma y_i}{n}$, $\bar{x} = \frac{\Sigma \bar{x}}{n}$

$$b = \frac{n\Sigma xy - \Sigma x \Sigma y}{n\Sigma x^2 - (\Sigma x)^2}$$

$\Sigma x = 24$, $\Sigma y = 16$, $\Sigma x^2 = 180$, $\Sigma y^2 = 90$

$\Sigma xy = 75$ and $n = 10$

$$b = \frac{(10 \times 75) - (24 \times 16)}{(10 \times 180) - (24)^2} = \frac{750 - 384}{1800 - 576}$$

$$= \frac{366}{1224} = 0.299$$

$\bar{x} = \frac{\Sigma x}{n} = \frac{24}{10} = 2.4$, $\bar{y} = \frac{\Sigma y}{n} = \frac{16}{10} = 1.6$

$a = 1.6 - 0.299(2.4)$

$a = 1.6 - 0.7176 = 0.8824$

$y = a + bx \Rightarrow y = 0.88241$

$y = 0.8824 + 0.299x$

when $x = 2$

$y = 0.8824 + 0.299(2)$

$y = 0.8824 + 0.598$

$y = 1.4804 \quad A$

(39) ${}^7C_2 (0.4)^2 (0.6)^5$

$21 \times 0.16 \times 0.07776$

$= 0.2613 \quad C$

(40) D

(41) $\mu = 15$, $\sigma = 9$, $SD = \sqrt{\sigma} = \sqrt{9} = 3$

$$P(x < 18) = P\left[\frac{x-15}{3} < \frac{18-15}{3}\right]$$

$$P\left[z < \frac{3}{3}\right] = P[z < 1]$$

$0.5 + 0.3413 = 0.8413 \quad B$

(42) $P[-2 < z < -1]$

$0.4472 - 0.3413$

$= 0.1359 \quad B$

(43) C
(44) A
(45) B
(46) C
(47) B
(48) D
(49) A
(50) Range = Max − Min
57 = Max − 9
Max = 57 + 9 = 66 B
(51) A
(52) B
(53) $\sigma = \left[\frac{1}{9} \times \frac{4}{25}\right]^{\frac{1}{2}} = \left[\frac{4}{225}\right]^{\frac{1}{2}}$
$= \frac{2}{15} = 0.133$ D
([illegible]) A
([illegible]) B
([illegible]) A
([illegible]) C
(58) $(0.61)^2 \times 100 = 0.3721 \times 100$
$= 37.21 = 37\%$ C
(59) C
(60) A
(61) C
(62) C
(63) From standard normal table
$-0.25 = 0.0987,\ 0.45 = 0.1736$
$P(-0.25 < Z < 0.45)$
$0.0987 + 0.1736 = 0.2723$ C
(64) $\mu = 4.35,\ \sigma = \sqrt{\sigma^2} = \sqrt{0.59} = 0.768$
$\frac{Y-\mu}{\sigma} = Z \Rightarrow \frac{Y-4.35}{0.768} = 1.44$
$Y - 4.35 = (1.44)(0.768) \Rightarrow$
$Y - 4.35 = 1.11 \Rightarrow Y = 1.11 + 4.38$
$Y = 5.46$, closest Answer C

(65) D
(66) $p = 0.00005,\ n = 10{,}000$
Poison distribution $= \frac{(np)^x e^{-np}}{x!}$
$np = 10{,}000 \times 0.00005 = 0.5$
P(exactly 2) = $P(x=2)$
$= \frac{(0.5)^2 e^{-0.5}}{2!} = \frac{0.25 \times 0.6065}{2}$
$= 0.07581$ C
(67) 0, 1, 2, 3 ⇒ Random variables
0, appeared once $= \frac{1}{8}$
1, appeared 3 times $= \frac{3}{8}$
2, appeared 3 times $= \frac{3}{8}$
3, appeared once $= \frac{1}{8}$
∴ required values are 0,1,1,1,2,2,2,3
mean $= \frac{0+1+1+1+2+2+2+3}{8}$
$= \frac{12}{8} = \frac{3}{2} = 1.5$ B
(68) Var. of $Y = \frac{\Sigma(x-\bar{x})^2}{n}$
$\frac{(0-1.5)^2}{8} + \frac{(1-1.5)^2}{8} + \frac{(1-1.5)^2}{8} + \frac{(-1.5)^2}{8}$
$+ \frac{(2-1.5)^2}{8} + \frac{(2-1.5)^2}{8} + \frac{(2-1.5)^2}{8} + \frac{(3-1.5)^2}{8}$
$= 0.28125 + 3(0.0625) + 3(0.03125)$
$+ 0.2815 = 0.8424$
closest answer C
(69) D
(70) Probability of picking a defective television $= \frac{15}{240} = 0.0625$.
Since we are considering two television
$\Rightarrow 0.0625 \times 0.0625 = 0.0039$
$\simeq 0.004$ C

FEDERAL UNIVERSITY OF TECHNOLOGY, OWERRI
DEPARTMENT OF STATISTICS
HARMATTAN SEMESTER EXAMINATIONS 2012/2013 SESSION
STA. 211: INTRODUCTION TO STATISTICS AND PROBABILITY - MULTIPLE CHOICE QUESTIONS
INSTRUCTIONS ANSWER ALL THE QUESTIONS TIME 1HR 45 MINS

Each question is followed by options lettered A to D.
Find out the correct option for each question and shade in pencil on your answer sheet the answer space which bears the same letter as the option you have chosen. Give only one answer to each question.

1. For the set of values 2,4,7,2,1,8,9,10,9,6 find the sum of deviations from the median
A -7 B. -7 C. 10 D. -10 E. None of the above.
2. In a dataset, the sum of deviations from the mean is always equal to
A. 1 B. -1 C. 0 D. number of the dataset E. None of the above
3. A tabular summary of a set of a data showing the fraction of the total number of items in sever classes is a.
A. frequency distribution B. Relative frequency distribution C. Frequency
D. Cumulative frequency distribution E. None of the above
4. Qualitative data can be graphically represented by using (a)
A. Histogram B. Frequency polygon C. Ogive D. Bar graph E. None of the above
5. The probability that event A will happen is 0.5 and the B will happen is 0.4. What is the probability that neither A nor B will happen? A. 0.5 B. 0.6 C. 0.1 D. 0.9 E. None of the above
6. A bag contains 2 red, 3 green and 2 blue balls. Two balls are drawn at random. What is the probability that none of the balls drawn is blue? A. 11/21 B. 10/20 C. 2/21 D. 5/21 E. none of the above
7. Fifteen percent of the students in a school of mathematical sciences are majoring in pure mathematics. 20% in industrial mathematics, 35% in computer science, and 30% in statistics. The graphical device(s) which can be used to present these data is (are)
A. A line graph B. Only a bar graph C. Only a pie chart D. Both a bar graph and a pie chart
E. None of the above
8. A researcher is gathering data from four geographical areas designated: South = 1, north = 2 ; East =3; West = 4 .
The designated geographical region represent A. Qualitative data B. Quantitative data
C. Label data D. either quantitative or qualitative data E. None of the above.
9. Tickets numbered 1 to 20 are mixed up and then a ticket is drawn at random. What is the probability that the ticket drawn has a number which is a multiple of 3 or 5? A. ½ B. 2/5 C. 8/15 D. 9/20 E. None of the above
10. A cumulative relative frequency distribution shows
A. The proportion of data, items with values less than or equal to the upper limit of each class
B. The proportion of data items with value less than or equal to the lower limit of each class
C. the percentage of data items with value less than or equal to the upper limit of each class
D. the percentage of data items with value less than the or equal to the lower limit of each class.
E. None of the above
11. If several frequency distribution are constructed from the same data set, the distribution with the widest class width will have the
A. Fewest classes B. Most classes C. Same number of classes as the other distribution since all are constructed from the same data D. Both fewest and most classes E. None of the above
12 . The sum of the relative frequencies for all classes will always equal.
A. The sample size B. The number of classes C. One D. Any value larger than one E. None of the above
13. In a box, there are 8 red, 7 blue and 6 green balls. One ball is picked up randomly. What is probability that it is neither red nor green? A. 8/21 B.3/4 C. 7/19 D. 1/3 E. None of the above
14. The most common graphical presentation of quantitative data is a smallest
A. Histogram B. Bar graph C. Relative frequency D. Pie chart E. None of the above
15. In constructing a frequency distribution, the approximate class width is computed as A. (largest data value-smallest data value) / number of classes B. (Largest data value - data value) / sample size C. (smallest data value - largest data value) / sample size D. Largest data value / number of classes E. None of the above
16. If the correlation between age of an auto and money spent for repairs is +0.90, this means the
A. 81% of the variation in the money spent for repairs is explained by the age of the auto. B. 81% of the money spent for repairs is unexplained by the age of the auto. C. 90% of the money spent for repairs is explained by the age of the auto. D. None of the above E. None of the above

17. A large oil company is studying the number of liters of kerosene purchased per bustomer at self-service pumps. The mean number of litters is10.0 with a standard deviation of 3.0 litters. The median is 10.75litters. What is the coefficient of skewness? A. -1.00 B. -0.75 C. +0.75 D. +1.00. E. None of the above
18. If the coefficient of correlation of the variation equals 0.61, it indicates that the proportion of the variation in the dependent variable explained by the variation in the independent variable is
A. 61% B. 98% C. 37% D. Cannot be determined E. None of the above
19. Suppose the correlation between x and y is -0.95. Which of the following conclusions is correct?
A. The linear relation between x and y is strong, y decreases when x increases. B. the linear relation between x and y is weak, y is decreases when x increases. C. The linear relation between x and y is strong, y increases when x. increases. D. The linear relation between x and y is weak, y increases when x increases. E. None of the above.
20. Consider a sample least squares regression analysis between a dependents variable, (Y) and an independent variable (X). A sample correlation coefficient of -1 (minus one) tells u that
A. There is no relationship between y and x in the sample B. There is no relationship between y and x in the population. C. There is a perfect negative relationship between y and x in the population D. There is a perfect negative relationship between y and x in the sample E. None of the above
21. Data that provide labels or names for categories of like items are known as
A. Qualitative data B. Quantitative data C. Label data D. Category data E. None of the above
22. A tabular method that can be used to summarize the data on two variables simultaneously is called A. Simultaneous equations B. Cross tabulation . C Histogram D. An ogive E. None of the above
23. A graphic presentation of the relationship between two variables is A. An ogive B. A histogram C. Either an ogive or a histogram, depending on the type of data D. A scatter diagram E. None of the above
24. A histogram is said to be skewed to the left if it has a A. Longer tail to the right B. Shorter tail to the right C. Shorter tail to the left D. Longer tall to the Left ' E. None of the above.
25 When a histogram has a longer tail to the right. It is said to be A symmetrical. B. Skewed to the left. C. Skewed to the right D. None of these alternatives is correct. E none of the above.
26. In a scatter diagram, a line that provides an approximation of the relationship between the variables is known as
A. Approximation line. B. Trend line C. Lines of zero intercept D. Lines of zero slope E. none of the above
Exhibit 1-1 The numbers of hours worked (per week) by 400 statistics students are shown below:

Number of hours	Frequency
0-9	20
10-19	80
20-29	200
30-39	100

27. Refer Exhibit 1-1. The class width for this distribution
A. is 9. B is 10. C. is 39, which is the largest value minus the smallest value or 39-0 =39.
D varies from class to class. E. none of the above.
28. Refer to Exhibit 1-1 students working 19 hours or less
A is 80. B is 100. C 180. D 300. E. none of the above.
29. Refer to Exhibit 1-1. The relative frequency of students working 9 hours or less
A is 20. B is 100. C is 0.95. D. 0.05 E. none of the above
30. Refer to Exhibit 1-1 The percentage of students working 19 hours or less is.
A 20%. B 25%. C 75%. D 80%. E. none of the above
31. Refer to Exhibit 1-1. The cumulative relative frequency for the class of 20-29
A is 300. B is 0.25. C is 0.75. D is 0.5. E none of the above
32. Refer to Exhibit 1-1 The cumulative percent frequency for the class 30-39 is,
A 100% B 75% C 50% D 25% E. none of the above
33. Refer to Exhibit 1-1 The cumulative frequency for the class of 20-29
A is 200. B is 300. C is 0.75. D is 0.5. E. none of the above
34. Refer to Exhibit 1-1. If a cumulative frequency distribution is developed for the above-data, the last class will have a cumulative frequency of A 100. B 1. C 30-39. D 400. E. none of the above
35. Refer to Exhibit 1-1. The percentage of students who work at least 10 hours per Week is
A 50%. B 5%. C 95%. D. 100% E. none of the above
36. Refer to Exhibit 1-1. The midpoint of the last class is
A 50. B 34. C 35. D 34.5 E. none of the above

Exhibit 1-2. A survey of 800 graduation-students in SOSC resulted in the following cross tabulatic regarding their undergraduate major and whether or not they plan to go graduate school.

	Undergraduate major			
Graduate School	MCS	GEO	Others	Total
Yes	70	84	126	280
No	182	208	130	520
Total	252	292	256	800

37. Refer to Exhibit 1-2. What percentage of the students does not plan to go to graduate school?
A. 280 B. 520. C. 65. D 32 E none of the above
38. Refer to Exhibit 1-2 What percentage of the students' undergraduate major is GEO?
A. 292. B 520. C 65. D 36.5 E none of the above
39. Refer to Exhibit 1-2. Of those students who are majoring in MCS, what percentage plans to go to graduate school? A 27.78. B 8.75 C 70. D 72.22. E none of the above
40. Refer to Exhibit 1-2. Among the students who plan to go to graduate school, what percentage indicated "other" majors? A 15.75. B 45. C. 54. D. 35. E none of the above
41. When an investigator uses data, which have already been collected by others, such data are called
A double data. B primary data. C secondary data. D secondary data. E none of the above
42. One of the following is not a method of data collection
A interview. B sample. C. observation D. questionnaire E. none of the above
43. How is a cross-sectional set of data collected from the units in the population?
A. once B twice. C. thrice D. many times. E. none of the above.
44 Which of the following is not an advantage of grouped frequency distribution?
A. economy of data. B organization. C. loss of information. D condensing of data. E. none of the above
45. When the mean gives a decimal value for discrete data, one of the following is Best used as a measure of central tendency. A mean. B mode. C. media. D. quartile. E. none of the above
46. One of the following is not true about histograms:
A. The spaces between adjacent bars must be equal B. The bars are joined together
C. The width of the bar may not be equal D. class boundaries are necessary E. none of the above.

Exhibition 1-3: given the data set below, use the information to answer the question that follows

Class Intervals	Frequency	Class Boundaries	Cumulative Frequency	Class Mark
50-57	8	49.5-57.5	8	53.5
58-65	14	57.5-65.5	22	61.5
66-73	26	65.5-73.5	48	69.5
74-81	18	73.5-81.5	66	77.5
82-89	12	81.5-89.9	78	85.5
90-97	2	89.5-97.5	80	93.5

Note that class width = upper class boundary - lower class boundary not upper class limit - lower class limit

47. Obtain the mean of the data set.
A 71.3 B 73.3. C 70.3. D 71 E. none of the above
48. Obtain the media of the data set.
A 73.03. B 75.04. C 71.04. D 76. E none of the above.
49. Obtain the model value.
A 65.5 B 66.5. C. 67.1 D. 66.1. E. none of the above
50. Obtain the standard deviation.
A 9.03. B 10 C. 10.04. D 71.3. E none of the above
51. Obtain the coefficient of variation.

A 10.04. B 14.08. C 13.08. D. 13. E. none of the above

52. Obtain the first quartile.
A 64.36 B 64. C 64.5. D 62.36. E none of the above

53. Obtain the upper quartile.
A 73.5. B 78.5. C 78.08. D 78.83. E. none of the above

54. Obtain the quartile coefficient of skewness.
A 7.7. B 0.06. C 0.077. D 1.077. E. none of the above

55. Obtain the Bowley's coefficient of skewness
A 0.88. B 0.77. C 0.78. D 0.078 E. none of the above

56. If A and B are not mutually exclusive, one of the following is true.
A. $P(A \cup B) = P(A)+P(B) - F \cap B)$ B. $P(A) + P(B)$. C. $P(A \cap B) = P$.
D. $P(A \cup B) = P(A)+P(B)+P(A \cap B)$ E. None of the above

57. Let x be a Binomial random variable with parameters, n and p. Determine n if p= 0.25.
A. 5 B.3 C. 4 D. 6 E. None of the above

58. Let x be a Poisson random variable with parameter μ. If the probability $P(0)= 0.04979$, findμ.
A 0.04. B 0.049. C 3. D 1. E. none of the above

59. Which of the following is not true about correlation coefficient between x and y?
A. it is related to the covariance between x and y B. it is related to the standard deviation between x and y.
C it lies between -1 and 1 D. It lies between 0 and 1 E. none of the above

60. Which of the formulas can be used to predict β_0 in regression problem?
A $y + \beta_1 x$. B. $y\ \beta_1 (x_t - x)$. C. $\sum_{t=1}^{n} x_t y_t - \sum_{t=1}^{n} X_1 \sum^{n} = 1$. D. $y - \beta_1 X)$ E. none of the above

61. To determine how well a linear regression model describes relationship between a dependent and an independent variable, one of the following may be used; A. scatter plot. B. line of best fit. C. coefficient of determination. D. coefficient of variation. E. none of the above

62. A question on multiple - choice test has five options.
What is the probability that you guess the correct option to the question?
A 1/4 B 2/5 C 1/15 D 1/5 E. None of the above

63. In a set of nine different numbers, which of the following cannot affect the value of the media
A. doubling each number. B. increasing each number by 10 C. increasing the smallest number only. D. increasing the largest number only E. none of the above

64. Okoro has a bag that contains 4 red, 6 yellow,2 blue, and 4 green balls,) If he reaches into the bag and removes a ball without looking, what is the probability that it will not be yellow?
A 1/8 B ¼ C 3/8 D 5/8 E. None of the above

65. A normal distribution of data has a mean of 34 and standard deviation of 5. Find the probability that a random value x is greater than 24, A. 97.5% B. 13.5% C. 0.34 D. 0.02 E. None of the above.

Exhibit 1-4: A chocolate company makes boxes of assorted chocolates, 40% of which are dark chocolate on average. The production line mixes the chocolates randomly and packages 10 per box.

66. Using Exhibit 1-4, what is the probability that at least 3 chocolates in a box are dark
A. 0.1673 B. 0.8327 C. 0.4000 D. 0.6000 E none of the above

67. Use Exhibit 1-4 to obtain the expected number of dark chocolates in a box?
A.2 B.3 C.5 D.4 E. none of the above

68. According to a survey, 64% of lecturers in a certain university feel that their college years were the most exciting. A Survey of 300 random lecturers was conducted. What is the portability that least 200 of the responses will agree?
A 94% B 84% C 16% D 6% E None of the above

69. The following are grades earned by students on a mathematics test:
76, 84, 91, 75, 83, 82, 65, 94, 90, 71, 92, 84, 83, 88, 80, 78, 84, 89, 95, 93.
Calculate the standard deviation of the test scores
A 7.82 B 8.03 C 8.23 D 8.75 E None of the above

70. A random variable Y follows a normal distribution with a mean of 8 and a standard deviation
The probability that Y is between 1.48 and 15.56 is
A 0.0222 B 0.4190 C 0.5222 D 0.9190 E None of the above

STA 211 SOLUTION 2012/2013

① To find the sum of deviations from the median, we have:

$$\text{Median} = \frac{X\left(\frac{n}{2}\right) + X\left(\frac{n}{2}+1\right)}{2}$$

Array data; 1,2,2,4,6,7,8,9,9,10
where n=10

$$X_{med} = \frac{X_{(5)} + X_{(6)}}{2} = \frac{6+7}{2} = 6.5$$

Sum of deviation $= \Sigma(X - X_{med})$
$= (1-6.5)+(2-6.5)+(2-6.5)+(4-6.5)+(6-6.5)+(7-6.5)+(8-6.5)+(9-6.5)+(9-6.5)+(10-6.5)$
$= -5.5+(-4.5)+(-4.5)+(-2.5)+(-0.5)+0.5+(1.5)+2.5+2.5+3.5$
$= -7$ B

② C
③ B
④ E
⑤ $P(A) = 0.5$, $P(B) = 0.4$
Probability that neither A nor B will happen $\Rightarrow P(\overline{A \cup B}) = 1 - P(A \cup B)$
But $P(A \cup B) = P(A) + P(B)$ {mutually exclusive}
$= 0.5 + 0.4 = 0.9$
$P(\overline{A \cup B}) = 1 - 0.9 = 0.1$ C

⑥ A bag contain 7 balls
Red = 2 , Green = 3, Blue = 2
All balls drawn are $= {}^7C_2 = 21$
The probability of 2 balls drawn not blue $\frac{{}^2C_2 + {}^3C_2 + {}^2C_0}{{}^7C_2}$
$= \frac{1+3+1}{21} = \frac{5}{21}$ D

⑦ D
⑧ A
⑨ $S = \{x : 1 \le x \le 20\}$
let A = multiples of 3
B = multiples of 5
$A = \{3, 6, 9, 12, 15, 18\}$
$B = \{5, 10, 15, 20\}$
$A \cap B = \{15\}$
Probability of ticket drawn is a multiple of 3 or 5 will be;
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
$= \frac{6}{20} + \frac{4}{20} - \frac{1}{20}$
$= \frac{9}{20}$ D

⑩ A
⑪ A
⑫ C
⑬ A bag contains 21 balls
Red = 8 , blue = 7 , Green = 6
Probability that one ball selected is neither red nor green
$P(\overline{R \cup G}) = P(B) = \frac{7}{21} = \frac{1}{3}$ D

⑭ A
⑮ A
⑯ C
⑰ Pearsons coefficient of skewness
$= \frac{3(\text{mean} - \text{median})}{\text{standard deviation}}$
$= \frac{3(10 - 10.75)}{3.0}$
$= \frac{3(-0.75)}{3} = -0.75$ B

⑱ A
⑲ B
⑳ A
㉑ A

22 B
23 D
24 D
25 C
26 E
27 B

28 $n(x \le 19 \text{hours}) = 80+20 = \underline{100}$ B

29 Relative frequency $Rf(x \le 9\text{hours})$
$= \frac{20}{400} = 0.05$ D

30 Percentage of the student working:
$(x \le 19) = \frac{100}{400} \times 100 = 25\%$ B

31 The cummulative frequency of class (20 − 29) will be
$\frac{200+80+20}{400} = \frac{300}{400}$
$= 0.75$ C

32 The cummulative percent frequency of the class (30−39) is:
$\left(\frac{200}{400} + \frac{80}{400} + \frac{20}{400} + \frac{100}{400}\right) \times 100$
$= \frac{400}{400} \times 100 = 100\%$ A

33 The cummulative frequency for the class of (20−29) is
$20 + 80 + 200 = 300$ B

34 The cummulative frequency for the last class (30−39) is
$100 + 200 + 80 + 20 = 400$ D

35 The percentage of student who work at least 10 hours $(x \ge 10)$
$= \frac{80+200+100}{400} \times 100$
$= 95\%$ C

36 The mid point of the last class (30 − 39) is : $\frac{30+39}{2} = 34.5$ D

37 The percentage of students that does not plan to go to graduate school $= \frac{520}{800} \times 100 = 65\%$ C

38 % student undergraduate major in GEO $= \frac{292}{800} \times 100 = 36.5\%$ D

39 % majoring in MCS and plans to go to graduate school $= \frac{70}{800} \times 100 = 8.75\%$ B

40 % who plans to go to graduate school indicated "other" majors
$\frac{126}{800} \times 100 = 15.75\%$ A

41 C
42 B
43 B
44 C
45 B
46 E

47

Class Interval	f	Class Boundary	Cf	x	fx	fx²
50 − 57	8	49.5−57.5	8	53.5	428	22998.0
58 − 65	14	57.5−65.5	22	61.5	861	52951.5
66 − 73	26	65.5−73.5	48	69.5	1807	126586.5
74 − 81	18	73.5−81.5	66	77.5	1395	108112.5
82 − 89	12	81.5−89.5	78	85.5	1026	87723.0
90 − 97	2	89.5−97.5	80	93.5	187	17484.5
	80				5704	415756.0

Mean $= \frac{\Sigma fx}{\Sigma f} = \frac{5704}{80} = 71.3$ A

48 Median $= l_i + \left[\frac{\frac{N}{2} - C_{f-1}}{f_m}\right] \times C$
$l_i = 65.5, N = 80, C_{f-1} = 22, f_m = 26, C = 8$
Median $= 65.5 + \left[\frac{\frac{80}{2} - 22}{26}\right] \times 8 = 71.04$ C

49 Mode $= l_i + \left[\frac{D_1}{D_1 + D_2}\right] \times C$
$D_1 = f - f_1 = 26 - 14 = 12$
$D_2 = f - f_1 = 26 - 18 = 8$
$l_i = 65.5, \ C = 8$
Mode $= 65.5 + \left[\frac{12}{12+8}\right] \times 8 = 70.3$ E

50 Standard deviation =
$SD = \sigma = \sqrt{\frac{\Sigma fx^2}{\Sigma f} - \left(\frac{\Sigma fx}{\Sigma f}\right)^2}$
$= \sqrt{\frac{415,756.0}{80} - \left(\frac{5704}{80}\right)^2}$
$= 10.64$ E

51 Coefficient of variation
$\frac{S.D}{\bar{x}} \times 100 = \frac{10.64}{71.3} \times 100$
$= 14.92\%$ E

52 $Q_1 = L_i + \left[\frac{\frac{N}{4} - C_{f-1}}{f_{Q_1}}\right] \times C$
$Cf_{-1} = 8, \ fQ_1 = 14, \ l_i = 57.5, \ C = 8$
$Q_1 = 57.5 + \left[\frac{\frac{80}{4} - 8}{14}\right] \times 8$
$= 64.36$ A

53 $Q_3 = l_i + \left[\frac{\frac{3N}{4} - C_{f-1}}{fQ_3}\right] \times C$
$l_i = 73.5, \ C_{f-1} = 48, \ f_{Q_3} = 18, \ C = 8$
$Q_3 = 73.5 + \left[\frac{\frac{3(80)}{4} - 48}{18}\right] \times 8$
$= 78.83$ D

54 Quartile coeff of skewness $= \frac{Q_3 - 2Q_2 + Q_1}{Q_3 - Q_1}$
$Q_3 = 78.83$, $Q_2 = \text{Median} = 70.3$, $Q_1 = 64.36$
$\therefore Q_{SK} = \frac{78.83 - 2(70.3) + 64.36}{78.83 - 64.36} = 0.179$ E

55 Bolwley's coefficient of skewness is the same as quartile coefficient of skewness, thus $= 0.179$ E

56 E

57 $1 = np \Rightarrow 1 = n(0.25) \Rightarrow n = \frac{1}{0.25} = 4$ C

58 For poisson distribution, $P(x) = \frac{\mu^x e^{-\mu}}{x!}$
if $P(0) = 0.04979 = \frac{\mu^0 e^{-\mu}}{0!}$
We look for μ such that:
$0.04979 = \frac{1 \cdot e^{-\mu}}{1}$
$0.04979 = e^{-\mu}$
$-\mu = \ln[0.04979]$
$-\mu = -2.9999$
$\mu = 2.9999 \simeq 3$ C

59 D
60 E
61 B

62 The probability that you guess the correct option; $P(x:K) = \frac{1}{K}$
if $K = 5$, then $P(x:K) = \frac{1}{5}$ D

63 B

64 Okoro'b bag contains 16 balls
Probability of not picking a yellow ball is $P(\bar{Y}) = 1 - P(Y)$
$= 1 - \left(\frac{6}{16}\right) = \frac{10}{16} = \frac{5}{8}$ D

65 $Z = \frac{X - \mu}{\sigma} \Rightarrow P[X \ge 24]$
$= P\left[\frac{X-34}{5} \ge \frac{24-34}{5}\right] = P[Z \ge -2]$
$= 0.5 - (-0.4773)$
$= 0.9773$ or 97.73% A.

66 The probability that at least 3 chocolates in a box are dark,
$\lambda = 40\% = 0.4$
$n = 10$ chocolates per box
$P(x \ge 3) = 1 - P(x < 3)$
$= 1 - [P(x=0) + P(x=1) + P(x=2)]$

But $P(x:\lambda) = \dfrac{\lambda^x e^{-\lambda}}{x!}$

thus,

$P(x \geq 3) = 1 - \left[\dfrac{0.4^0 e^{-0.4}}{0!} + \dfrac{0.4e^{-0.4}}{1!} + \dfrac{0.4^2 e^{-0.4}}{2!}\right]$

$1 - (0.6703 + 0.2681 + 0.0536)$

$= 0.00797$ E

67 To obtain the expected number of dark chocolate in a box will be ;

$E(X) = \lambda$ and $\lambda = np$

such that $E(x) = np$

$= 10 \times 0.4$

$= 4$ D

68. Probability that at least 200 responses will agree

$= \dfrac{200}{300} \times 100 = 66.7\%$ E

69. $S \cdot D = \sqrt{\dfrac{\Sigma f(x-\bar{x})^2}{\Sigma f}}$

State the array data ;

65, 71, 75, 76, 80, 82, 83, 83, 84, 84, 84, 88, 89, 90, 91, 92, 93, 94, 95

put in frequency distribution table

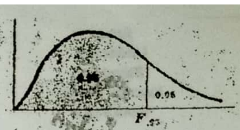
x	f	fx	$(x-\bar{x})$	$(x-\bar{x})^2$	$f(x-\bar{x})^2$
65	1	65	−18.85	355.3225	355.3225
71	1	71	−12.85	165.1225	165.1225
75	1	75	−8.85	78.3225	78.3225
76	1	76	−7.85	61.6225	61.6225
78	1	78	−5.85	34.2225	34.2225
80	1	80	−3.85	14.8225	14.8225
82	1	82	−1.85	3.4225	3.4225
83	2	166	82.15	6748.6225	13497.2450
84	3	252	168.15	28274.4225	84823.2675
88	1	88	4.15	17.2225	17.2225
89	1	89	5.15	26.5225	26.5225
90	1	90	6.15	37.8225	37.8225
91	1	91	7.15	51.1225	51.1225
92	1	92	8.15	66.4225	66.4225
93	1	93	9.15	83.7225	83.7225
94	1	94	10.15	103.0225	103.0225
95	1	95	11.15	124.3225	124.3225
	20	1677			100053.7325

$\bar{x} = \dfrac{\Sigma fx}{\Sigma f} = \dfrac{1677}{20} = 83.85$

$S \cdot D = \sqrt{\dfrac{100053.7325}{20}} = 7.82$ A

70. Incomplete information. (SD not given)

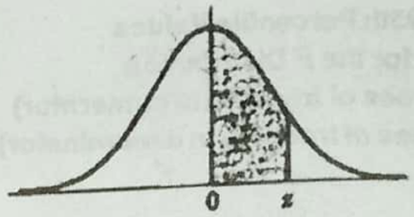
95th Percentile Values for the F Distribution
(ν_1 degrees of freedom in numerator)
(ν_2 degrees of freedom in denominator)



ν_2 \ ν_1	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
1	161	200	216	225	230	234	237	239	241	242	244	246	248	249	250	251	252	253	254
2	18.5	19.0	19.2	19.2	19.3	19.3	19.4	19.4	19.4	19.4	19.4	19.4	19.4	19.5	19.5	19.5	19.5	19.5	19.5
3	10.1	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.37
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.54
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.40
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25	2.21
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.04
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.19	2.15	2.10	2.06	2.01	1.96
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.84
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.92	1.87	1.81
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84	1.78
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05	2.01	1.96	1.91	1.86	1.81	1.76
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79	1.73
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.82	1.77	1.71
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.07	1.99	1.95	1.90	1.85	1.80	1.75	1.69
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.06	1.97	1.93	1.88	1.84	1.79	1.73	1.67
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.04	1.96	1.91	1.87	1.82	1.77	1.71	1.65
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.03	1.94	1.90	1.85	1.81	1.75	1.70	1.64
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.89	1.84	1.79	1.74	1.68	1.62
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84	1.79	1.74	1.69	1.64	1.58	1.51
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75	1.70	1.65	1.59	1.53	1.47	1.39
120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96	1.91	1.83	1.75	1.66	1.61	1.55	1.50	1.43	1.35	1.25
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.52	1.46	1.39	1.32	1.22	1.00

Source: E. S. Pearson and H. O. Hartley, *Biometrika Tables for Statisticians*, Vol. 2 (1972), Table 5, page 178, by permission.

Areas Under the Standard Normal Curve from 0 to z



z	0	1	2	3	4	5	6	7	8	9
0.0	.0000	.0040	.0080	.0120	.0160	.0199	.0239	.0279	.0319	.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0754
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2258	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2518	.2549
0.7	.2580	.2612	.2642	.2673	.2704	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2996	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.4484	.4495	.4505	.4515	.4525	.4535	.4545
1.7	.4554	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936
2.5	.4938	.4940	.4941	.4943	.4945	.4946	.4948	.4949	.4951	.4952
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	.4963	.4964
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.4979	.4980	.4981
2.9	.4981	.4982	.4982	.4983	.4984	.4984	.4985	.4985	.4986	.4986
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990
3.1	.4990	.4991	.4991	.4991	.4992	.4992	.4992	.4992	.4993	.4993
3.2	.4993	.4993	.4994	.4994	.4994	.4994	.4994	.4995	.4995	.4995
3.3	.4995	.4995	.4995	.4996	.4996	.4996	.4996	.4996	.4996	.4997
3.4	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4998
3.5	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998
3.6	.4998	.4998	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999
3.7	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999
3.8	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999
3.9	.5000	.5000	.5000	.5000	.5000	.5000	.5000	.5000	.5000	.5000